HEAVY METAL TESTED & CERTIFIED STANDARDS

Heavy Metal Tested & Certified

Seafood Program Manual

2026 Edition

Karen Pendergrass
Standards Architect, The Paleo Foundation

Paleo Certified, Inc. dba The Paleo Foundation

Heavy Metal Tested & Certified (HMTc)

Seafood Program Manual

2026 Edition

Karen Pendergrass Standards Architect, Paleo Foundation ORCID: 0000-0002-2348-7259

Paleo Certified, Inc. dba The Paleo Foundation Orlando, Florida

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Categories covered: Predatory Marine Fish · Non-Predatory Marine Fish · Freshwater Fish · Shellfish · Seaweed & Kelp Foods

Edition

2026 Edition. First bound publication of the HMTc Seafood Program Manual.

The HMTc methodology has evolved through documented revisions since 2024. The current edition reflects refinements to the per-analyte percentile calibration, the addition of the row-pass framework, the operationalization of the Standards Ratchet Mechanism with explicit trigger conditions and step-size caps, and the formalization of the Confidential Remediation Track. Version history is maintained under document control. The 2026 Edition is the first to bind the program's foundational framework, governance, standards, and operational schedule into a single volume.

Operating documents controlled separately

The Program Manual reproduces the program's foundational documents at the time of binding. Two of those documents are under independent version control and may be revised between editions of the manual:

- **HMTc Program Governance Policy.** The operational constitution. Reproduced in full as Part 2 of this manual. Revised version available at the Paleo Foundation's controlled-document registry.
- **HMTc Lot Testing Schedule.** The operational testing protocol. Reproduced in full as Part 4 of this manual. Revised version available at the controlled-document registry.

When discrepancies arise between this manual and the controlled documents, the more recent controlled document governs for operational purposes. The Program Operator will issue a formal Technical Clarification to resolve any ambiguity. This manual edition's reproductions are authoritative as of the publication date below.

Methodology continuity

The 2026 Edition rests on a body of published research describing the HMTc framework's design rationale. Three peer-track preprints are cited throughout this manual and are available on Zenodo:

- *The Counterproductive Consequences of Public Exposé Testing* (doi: 10.5281/zenodo.19470572). The structural argument for why the program exists.
- *The Cost of Operating Without Credible Third-Party Heavy-Metal Certification* (doi: 10.5281/zenodo.18903738). The financial argument for the cost of inaction.
- *Certification as a Framework for Reducing Heavy Metal Exposure in Infant and Child Foods* (doi: 10.5281/zenodo.18905821). The design rationale for the certification architecture.

The structural design principles in these preprints remain canonical. Where specific calibration values in the preprints have evolved (notably the 80th-percentile achievability framing in *Certification as a Framework* Section 11), the current operational values appear in this manual and in the Governing Principles, which supersede the earlier framing for operational purposes.

Evidence base

Every numeric claim, regulatory reference, and toxicological framing in this manual traces to specific source pages on heavymetalindex.com, the public curated corpus of the heavy-metals-in-food literature that the Paleo Foundation operates as a separate property. Heavymetalindex.com is architecturally distinct from the HMTc certification program. The one-way reference architecture (HMTc cites HMI; HMI does not endorse HMTc) is what gives the standards their authority. Adversarial readers can audit any limit in this manual by following the references to the underlying source pages. Part 1 chapter "How HMTc Knows What It Knows" describes this dependency in detail.

Publication metadata

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Revision 2026-06-18. This printing incorporates the Expedited Baseline via Prior-Data Credit pathway added to the foundational documents (Governance Policy §4.5, reproduced in Part 2.3; Lot Testing Schedule §4.3, reproduced in Part 4). The reproduced Governance Policy and Lot Testing Schedule are updated to version 2.1. A new version DOI is assigned to this printing on deposit.

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Intellectual property protection. The HMTc Certification Mark is pending registration with the United States Patent and Trademark Office (Serial No. 99307306). The heavymetalindex.com corpus that this manual references is protected under U.S. copyright law and U.S. trade-secret doctrine, with additional protection planned through an EU corporate entity in a future Edition; substantial extraction or re-utilization of the corpus without authorization is prohibited.

How to cite this document

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For citations to specific chapters or sections, append the part and section identifier (e.g., “Part 2.10 §12.3, *Standards Ratchet Trigger Conditions*”). For citations to specific HMTc limit values, cite the relevant Part 3 chapter

by its product slug (e.g., “Part 3.1, *Fish — Marine, Predatory*, Master Limit Table”) and the chapter’s Sources Cited section for the underlying source attribution. Numbered references in this manual (Part 5.4) follow IEEE-adjacent format; readers needing alternative bibliographic format (APA, Vancouver, Chicago) may request the reference list as a structured BibTeX or RIS file from the Program Operator.

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How to Read This Document

This Program Manual is structured to serve four audiences simultaneously. Each audience has a different reason for opening the book and a different reading path that leads to what they need. Reading the entire volume end-to-end gives the most complete picture; the reading orders below are for readers who want the relevant content first.

For Brand Legal Teams

You are reading this manual to evaluate whether HMTc is defensible enough to recommend that your client enter the program. You will be asked, by a CEO or general counsel, whether participation creates litigation exposure or reduces it. Your conclusion needs to survive a hostile reviewer at a competitor or a plaintiff's expert in a deposition.

Recommended reading order:

1. **Part 1** in full. The structural argument explains why the program exists; the financial argument quantifies what your client faces without it.
2. **Part 2.4 (The Confidential Remediation Track)**. The most important single chapter for legal defensibility. The track exists so that brands not yet at certification standards can enter, build a documented improvement record, and shift the legal narrative from "knew and did nothing" to "knew, enrolled, and has third-party-verified evidence of continuous improvement."
3. **Part 2.6 (Decision Rules under Measurement Uncertainty)** and **Part 2.7 (Data Governance and Integrity)**. These chapters describe the analytical and documentary discipline that makes the program's evidence record admissible.
4. **Part 5.2 (Program-Level Adversarial Read)**. The structured response to every category of attack you might face against the

program. Read this chapter if your client is being asked to defend their participation against a hostile inquiry.

5. **The product chapter in Part 3** matching your client's category. This is the chapter that will be attached as an exhibit to their licensing agreement.

For Retailer Compliance and QA Teams

You are reading this manual to decide whether to recognize HMTc as a basis for shelf access or assortment approval. You need to confirm that the program is rigorous enough to substitute for your own ingredient-level surveillance, and that participating brands meet a meaningfully higher bar than non-participating brands.

Recommended reading order:

1. **Part 2.0 (Program Flow Diagram)** through **Part 2.3 (Certification Status Framework)**. The architecture of the program and the meaning of the certification mark at each status level.
2. **Part 2.6 (Decision Rules)** and **Part 4 (Lot Testing Schedule)**. The analytical and operational rigor that backs the mark.
3. **Part 2.9 (Supplier Intelligence Sharing)**. The cross-brand collective intelligence system that detects supply-chain problems before they reach finished products.
4. **Part 2.10 (Standards Ratchet Mechanism)**. The mechanism that tightens limits over time as the certified market improves, which is what makes HMTc a forward-moving signal rather than a static gate.
5. **Part 5.5 (Recognition Framework)**. How HMTc compares to USDA Organic, Non-GMO Project, NSF/ANSI 173, and other adjacent programs you may already recognize.

For Regulators and Public Health Officials

You are reading this manual to assess whether HMTc represents legitimate industry self-regulation that complements federal action levels and EU maximum levels, or whether it is a marketing program dressed as a standard. You need to confirm that the methodology is traceable to

documented inputs and that the program drives measurable reduction over time.

Recommended reading order:

1. **Part 1.5 (How HMTc Knows What It Knows)**. The evidence base lives at heavymetalindex.com, a public curated corpus of the open literature, architecturally separate from the certification program. Every limit in this manual traces to specific source pages on that wiki.
2. **Part 2.1 (Purpose and Core Principles)**, particularly the framing of action levels as risk-management tools and not safety thresholds.
3. **Part 2.2 (Scope and Analyte Tiering)**. The Tier 1 / Tier 2 metal classification with toxicological rationale for each metal. Includes the aggregate-exposure rationale showing why limits below regulatory action levels are warranted even when a single product complies with federal guidance, because consumers eat across many products and exposures sum.
4. **Part 2.10 (Standards Ratchet Mechanism)**. The data-driven mechanism that commits the program to tighten limits as industry capability improves. Trigger conditions, step-size cap, transition rules, and forward-forecast obligation are all specified.
5. **Part 5.1 (How HMTc Standards Are Set)** and **Part 5.4 (Consolidated Sources Index)**. The methodology described in principle, with the full source attribution.

For Adversarial Readers

You are reading this manual to find the gaps. You are looking for methodology underspecification, missing toxicological anchors, weak chain-of-custody language, ratchet gaming opportunities, conflict-of-interest framings, or claims that overreach what the literature supports. You will brief your conclusions to a court, a state attorney general's office, or a competing program's leadership.

Recommended reading order:

1. **Part 5.2 (Program-Level Adversarial Read)**. Start here. This chapter enumerates the categories of attack the program anticipates, locates each in the manual, and answers them on the merits. Reading

this first will save you time identifying which attacks have already been responded to and which (if any) you can advance further.

2. **Part 1.5 (How HMTc Knows What It Knows)**. The evidence base claim. Verify the heavymetalindex.com corpus, the architectural separation from HMTc, and the source-page traceability for limits.
3. **Part 2.10 (Standards Ratchet Mechanism)**. Check whether the trigger conditions, step cap, and transition rules can be gamed.
4. **Each product chapter in Part 3** for the per-product adversarial reads. The product-level attacks are answered chapter by chapter; the program-level attacks live in Part 5.2.
5. **Part 4 (Lot Testing Schedule)**. Look for chain-of-custody, lab-shopping prohibition, and data-integrity language. The program's discipline is operationalized here.

Reading the entire volume

If you have time to read the manual end-to-end, the structure is:

- **Part 1** makes the case for why the program exists, financially and structurally, and introduces the public evidence base.
- **Part 2** is the program's operational constitution.
- **Part 2.5** demonstrates how brands and QA teams use heavymetalindex.com day-to-day.
- **Part 3** publishes the per-category standards. Each chapter is detachable as a licensing exhibit.
- **Part 4** specifies the testing schedule and procedures.
- **Part 5** documents methodology, anticipates adversarial attacks, glosses terms, indexes sources, compares to adjacent programs, summarizes licensing, and closes with the colophon.

Front matter is short. Part 1 sets up the reader; Part 5 should leave them with no open questions. If you find one, the Program Operator's contact is in the colophon.

Part 1, Why HMTc Exists

1.1 Introduction

Before this manual specifies a single concentration limit, a single status threshold, or a single testing protocol, the reader is owed an answer to a prior question: why does this program need to exist? Federal action levels are public. State disclosure laws are operative. Plaintiffs' attorneys are organized. Retailer specifications are in force. Why should brand legal teams, retailer compliance staff, regulators, and adversarial reviewers spend time on a third-party certification program when the regulatory environment is already adversarial?

The answer is direct: the existing enforcement environment does not reduce heavy metal contamination in food. It punishes the brands trying to reduce it, rewards the brands generating no data, and converts every documented improvement effort into discoverable evidence for the next class action. The result is a regulatory and litigation system that produces alarm, settlements, and reputational damage without producing measurable contamination reduction. Part 1 of this manual explains, in detail, why that is true and what a working alternative requires.

This part has five chapters. Chapter 1.2 lays out the **structural problem**: the system of public exposé testing that has emerged as a de facto enforcement mechanism in the food-safety landscape is analytically incoherent, economically backward, and demonstrably ineffective at reducing contamination. The argument is drawn from the published preprint *The Counterproductive Consequences of Public Exposé Testing*. Chapter 1.3 lays out the **financial problem**: the cost of operating without a credible certification framework, modeled scenario by scenario, for brands of three different revenue tiers. Conservative modeling produces one-year exposure ranging from $1.6M for a small brand to $187M+ for

a large one. The argument is drawn from the published preprint *The Cost of Operating Without Credible Third-Party Heavy-Metal Certification*.

Chapter 1.4 names the **structural alternative**: a confidential, staged, ratcheting third-party certification framework, the design rationale of which is summarized from the published preprint *Certification as a Framework for Reducing Heavy Metal Exposure in Infant and Child Foods*. That framework is the HMTc program. Part 2 of this manual specifies it; Part 1 makes the case that it is needed.

Chapter 1.5 introduces the **evidence base**: heavymetalindex.com, the public curated corpus of the heavy-metals-in-food literature that the Paleo Foundation operates as a separate property. Every concentration limit in this manual traces to specific source pages on that wiki. An adversarial reader can audit any limit by following the references. The architectural separation between heavymetalindex.com and the HMTc certification program is what gives the standards their authority; that authority is described in detail in Chapter 1.5 and tested again in Part 5.2.

The structure of the argument is intentional. Brand legal teams reading only the standards in Part 3 cannot evaluate whether the program is rigorous enough to recommend; they need the structural and financial case in Part 1 first. Regulators reading only the methodology in Part 5 cannot place the program in the context of why it exists; they need the public-health failure analysis in Chapter 1.2 first. Adversarial readers attacking the program need to be told, before they encounter any specific number, what the program claims to know and where the evidence lives, so they can attack the right surface rather than wasting time on positions the program does not hold.

The reader who finishes Part 1 should understand three things. The current enforcement environment is structurally incapable of reducing contamination. The cost of inaction is large enough that even a small brand cannot rationally remain outside a program that reduces probability and severity of contamination events. And the evidence base for the program's standards is publicly auditable. With those three things established, Part 2′s operational specification can be read as the working specification of a program that exists for a reason, rather than as a marketing artifact dressed as a standard.

1.2 The Structural Problem: Why Public Exposé Testing Fails

The detection of heavy metals in consumer food products is not a question of *if* but of *how much, how often, and in what context*. The United States Food and Drug Administration's Total Diet Study reports lead in 15% of food samples, arsenic in 43%, cadmium in 33%, and mercury in 10% [1]. These are not anomalies attributable to negligent manufacturers; they reflect the geochemistry of agricultural soils, the hydrology of irrigation systems, and the industrial legacy of environmental contamination. Heavy metals are, in the strictest analytical sense, ubiquitous in the food supply.

Against this backdrop, a pattern of consumer advocacy has emerged that this manual calls *unstructured public exposé testing*. Independent actors, typically bloggers, consumer watchdog organizations, or self-described whistleblowers, privately commission heavy metal testing of commercially available products, then publicly disclose the results in formats designed to maximize consumer alarm and media attention. The tested brands are identified by name. The results are presented without population-level context, without category benchmarking, without measurement uncertainty disclosure, and without the analytical framework necessary to distinguish a meaningful public health concern from a statistically unremarkable detection.

This practice is counterproductive. It does not reduce heavy metal contamination. It does not improve food safety. It does not protect consumers. What it does, systematically and demonstrably, is create perverse incentives that suppress the very transparency, data sharing, and collaborative remediation that are prerequisites for actually driving contamination levels down. It punishes brands that may be the cleanest in their category. It rewards opacity. And it provides litigation scaffolding

that plaintiffs' attorneys can operationalize against the exact companies that are trying to do the right thing.

The argument that follows is structured in four parts: the anatomy of an exposé and what it omits, the game-theoretic and naming-and-shaming literature that predicts its failure, the supply-chain dynamics that make brand-level attribution analytically incoherent, and the litigation amplification pathway that converts published exposé results into class actions and Proposition 65 settlements. The full peer-track treatment appears in the published preprint cited at the close of this chapter.

What is present in an exposé, and what is missing

The typical unstructured exposé follows a recognizable pattern. An independent actor purchases a small number of products from retail channels, submits them to a laboratory (often an EPA-certified facility, which lends an aura of governmental authority to the results), and receives a certificate of analysis reporting detected concentrations of one or more heavy metals. The results are then published on a blog, social media platform, or advocacy website, frequently organized into categories such as "safest," "better," and "not our favorite", language that implies a consumer-protection framework without the methodological rigor to support one.

What is present in this methodology is a *detection*. What is absent is everything required to make that detection meaningful.

There is no **population-level benchmarking**. A chickpea product reporting 15 ppb lead may be the cleanest product in an entire category where the median is 40 ppb. Without population-level data, the distribution of results across all comparable products in the category, the reported number is analytically orphaned. It cannot be contextualized, and therefore it cannot inform a rational purchasing decision or a meaningful public-health assessment. The FDA's Total Diet Study[1] and its Closer to Zero initiative[7] are explicitly designed around population-level exposure modeling precisely because single-sample results are insufficient for risk characterization. The exposé model inverts this logic: it treats the single sample as the unit of analysis and asks the consumer to draw conclusions that the data cannot support.

There is no **measurement uncertainty disclosure**. Analytical chemistry at low parts-per-billion concentrations introduces inherent

measurement uncertainty. A reported result of 11 ppb against a 10 ppb threshold does not necessarily indicate that the true concentration exceeds the threshold; the measurement may reflect analytical noise. Accredited laboratory practice under ISO/IEC 17025[33] requires that measurement uncertainty be reported and incorporated into compliance decisions through formal decision rules[34, 35]. The exposé model treats reported values as exact, presenting 11 ppb as a definitive exceedance when it may not be one. This is not a minor methodological quibble; it is a fundamental analytical error that can make or break a brand's public reputation.

There is no **temporal or seasonal context**. Heavy metal concentrations in agricultural products are not static. They vary by growing season, soil conditions, irrigation source, geographic origin, and post-harvest processing. A single lot tested at a single point in time may not represent the product's typical contamination profile. Population-level trend analysis, which requires structured, repeated testing across multiple lots and seasons, is the methodological standard for characterizing a product's heavy metal performance[8]. A single snapshot cannot distinguish a systemic problem from a seasonal outlier, yet the exposé model treats every detection as a stable characteristic of the brand.

There is no **distinction between detection and risk**. Detection is not synonymous with danger. The toxicological significance of a detected concentration depends on the serving size, consumption frequency, body weight, co-exposure pathways, and the specific metal's dose-response relationship. EFSA, JECFA, and the FDA all employ exposure-based risk-assessment frameworks that integrate these variables[8, 13, 7]. The exposé model collapses this complexity into a binary: detected or not detected, safe or unsafe. This framing is not merely oversimplified, it is actively misleading, because it conflates the presence of a contaminant with the presence of a health risk.

The chilling effect: how public shaming suppresses transparency

The compliance theory literature identifies a well-documented phenomenon: when disclosure is weaponized against the discloser, rational actors respond by reducing disclosure. This is the *chilling effect*, and it operates

with particular force in the heavy metals domain because the underlying contamination is largely not within the brand's direct control.

Consider the decision facing a food brand that is actively working to reduce heavy metal contamination in its products. The brand has two strategic options: invest in testing, data collection, and transparency (Option A), or minimize testing and avoid generating discoverable data (Option B). In the absence of public exposé risk, Option A dominates: better data enables better supplier management, better regulatory defense, and better consumer trust. But the introduction of exposé risk changes the payoff matrix. Under Option A, the brand generates data that can be captured, decontextualized, and published by an external actor in a format that maximizes reputational damage. Under Option B, the brand generates no data, cannot be targeted, and faces only the background regulatory risk that all brands share. The rational strategic response to public exposé risk is therefore to reduce testing, reduce data generation, and reduce transparency, precisely the opposite of what the food-safety system requires[45, 46].

A substantial body of research across international relations, regulatory theory, and criminology examines the conditions under which naming and shaming improves or degrades compliance. The findings are consistent: shaming is most likely to backfire when the target perceives the shaming agent as an out-group actor lacking legitimate authority, when the shaming lacks proportionality to the offense, and when the target has no clear pathway to redemption[43, 44, 45]. All three conditions are satisfied in the public exposé model. The shaming agent is typically a blogger or advocacy organization with no regulatory authority, no accreditation, and no formal relationship with the brand. The shaming is disproportionate because it treats the presence of a ubiquitous environmental contaminant as if it were evidence of corporate malfeasance. And there is no pathway to redemption: once results are published, the brand has no mechanism to demonstrate improvement, no graduated framework for compliance, and no way to contextualize its performance against category peers.

Snyder (2020) documents that naming and shaming in international contexts frequently produces a "backlash narrative" that locks in the targeted behavior rather than correcting it[44]. Terman's research demonstrates a "politicization paradox" in which enforcement mechanisms

designed to punish violations can operate in ways that encourage, reward, and perpetuate them. Van Erp (2011) finds that regulatory disclosure influences compliance only when it connects to firms' existing motivations for maintaining a good reputation, not when it is perceived as arbitrary or punitive[43]. The exposé model satisfies none of these conditions for productive shaming. The full peer-track treatment of these compliance-theory mechanisms appears in[46].

Supply chain misattribution: shaming the wrong node

A fundamental analytical error in the exposé model is the attribution of heavy metal contamination to the brand rather than to the supply chain from which the brand sources. This misattribution is not merely unfair, it is structurally incapable of driving contamination reduction because it targets the wrong node in the causal chain.

Heavy metals in food products originate primarily from soil geochemistry, irrigation water, atmospheric deposition, and industrial contamination of agricultural land. These are supply-side factors over which the finished-product brand exercises limited direct control. A brand purchasing chickpeas from a regional commodity market is drawing from a supply pool that reflects the geochemical characteristics of the growing region, the agricultural practices of the growers, and the seasonal variation in environmental conditions. The brand can test incoming ingredients, set specifications, and reject non-conforming lots. But it cannot unilaterally remediate the soil in which the chickpeas were grown or change the hydrology of the irrigation system. **Shaming a brand for the geochemistry of its supply region is like shaming an airline for turbulence.**

Multiple brands frequently source from the same supply pool. When seasonal contamination elevates heavy metal levels in a particular growing region, all brands sourcing from that region will be affected. But the exposé model tests a single brand, publishes the result, and allows consumers to infer that the named brand is uniquely contaminated. The untested brands sourcing from the same supply pool escape scrutiny entirely, not because their products are cleaner, but because they were not selected for testing. This selection bias is not a minor limitation; it is a

systematic distortion that rewards the untested and punishes the tested, regardless of actual contamination levels.

The selection dynamics are particularly damaging for small and mid-sized brands that are disproportionately likely to be investing in quality and transparency and unable to absorb the reputational and legal costs of being publicly named. A small brand with $5 million in annual revenue that has invested in ISO/IEC 17025-accredited testing, supplier qualification, and heavy metal reduction may be the cleanest brand in its category, and yet, if that brand is the one selected for exposé testing, it will bear the full reputational cost while its larger, less transparent competitors escape notice. The financial scenario modeling in Chapter 1.3 estimates one-year losses on this kind of event at $0.5M–$5M for a small brand at $5M revenue, enough to impair growth or threaten solvency.

This creates a deeply perverse outcome: the brands most committed to quality improvement are the most vulnerable to reputational destruction by an exposé actor, while the brands least committed to quality improvement are the least vulnerable because they generate no data, participate in no programs, and attract no scrutiny.

Litigation amplification: how published exposé results become class actions

Public exposé testing does not operate in a vacuum. It operates in a legal environment where plaintiffs' attorneys are actively monitoring consumer advocacy disclosures for actionable data. The connection between published testing results and litigation filings is direct and documented.

A Reuters report described a proposed class action suing Amazon.com over alleged sale of rice products contaminated with arsenic and other heavy metals, seeking at least $5 million in damages and citing a published third-party testing report as the precipitating trigger[26, 27]. This is a prototypical pattern: a published testing report becomes the complaint's factual scaffold, and the brand is named even if it did not manufacture the product. The exposé actor may intend consumer education, but what they produce is plaintiff-ready evidence.

Heavy-metal product-liability litigation consolidated into multidistrict proceedings, alleging injury from detectable contamination, provides

a further demonstration of how testing disclosures become legal instruments. The MDL record reflects plaintiffs' allegations that any detectable amount could cause injury, illustrating how the absence of agreed-upon safe thresholds can be operationalized into litigation narratives. Defense costs in such matters routinely exceed $2 million per major case, with discovery costs alone often exceeding $600,000[38].

In California, Proposition 65 creates a private-enforcement regime where any private party can initiate litigation alleging exposure to a listed chemical without a government enforcement action[28]. Published exposé results provide the factual predicate for such actions. Settlement economics in Proposition 65 cases are dominated by attorneys' fees: a $28,000 settlement may allocate $25,000 to attorneys' fees and only $3,000 to civil penalties[29]. A brand without a documented third-party testing and certification framework becomes a repeat defendant across SKUs, lots, and flavors, with each exposé publication providing a new cause of action. The irony is acute: the exposé actor generates the evidence that enables the litigation that imposes the costs that discourage the transparency that would enable contamination reduction.

The cycle is self-reinforcing. Public exposé testing generates data. The data is published without context. The publication triggers consumer alarm and media coverage. The media coverage attracts plaintiffs' attorneys. The litigation imposes costs on the named brand. The brand learns that transparency is punished. The brand reduces transparency. Less data is generated. Less data means less ability to identify and remediate contamination. Contamination levels do not decline. The exposé actor tests again and finds contamination. The cycle repeats.

At no point in this cycle does contamination actually decrease. The cycle is optimized for attention, alarm, and litigation, not for the structural changes in supply chain management, ingredient sourcing, and remediation technology that would actually reduce the metals in the food supply.

What this means for the rest of this manual

The structural problem is not that exposé actors are insincere. The intentions of most exposé actors are genuine. The problem is that the *mechanism* is incompatible with the *goal*. If the goal is to reduce heavy

metal contamination across the food supply, not to generate clicks, not to trigger litigation, not to perform consumer advocacy in public, then the mechanism must match the goal. Pressure must be applied at the supply chain level, through frameworks that reward transparency rather than punishing it, and through systems that build on collective data rather than isolating individual brands for public shaming.

That alternative framework is HMTc, and it is specified in detail in Part 2 of this manual. Before that specification can be evaluated, two more chapters of context are required: Chapter 1.3 quantifies the financial cost to a brand of operating in this environment without certification, and Chapter 1.5 introduces the public evidence base that HMTc rests on. Chapter 1.4 bridges to Part 2.

Full peer-track treatment. The argument summarized in this chapter is presented at peer-track length in *Pendergrass, K. The Counterproductive Consequences of Public Exposé Testing: How Unstructured Disclosure Undermines Heavy Metal Contamination Reduction. Journal of Food Metallomics. 2026. doi: 10.5281/zenodo.19470572*. The preprint includes the full literature review on chilling effects, naming and shaming, supply-chain attribution, and the litigation-amplification pathway. Citations are reproduced in Part 5′s Consolidated Sources Index.

1.3 The Financial Problem: The Cost of Operating Without Certification

A brand legal team or CFO reading the previous chapter understands the structural problem. The next question is operational: what does the absence of credible third-party certification cost the company in dollar terms? This chapter answers that question with scenario modeling at three revenue tiers, conservative assumptions, and an expected-loss framework that brand finance teams can use directly. The argument is drawn from the published preprint *The Cost of Operating Without Credible Third-Party Heavy-Metal Certification*, which provides the full evidence anchors, citations, and references.

The core finding: a single contamination-triggered event, recall, delisting, or major lawsuit cluster, can plausibly impose one-year losses of $1.6M–$6.6M for a small brand at $5M revenue, $8M–$33M for a mid-size brand at $50M revenue, and $57M–$187M+ for a large brand at $500M revenue. Expected-loss modeling suggests that third-party certification can break even on downside-risk avoidance alone at well under $200,000 per year for a typical mid-brand exposure profile, before considering the channel-access benefits that certification provides as a gating mechanism for retailer and marketplace approval.

Prevalence: why "we never test positive" is not a stable operating assumption

The relevant business question is not whether heavy metals will be detected in a brand's products, but whether the brand's controls are credible, documented, and independently defensible when regulators, re-

tailers, or plaintiffs' counsel ask. Three independent surveillance datasets establish that detection is common enough that absence-of-data is not a defensible posture.

The FDA modernized Total Diet Study[1] for FY2018–2020 analyzed 3,241 food and beverage samples across 305 foods and beverages for 21 elements. Detection frequencies for the key toxic elements: lead detected in 15% of samples (non-detect to 164 ppb), arsenic detected in 43% (non-detect to 10,900 ppb), cadmium detected in 33% (non-detect to 485 ppb), and mercury detected in 10% (non-detect to 2,890 ppb). For seafood specifically, surveillance of fish and shellfish consistently reports detectable mercury, cadmium, lead, and arsenic in the large majority of samples, with the toxic burden concentrated in predatory species, detection is *common* even though distribution varies sharply by species and trophic level.

Seafood surveillance programs report the same pattern. Routine monitoring of fish and shellfish for arsenic, cadmium, lead, and mercury finds detectable levels in the overwhelming majority of samples, with predatory finfish carrying the highest mercury burden. Detection is the normal state of affairs, not the anomaly.

For botanicals, a major input across foods, supplements, and functional products, a peer-reviewed synthesis analyzing 1,773 crude plant extract samples across 86 herbal types found 30.51% had at least one over-limit metal under referenced pharmacopoeial criteria, with over-limit ratios of Pb 5.75%, Cd 4.96%, As 4.17%, and Hg 3.78%. A U.S.-based study of As, Cd, and Pb in spices reported detection of these metals in more than 90% of samples, with 40.5% exceeding a New York State Pb limit.

These surveillance findings support a practical inference: detection is common, exceedances depend on product class and thresholds, and systematic control with defensible specifications is the rational posture for brands seeking to minimize adverse-event probability. "We never test positive" is not a stable operating assumption; it is an operating assumption waiting to be falsified.

Regulatory pressure: from quality concern to direct cost

Regulatory benchmarks have hardened from guidance into enforcement, transforming heavy metals from a quality concern into a board-level liability.

In the United States, FDA action levels and advisory guidance for toxic elements in seafood are explicit, with longstanding methylmercury advice for fish consumption and category-specific attention to predatory species. FDA's compliance program for toxic elements is formalized. Enforcement and advisory actions extend across the supply chain, from import alerts on contaminated seafood lots to consumer advisories on high-mercury predatory fish.

California Proposition 65 establishes a direct-cost compliance regime for seafood: products containing listed contaminants such as methylmercury, lead, and cadmium can trigger warning obligations and private enforcement, and seafood is among the categories where Prop 65 fish-consumption and methylmercury exposure questions have been actively litigated. The result is an enforcement posture in which a brand without documented controls becomes a recurring target.

In the EU, Regulation (EU) 2023/915[19] explicitly reiterates the ALARA principle and consolidates maximum levels across contaminants. The annex includes detailed maximum levels for lead, arsenic, and mercury across numerous commodities, including fishery products, with maximum levels for lead in fish muscle (0.30 mg/kg), cadmium in fish muscle (0.050 mg/kg), and mercury in predatory fish (1.0 mg/kg). Globally, the Codex Alimentarius General Standard for Contaminants and Toxins (CXS 193-1995) provides internationally recognized principles and lists maximum levels.

The convergence, FDA action levels hardening, California AB 899[21] mandating disclosure, EU 2023/915 consolidating maximum levels with ALARA expectations, means that the regulatory floor is rising under brands' feet at the same time that litigation centralization and retailer gatekeeping are tightening from above. A brand operating without a documented, third-party-verified program is increasingly exposed at every layer simultaneously.

Litigation vulnerability: economics of being an easy target

In heavy-metals disputes, being targeted is rarely only about the presence of metals; it is frequently about whether a brand can defend its controls, warnings, and claims using evidence that courts, regulators, retailers, and consumers accept as credible. When that credibility is weak, litigation can become cheaper to file and more expensive to defend.

Multidistrict centralization of heavy-metal product-liability cases concentrates common discovery, expert battles, and bellwether structures, amplifying legal spend and reputational exposure even before merits are adjudicated. Once a contaminant category attracts coordinated litigation, every brand in it inherits the centralized cost structure. From a cost standpoint, MDL centralization changes the incentive landscape: common discovery, expert battles, and bellwether structures can amplify legal spend and reputational exposure even before merits are adjudicated.

Reuters reported a proposed class action suing Amazon.com over alleged sale of rice products contaminated with arsenic and other heavy metals, seeking at least $5 million in damages under state consumer protection theories. Reuters later reported Amazon's attempt to dismiss, arguing among other points that no excessive levels beyond regulatory limits were alleged. Even when ultimate liability is uncertain, this pattern imposes a predictable cost stack: rapid expert engagement, e-discovery, motion practice, and brand-risk management, costs that are typically higher when a company cannot point to recognized third-party testing and certification as part of routine operations.

California Proposition 65[28] is structurally important because it can generate frequent, product-by-product disputes where attorney fees are significant relative to the underlying exposure issue. A Prop 65 settlement agreement involving a pre-workout product allocated $25,000 of a $28,000 total settlement to attorneys' fees and related costs, with only $3,000 in civil penalties[29]. The micro-economics imply that if a brand lacks an HMTc-like framework, it can become a repeat defendant across SKUs, flavors, and lots, with recurring settlements and ongoing testing and warning obligations that are more expensive than a proactive certification strategy.

A widely cited federal-courts survey of major companies[38] found mean outside legal fees per major case around $2.0 million, with discovery costs often well in excess of $600,000[38]. These benchmarks are not heavy-metals-specific, but they are critical for modeling: if a heavy-metals dispute becomes major litigation, **defense costs alone can plausibly exceed the annual cost of robust third-party certification, even before settlements, recalls, or sales impacts are included.**

Channel access: retailer and marketplace gatekeeping

A brand without credible third-party certification faces not only regulators and plaintiffs but also commercial gatekeepers. Retailers, marketplaces, and procurement teams increasingly outsource trust to recognized standards and accredited testing.

CVS[30] publicly states it requires third-party laboratory testing of all vitamins and supplements and explicitly lists heavy metals, inorganic arsenic, cadmium, chromium VI, mercury, and lead, as contaminants that cannot exceed the current NSF/ANSI 173[31] Standard, framing this as a consumer-facing trust commitment. For marketplace access, Amazon's dietary supplement seller documentation requirements include submission of a GMP certificate from an accredited third-party body (21 CFR 111/117) and a finished product COA issued by an ISO/IEC 17025[33] accredited laboratory, or evidence of enrollment in recognized third-party quality certification programs including NSF/ANSI 173[31] product certification and USP verification.

These requirements convert lack of certification into direct commercial risk: listing suppression, delayed launches, forced reformulation, or delisting, each of which has an immediate revenue impact independent of whether a regulator has acted.

Scenario modeling: one-year impact at three revenue tiers

Real-world costs vary by category, channel mix, and incident severity. The scenarios below use conservative assumptions and explicit ranges. The public event type is a public contamination event involving recall

or advisory, major news coverage, and retailer response. Assumptions: gross margin 40%; sales decline of 22% for 4 months (≈7.3% of annual revenue, based on the recall-sales response benchmark for meat and poultry recalls); delisting and channel disruption equivalent to loss of 15% of annual revenue for 6 months (≈7.5% of annual revenue, a conservative one-major-channel assumption); legal fees scaled relative to the U.S. courts survey mean of approximately $2.0 million per major case; recall direct cost scaled using empirical industry ranges.

Small brand at $5M annual revenue

Cost component	**Range**
Direct recall and market action	$0.5M–$3M
Legal defense and discovery	$0.25M–$1.5M
Settlement, penalties, plaintiff attorney fees (Prop 65 patterns)	$0.05M–$0.5M
Lost sales	$0.37M revenue ($0.15M gross profit)
Delisting and listing suppression	$0.38M revenue
Remediation and verification surge	$0.05M–$0.3M
Indicative total one-year cost	**$1.6M–$6.6M+**

For a brand at $5M revenue, this is sufficient to impair growth or threaten solvency. The $1.6M low end is 32% of annual revenue; the $6.6M high end exceeds annual revenue by 30%.

Mid-size brand at $50M annual revenue

Cost component	**Range**
Direct recall and market action	$3M–$15M
Legal defense and discovery	$1M–$7M
Settlement, penalties	$0.25M–$5M
Lost sales	$3.65M revenue ($1.46M gross profit)
Delisting and listing suppression	$3.75M revenue
Remediation and verification surge	$0.2M–$1M
Indicative total one-year cost	**$8M–$33M+**

Large brand at $500M annual revenue

Cost component	Range
Direct recall and market action	$15M–$75M
Legal defense and discovery	$3M–$20M
Settlement, penalties	$1M–$50M
Lost sales	$36.5M revenue ($14.6M gross profit)
Delisting and listing suppression	$37.5M revenue
Remediation and verification surge	$0.5M–$5M
Indicative total one-year cost	**$57M–$187M+**

These scenario anchors draw on industry recall-cost magnitudes, federal-courts legal-fee benchmarks, Proposition 65 settlement structure, peer-reviewed sales-decline benchmarks, and published market-value loss evidence. The estimates exclude long-tail brand impairment, shareholder-wealth reductions for publicly traded firms (one peer-reviewed study found $109M reductions after severe recalls), and the cascade effects of repeat litigation across SKUs.

Note on what the expected-loss model quantifies. The model below quantifies *risk-event* reduction (probability and severity of recalls, delistings, and litigation triggers), not safety. A brand with HMTc certification is in a better defensive posture than a brand without it; both brands operate in a contamination environment that cannot be eliminated by any program. The certification mark is an action-level compliance signal under surveillance, not a safety guarantee, see Part 2.1 §1.2 and Part 2.5 §6.2.

ROI: the expected-loss framework

The expected-loss framework for evaluating certification ROI proceeds as follows. Let L denote the one-year loss if an event occurs (from the scenarios above). Let p denote the annual probability of a public metals event for the specific brand (including recall, major delisting, or major litigation trigger). Let r denote the proportional loss reduction due to certification (combined probability plus severity effect, conservative range 10–40%). Let C denote the annual cost of certification and associated testing and audits.

Expected annual loss without certification: $Eo = p \times L$. Expected annual loss with certification: $E1 = p \times (1 - r) \times L$. Expected annual benefit: $\Delta E = Eo - E1 = p \times r \times L$. Break-even condition: $C \leq p \times r \times L$.

As a worked illustration for a mid-size brand: if $L = \$15M$, $p = 3\%$, and $r = 25\%$, then $\Delta E \approx \$112{,}500$ *per year*. Under those conservative assumptions, certification must cost $112,500 per year or less to break even on downside-risk avoidance alone. If the brand's channel access depends on third-party testing (such as marketplace COA plus ISO/IEC 17025[33] requirements) or if the probability or severity is higher (such as predatory-fish methylmercury exposure or other high-bioaccumulation product classes), the break-even rapidly improves.

This expected-loss model undervalues certification whenever certification is a prerequisite for market access, a gating benefit, because avoided delisting is effectively an earned revenue protection rather than avoided loss. Retailer policies like those of CVS[30] explicitly position third-party testing as a standard for supplements sold in their channel, meaning certification can be a revenue-enabling investment, not merely an insurance-like expense.

Four mechanisms by which certification reduces expected losses

A credible HMTc-like framework mitigates losses through four mechanisms, each tied to evidence presented in this chapter.

The first is **reducing probability through preventive controls and supplier qualification**. Surveillance shows frequent detection in foods and botanicals. A certification program that enforces specifications and routine verification reduces the probability of an outlier lot reaching retail undetected.

The second is **severity reduction via faster containment and smaller scope**. EU and FDA frameworks are increasingly commodity- and child-category-specific, implying that a program capable of rapid traceability and documented compliance can reduce recall breadth and delisting duration.

The third is **litigation defensibility, or the due-diligence premium**. MDL centralization and consumer class actions illustrate

that heavy-metals claims can become scalable. Third-party certification creates contemporaneous, independently validated evidence that can narrow claims, support motions, and reduce settlement leverage.

The fourth is **channel access and procurement de-risking**. Retailers and marketplaces increasingly require third-party testing, recognized standards such as NSF/ANSI 173[31], and ISO/IEC 17025[33] documentation. Certification therefore reduces the probability of delisting even absent a recall.

What this means for the rest of this manual

The financial picture is unambiguous. The cost of a contamination event for a brand at any revenue tier is large enough that the expected-loss reduction from certification can exceed certification costs even under conservative probability assumptions. The channel-access value of certification, increasingly required by major retailers and marketplaces, provides additional return that the expected-loss model does not fully capture. The convergence of hardening regulatory frameworks, litigation centralization in MDL No. 3101[24], Proposition 65[28] economics, and retailer gatekeeping means that the cost of inaction is not a static compliance risk but an accelerating commercial liability.

What remains is to specify the program itself. Chapter 1.4 bridges the structural argument from Chapter 1.2 and the financial argument from this chapter to the operational program in Part 2.

Full peer-track treatment. The argument summarized in this chapter is presented at peer-track length in *Pendergrass, K. The Cost of Operating Without Credible Third-Party Heavy-Metal Certification. Journal of Food Metallomics. 2026. doi: 10.5281/zenodo.18903738*. The preprint includes the full surveillance-data tables, regulatory-enforcement examples, retailer-requirement documentation, and recall-economics evidence anchors. Citations are reproduced in Part 5′s Consolidated Sources Index.

1.4 The Structural Alternative: HMTc

The structural problem in Chapter 1.2 and the financial problem in Chapter 1.3 converge on the same conclusion: the existing enforcement environment does not reduce heavy metal contamination, and the cost of remaining outside a credible certification program is unsustainable for any brand serious about its category position. This chapter names the alternative, the Heavy Metal Tested & Certified (HMTc) program, and previews the design principles that Part 2 specifies in operational detail.

The HMTc framework is built around six design principles, each of which responds directly to a failure mode identified in the preceding chapters. The principles are summarized here; the operational specification appears in Part 2.

Design principle 1: Staged compliance

A program with unachievable standards attracts no participants and protects no consumers. A program with loose standards lacks credibility and provides no meaningful differentiation. The staged-compliance architecture resolves this tension by setting rigorous standards, meaningfully below regulatory action levels, calibrated against the published achievability of the open literature, while giving brands structured pathways to reach full compliance rather than requiring every threshold be met on day one.

Each tested product lot receives a certification status based on its analytical results: Status A (full compliance), Status B (transitional compliance with corrective action obligations), Status C (probation with mandatory hold-and-release and independent sampling), Status D (suspension), or Status E (revocation). The framework keeps brands

continuously improving without converting normal lot-to-lot analytical variability into a commercial catastrophe. A brand whose typical lead level is 4 ppb against a 5 ppb limit will occasionally produce a lot at 6 or 7 ppb through normal variation, not negligence; the staged system absorbs that variability while maintaining clear escalation triggers for genuine control failures.

The five-status framework, eligibility criteria, mark-usage rules, and escalation paths are specified in Part 2.3.

Design principle 2: Risk-tiered metal classification

Not all heavy metals present the same toxicological profile, and a defensible certification framework reflects this. The HMTc program classifies covered metals into two tiers based on their toxicological characteristics, with governance consequences flowing directly from the classification.

Tier 1 metals are carcinogenic with no established safe threshold, neurotoxic with no established safe threshold, or bioaccumulative with long biological half-lives. No transitional allowance is permitted; any confirmed exceedance triggers immediate escalation to probationary status or higher. Lead, cadmium, inorganic arsenic, and methylmercury fall into this category.

Tier 2 metals have established tolerable daily intakes, shorter biological half-lives, or primarily acute rather than chronic toxicity profiles. Transitional exceedances up to 150% of the action level may qualify for conditional certification if the brand initiates corrective action and demonstrates an improving trend. Nickel, tin, aluminum, and chromium fall into this category.

The tiered approach means that every governance decision, every transitional allowance granted, every probation imposed, has a traceable rationale grounded in the toxicological literature. **If a regulatory body or plaintiff's attorney asks why a 130% nickel result received transitional certification while a 110% lead result triggered immediate probation, the answer traces directly to the comparative toxicological profiles of the two metals. That defensibility is the point.**

The full Tier 1 / Tier 2 specification, with toxicological rationale for each metal, appears in Part 2.2.

Design principle 3: Exposure-based testing frequencies

Testing frequency scales with exposure risk rather than being applied uniformly across all product categories. The relevant variables are consumption volume relative to body weight, frequency of consumption, and the degree to which the product may serve as a sole or dominant food source.

Predatory finfish such as tuna, swordfish, shark, and marlin, which accumulate methylmercury across long lifespans and high trophic positions, present the highest exposure risk and warrant the most frequent surveillance testing with independent sampling. Non-predatory marine fish such as cod, herring, sardine, and mackerel carry a lower but non-trivial burden and receive the next testing tier. Shellfish, where cadmium and inorganic arsenic concentrate in bivalves and crustaceans, receive surveillance scaled to their occurrence profile. Seaweed and kelp foods, which can concentrate inorganic arsenic, receive surveillance during full compliance proportionate to consumption.

A regulator or attorney asking why a predatory species such as swordfish is tested more frequently than a low-trophic species such as sardine receives an answer grounded in bioaccumulation data and exposure mathematics, not in administrative convenience. Testing intensifies during transitional and probationary status: every-lot testing applies to all products during Status C, with hold-and-release procedures.

The full testing matrix appears in Part 4 of this manual (reproducing the Lot Testing Schedule).

Design principle 4: Measurement uncertainty and decision rules

The analytical chemistry of heavy metal testing at low parts-per-billion concentrations introduces inherent measurement uncertainty. A reported result of 11 ppb against a 10 ppb limit does not necessarily

indicate that the true concentration exceeds the limit; the measurement could reflect analytical noise. Certification decision rules must account for this uncertainty to be both scientifically valid and legally defensible.

The HMTc decision rule is two-sided. A lot **passes** if the reported result plus the laboratory's expanded measurement uncertainty is at or below the limit, compliance is confirmed even at the upper bound of uncertainty. A lot **fails** if the reported result minus the measurement uncertainty exceeds the limit, exceedance is confirmed even at the lower bound. Results falling between these conditions are treated as **borderline** and require confirmatory action before a status is assigned. The approach protects brands from being penalized for measurement noise while protecting consumers from borderline results being treated as passes. It follows established metrological principles used in accredited laboratory practice and international standards (ISO/IEC 17025, EURACHEM/CITAC, ILAC-G8).

Equally important is the prohibition on **testing into compliance**. Brands may not conduct repeated ad hoc retesting, switch laboratories, or selectively report results in an attempt to find a passing number. Any confirmatory testing must follow a predefined, program-approved protocol; undocumented resampling is treated as a data-integrity violation subject to suspension or revocation. This prohibition eliminates the appearance of gaming and ensures that the testing record can withstand discovery in litigation.

The full decision-rule specification appears in Part 2.6.

Design principle 5: Confidential remediation as a structural feature

The commercial and legal landscape around heavy metals in seafood creates a paradox: the brands most in need of a structured improvement program are the least likely to join one voluntarily if doing so creates public exposure. Any public acknowledgment that a brand's products contain elevated heavy metals, even in the context of a voluntary improvement program, could be weaponized in litigation or media coverage. Chapter 1.2 documented this dynamic at length.

The Confidential Remediation Track resolves the paradox. Brands whose products do not yet qualify for certification may enter the program, access remediation support and technical guidance, and begin building a documented improvement record, without public disclosure of their enrollment or test data. The brand's participation, testing results, corrective action plans, and improvement trajectory are treated as confidential business information under the license agreement.

This confidentiality is not a loophole; it is structural. It serves the program's public-health objectives by maximizing the number of brands actively working to reduce contamination. A brand that remains outside the program because it fears the consequences of joining is a brand whose products are not improving. A brand that enters confidentially and begins reducing heavy metal levels is a brand whose products are getting safer, even if the public does not yet know.

From a legal defensibility standpoint, the Confidential Remediation Track creates a documented record of proactive, voluntary, third-party-verified improvement that begins before any litigation or regulatory inquiry. This record, comprising timestamped test results, corrective action documentation, and verified declining concentration trends, constitutes contemporaneous evidence of due diligence. **The legal narrative shifts from "the brand knew and did nothing" to "the brand knew, enrolled in an independent certification program, and has documented evidence of continuous improvement." That shift is the difference between exposure and defense.**

These improvements reduce *probability* and *severity* of contamination events; they do not produce safety guarantees. The mark certifies action-level compliance under surveillance, not the absence of risk. Brands evaluating the Confidential Remediation Track should also read Part 2.4 on discovery dynamics, the Track produces a defensible evidence record, but the record itself does not carry attorney-client privilege and may be subject to subpoena.

Because the Confidential Remediation Track is the single most important differentiator of HMTc from the exposé-driven enforcement environment Chapter 1.2 attacked, the track receives its own chapter in Part 2 (Part 2.4) rather than appearing as a subsection of program-entry rules.

Design principle 6: Ratcheting standards with capped step size

A static certification standard is inherently self-defeating. If standards remain unchanged while industry improves, they eventually certify the status quo rather than driving further progress. Conversely, if standards are perceived as arbitrary or subject to surprise changes, brands lose the planning certainty necessary to invest in reformulation and supply-chain improvements. The Standards Ratchet Mechanism resolves this through a formal commitment to tighten standards over time, moving in only one direction, on data-driven trigger conditions, with a capped step size that bounds the per-cycle impact on any brand.

The ratchet for a given analyte fires when both of two conditions hold for two consecutive annual reviews: the 80th percentile of certified Status A lot results sits at or below 80% of the current limit, and at least 50% of certified Status A SKUs have lots at or below 60% of the current limit. The first condition tests aggregate market shift; the second tests breadth. When fired, the new limit equals the maximum of (a) the actual 80th percentile of Status A results or (b) 80% of the current limit. The second clause is the step cap, no brand faces more than a 20% reduction per ratchet event. Transition is 12 to 18 months from announcement to enforcement, with Status A certifications grandfathered during the transition period.

The forward forecast obligation converts the ratchet from a surprise into a planning parameter. At every annual review, the Program Operator publishes a projection of the next-cycle limit based on current improvement trends, so brands considering certification can model their next three-to-five-year obligation before they sign.

The full ratchet specification appears in Part 2.10.

A seventh principle: supplier intelligence as collective action

Individual brands testing their own finished products can identify contamination after it has already occurred. A certification program operating across multiple brands has the opportunity to detect contamination pat-

terns at the supply chain level and alert affected participants before the contamination reaches finished products.

The HMTc Supplier Intelligence System aggregates findings from corrective-action investigations across all participating brands. When one brand's root-cause analysis identifies a supplier, ingredient source, or geographic origin with elevated heavy metal levels, the program issues anonymized alerts to other brands using the same supply-chain inputs. The alerting brand's identity is never disclosed; only the ingredient category, geographic origin, and nature of the concern are communicated.

This system creates a collective-intelligence network that increases in value as the program grows. Each additional brand contributes data that benefits all other participants, creating a positive network effect that strengthens retention and attracts new entrants. From a public-health perspective, it accelerates the identification and remediation of contamination sources across the entire supply chain rather than within individual brands operating in isolation. From a competitive-strategy perspective, supplier intelligence is the durable moat that distinguishes mature HMTc certification from any single-brand internal-testing program.

The full supplier-intelligence specification appears in Part 2.9.

What Part 1 has established

Part 1 has now made three arguments. The structural argument (Chapter 1.2) demonstrated that public exposé testing fails as an enforcement mechanism because it punishes transparency, misattributes contamination to brands instead of supply chains, and produces a self-reinforcing cycle that does not reduce contamination. The financial argument (Chapter 1.3) quantified what brands face in the existing environment: $1.6M–$6.6M one-year exposure for small brands, $8M–$33M for mid-size, and $57M–$187M+ for large brands. This chapter (1.4) named the alternative: a confidential, staged, ratcheting, third-party certification framework that responds to each failure mode of the current environment with a specific design feature.

What remains in Part 1 is to introduce the evidence base. The HMTc methodology rests on a public, curated corpus of the heavy-metals-in-food literature, heavymetalindex.com, that is architecturally separate

from the certification program. Every standard in this manual traces to specific source pages on that wiki. Chapter 1.5 describes that dependency in detail. With that final piece of context, Part 2 can be read as the operational specification of a program that exists for a reason, against a literature evidence base that any adversarial reader can audit.

Full peer-track treatment. The design rationale summarized in this chapter is presented at peer-track length in *Pendergrass, K. Certification as a Framework for Reducing Heavy Metal Exposure in Infant and Child Foods: Integrating Legal Defensibility, Scientific Rigor, and Commercial Viability. Journal of Food Metallomics. 2026* [48]. The preprint's calibration framing (an 80% achievability threshold) reflects the program's March 2026 state; the current operational calibration is the 90/30 row-pass intent specified in the Governing Principles and Part 2 of this manual. The preprint's structural design principles remain canonical.

1.5 How HMTc Knows What It Knows: heavymetalindex.com

Before this manual specifies a single concentration limit, the reader is owed an answer to the question: where do the numbers come from? Not in general terms, *literature, regulatory authorities, peer-reviewed studies*, but specifically. The 300 ppb lead limit for predatory marine fish. The 200 ppb inorganic arsenic limit for the same. The methylmercury threshold for predatory fish. Where, exactly, is the evidence for each of these, and can an adversarial reader audit it without taking the certifier's word?

The answer is heavymetalindex.com.

The architectural separation

The Paleo Foundation operates two architecturally separate properties. The certification program, HMTc, lives at heavymetaltested.com. The literature evidence base, Heavy Metal Index, hereafter HMI, lives at heavymetalindex.com. These are separate properties by design, and the separation is structural rather than cosmetic.

HMI is the public, curated wiki of the heavy-metals-in-food literature. It contains source pages for peer-reviewed studies, agency reports (FDA, EFSA, JECFA, EPA, Codex, ATSDR), industry surveillance data, and primary regulatory documents. Each source page summarizes a single document's findings, preserves its numeric claims with full provenance, and links to ingredient pages, metal pages, product-category pages, and regulation pages. The corpus is structured so that a reader can begin with any single concentration claim on any product page and follow the references back through the routing layer to the original source documents.

HMTc references HMI for the literature baseline that informs every certification standard. The reference is **one-way**: HMTc cites HMI; HMI does not endorse HMTc, does not contain percentile arithmetic, does not name HMTc-certified brands, and does not advocate for the certification program. The wiki/HMTc firewall is operative policy under the HMTc Governing Principles (specifically Principle 4 on legal defensibility), and is enforced through the program's automated lint infrastructure. The wiki is the public literature evidence layer; the Standards Workbench carries the threshold math, which remains workbench-side and is not published in the wiki.

This separation is what gives the standards their authority.

Why the separation matters

A certifying body that operates its own evidence base has a structural conflict of interest: the body that decides what the literature supports is also the body that decides what to certify. An adversarial reader can attack the evidence base on the grounds that it was selected to justify a pre-determined commercial position. Single-source certifiers, programs where the evidence and the certification live in the same property under the same control, cannot fully defend against this attack because the conflict is real.

The HMI/HMTc separation defeats this attack. HMI is a curated literature wiki published as a public resource. It exists whether HMTc does or does not exist. It will continue to exist when the methodology in this manual is replaced by a future revision. It contains studies whose findings make HMTc thresholds look conservative and studies whose findings make HMTc thresholds look aggressive. It does not adjust its evidence to fit the certification program. The Paleo Foundation operates HMI as a public-good resource because curating the complete literature is the only durable foundation for any defensible standards program built on top.

This is the **Cochrane-for-heavy-metals** positioning. Cochrane systematic reviews[41] are the gold standard in medicine for a single reason: they aggregate the complete relevant literature, not selected papers, and they publish their methodology so that any reviewer can verify their conclusions. HMI is structured on the same principle. Aggregating the complete literature, not selected papers, is what produces durable

epistemic authority, and durable epistemic authority is what allows the standards program built on top to withstand adversarial scrutiny.

Provenance: how a limit in this manual traces to the corpus

Every concentration limit in Part 3 of this manual was derived from a pooled distribution of source pages on HMI for the relevant product-by-metal cell. The pooling math (the per-analyte percentile calculations, the source-level bootstrap, the sensitivity-band analysis) lives workbench-side, per the firewall. But the **inputs** to the pooling, the source pages contributing to each cell, are public.

A reader who wants to audit a specific HMTc limit follows this path:

1. **Locate the product chapter** in Part 3 of this manual. The Master Limit Table gives the HMTc limit, the applicable regulatory cap, and the basis note for each of the ten priority analytes.
2. **Read the chapter's Sources Cited section.** Each citation resolves to a source page on heavymetalindex.com.
3. **Read the source page.** Each source page summarizes the underlying document's findings with concentrations, sample sizes, methods, and basis (as-sold, as-consumed, dry weight). The source page links to the original document where available.
4. **For the routing layer**, the corpus also maintains a structured routing audit at `data/evidence/product_source_routing_audit.csv` that maps every source to every product-category, ingredient, metal, and regulation page it touches. Adversarial readers who suspect a source has been omitted from a cell can check the routing audit directly.
5. **For the pooled distribution itself**, qualified reviewers, regulators with formal inquiries, plaintiff's experts requesting access through counsel, sampling auditors verifying program operation, may request the workbench-side aggregate distribution data under controlled terms. The pooling engine is proprietary; the pool composition is not.

This design preserves the wiki/HMTc firewall (the engine arithmetic stays workbench-side, where it cannot be reverse-engineered by competing certifiers) while making the evidence inputs auditable (every limit traces to public source pages on HMI).

The failure mode this design prevents is the "we set this limit because we said so" attack. HMTc cannot make that argument because the source attribution is public. An adversarial reader who believes a limit is too tight can find the source pages that drove it; an adversarial reader who believes a limit is too loose can find the source pages that justified the floor. The argument moves from *whether the certifier is biased* to *what the literature supports*, which is the right level for the argument to occur.

What the corpus contains, in scope and structure

HMI's scope is deliberately broad: heavy-metals-in-food literature globally, across all product categories, in all jurisdictions, across regulatory and peer-reviewed sources. The corpus uses three structural layers.

The **source layer** (`wiki/sources/`) is one page per source document, with frontmatter recording the document's identifiers (DOI, raw filesystem handle, cite-key), evidence tier (A for peer-reviewed primary studies and government reports; B for industry white papers and NGO reports; C for news articles and press releases), and routing metadata (which products, ingredients, metals, and jurisdictions the source covers). Every numeric claim in the corpus traces to a source page; the corpus does not contain values not anchored to a source.

The **substrate layer** (`wiki/metals/`, `wiki/ingredients/`, `wiki/products/`, `wiki/regulations/`, `wiki/testing/`, `wiki/health/`, `wiki/microbiome/`) is one page per substantive subject. Metal pages summarize toxicology, exposure routes, regulatory limits, testing methods, and microbiome effects for each of the ten priority analytes and several others. Ingredient pages carry the structured `contamination_profile` block (typical and 95th-percentile concentrations, sample counts, geographic and seasonal variance, regulatory limits) for every commodity in scope. Product-category pages aggregate evidence at the consumer-product level. Regulation pages document each operative rule with exact limit values, units, and effective dates from the original agency source.

The **routing layer** (`data/evidence/product_source_routing_audit.csv`) is the mapping from each source to every page it should appear on. It is regenerated automatically when source pages change. The routing audit's transparency is what guarantees that no source has been silently omitted from a cell.

The corpus also maintains an evidence register (`data/evidence/`) of structured concentration values, regulatory facts, and review events. Bulk ingest of the literature is ongoing; the corpus today is sufficient to support Category 6 (seafood) certification standards, with additional categories expanding under continuous ingest.

What this means for adversarial readers

A reader looking to attack an HMTc limit has three structural choices:

1. **Attack the source page.** If the reader believes a specific source has been misread or misrepresented in the corpus, they can compare the source page on HMI to the original document and identify the discrepancy. The Paleo Foundation maintains a public correction process for source-page errors.
2. **Attack a routing decision.** If the reader believes a relevant source has been omitted from a product-by-metal cell (or that an irrelevant source has been included), they can check the routing audit and the source's frontmatter. Routing decisions are deterministic and reproducible from the source pages' metadata.
3. **Attack the methodology.** If the reader believes the per-analyte percentile selection, the pooling math, or the regulatory-cap interaction produces an indefensible result, the methodology in Part 5.1 and the Governing Principles supersede this argument. The methodology is described in this manual; the engine arithmetic is workbench-side but the design is auditable.

What the reader cannot do is argue that HMTc invented its evidence base. The corpus exists publicly. Anyone can verify any source page.

What this means for brand legal teams

A brand legal team reading this chapter understands that HMTc standards are not opaque. They are calibrated against a public corpus that the brand's own technical experts can review. A brand contemplating certification can ask its toxicology consultant to read the HMI source pages for the relevant product category and confirm that the corpus reflects the state of the literature as the consultant understands it. If the consultant

finds gaps, the gaps are flagged through the same public correction process available to any reader.

This is part of what HMTc certification buys for the brand: not just a mark, but **a defensible evidence trail to the certifier's standards-setting methodology**. When a plaintiff's expert in a future deposition asks "why did you trust the HMTc standards?" the brand's answer is "because the evidence base is public, our toxicology team verified it, and the methodology is documented." That answer is much stronger than the answer available to a brand operating against internal standards or marketing-program certifications.

What this means for regulators

A regulator reading this chapter understands that HMTc operates with public evidence in a way that complements rather than substitutes for federal regulatory action. The HMI corpus is available to FDA, EFSA, state attorneys general, and any other regulatory body that wants to audit how a specific HMTc standard was derived. The structured routing audit allows targeted regulatory inquiries: a reviewer who asks "what does HMI's evidence base say about cadmium in cocoa from Ecuador?" can pull the ingredient page's geographic-variance block and the contributing source pages directly. This is the kind of structured, machine-readable data that the National Academies' 2011 report on targeted transparency identified as the precondition for productive regulatory engagement with industry self-regulation.

What this means for the rest of this manual

The reader has now encountered the program's foundational claim: every numeric assertion in Part 3 traces to a public corpus that any adversarial reviewer can audit. With that claim established, the operational specification of the program in Part 2, the per-product standards in Part 3, and the testing schedule in Part 4 can be read against the right backdrop. The engine arithmetic that produces each specific limit lives workbench-side; the inputs to that arithmetic are public. The methodology that the engine implements is documented in Part 5.1. The adversarial-read chapter in Part 5.2 enumerates the categories of attack that the program anticipates and answers them on the merits.

This concludes Part 1. The reader who has read all five chapters now understands:

- Why the program exists (the existing enforcement environment fails structurally);
- What the cost of inaction is (large enough that certification is a rational investment under conservative probability assumptions);
- What the program's design principles are (staged compliance, risk-tiered metal classification, exposure-based testing, measurement-uncertainty decision rules, confidential remediation, ratcheting standards, supplier intelligence);
- And where the evidence base lives (heavymetalindex.com, a public curated corpus architecturally separate from the certification program).

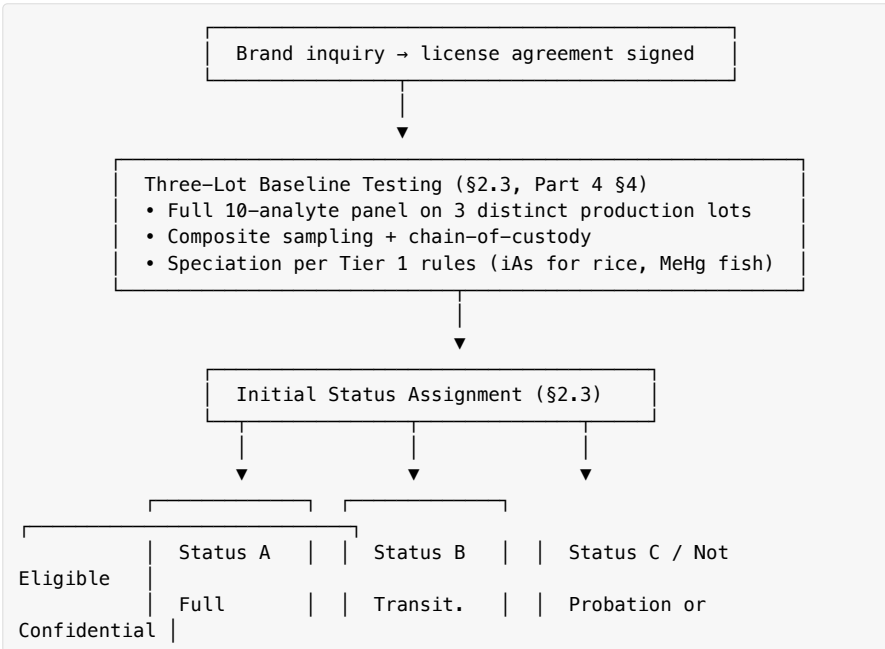
The next chapter, Part 2.0 Program Flow Diagram, opens Part 2 with a single-page visualization of the certification lifecycle before the operational chapters specify the program in detail.

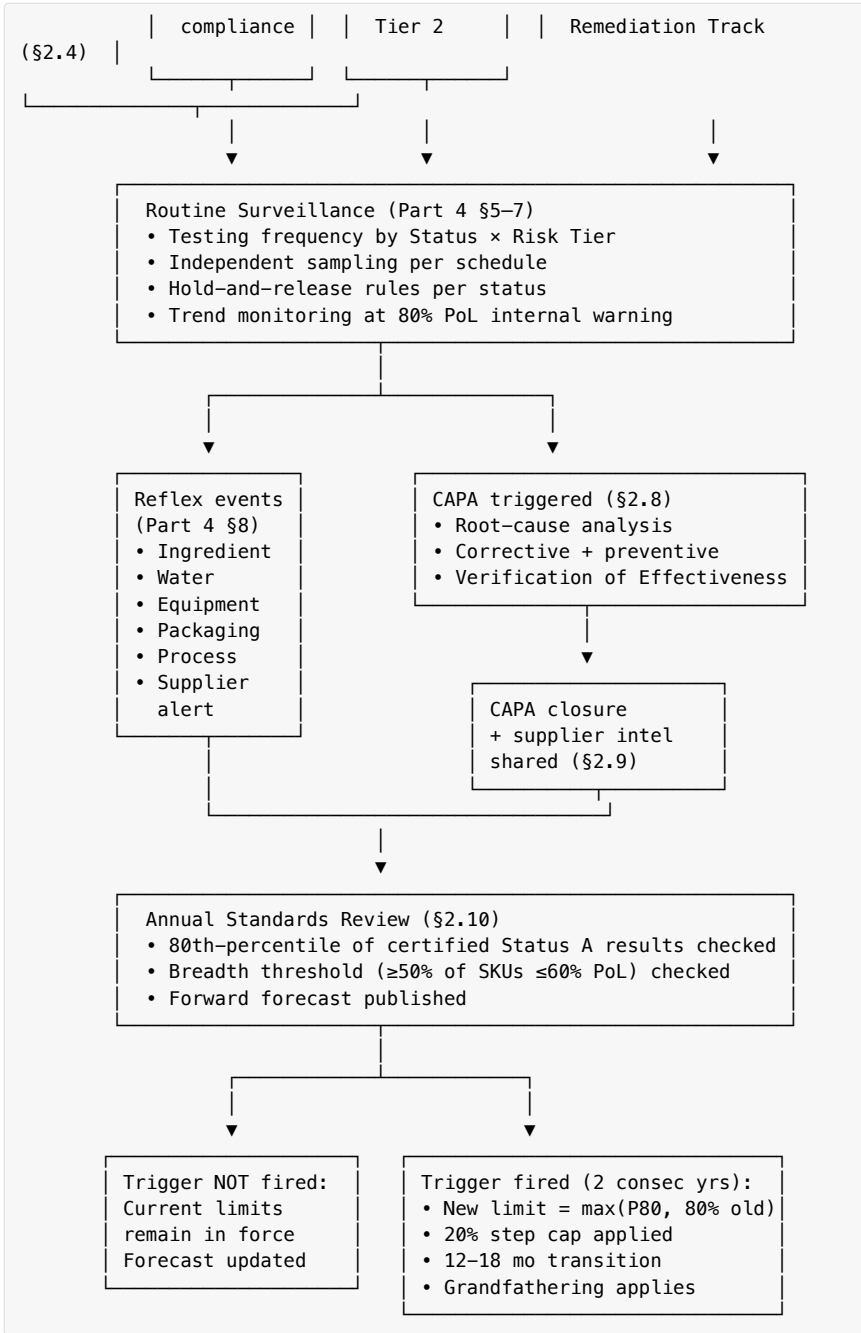
Part 2, The HMTc Program

2.0 Program Flow Diagram

A brand legal team or QA reader who has finished Part 1 understands why the program exists. Part 2 specifies what the program *is*: the operational constitution of HMTc. Before the operational chapters that follow, this single-page diagram shows how a brand and a SKU move through the program from inquiry to ratchet review. The numbered nodes correspond to chapters in Part 2 where each phase is specified.

Lifecycle




compliance
Tier 2
Remediation Track (§2.4)
Routine Surveillance (Part 4 §5–7)
• Testing frequency by Status × Risk Tier
• Independent sampling per schedule
• Hold-and-release rules per status
• Trend monitoring at 80% PoL internal warning
Reflex events (Part 4 §8)
• Ingredient
• Water
• Equipment
• Packaging
• Process
• Supplier alert
CAPA triggered (§2.8)
• Root-cause analysis
• Corrective + preventive
• Verification of Effectiveness
CAPA closure + supplier intel shared (§2.9)
Annual Standards Review (§2.10)
• 80th-percentile of certified Status A results checked
• Breadth threshold (≥50% of SKUs ≤60% PoL) checked
• Forward forecast published
Trigger NOT fired: Current limits remain in force Forecast updated
Trigger fired (2 consec yrs):
• New limit = max(P80, 80% old)
• 20% step cap applied
• 12–18 mo transition
• Grandfathering applies

The diagram is the operational map for the rest of Part 2. Brand legal readers can return to it as a reference; each chapter that follows specifies one or more of the nodes shown above. The first operational chapter, Part 2.1, establishes the Program's purpose and core principles. The Confidential Remediation Track for brands whose baseline does not yet qualify for Status A or B is specified in Part 2.4.

2.1 Purpose and Core Principles

Narrative

The HMTc Program is designed to achieve three simultaneous objectives: protect participating brands from litigation exposure through documented, defensible testing and improvement protocols; protect consumers by incrementally driving down heavy metal exposure in seafood products; and create commercial value through a certification mark that signals verified, ongoing compliance under independent surveillance. These three objectives are not in tension. A program that fails any one of them fails all three, a program that does not reduce brand legal exposure cannot recruit participants; a program that does not drive consumer-exposure reduction has no defensible public-health basis; and a program that does not create commercial value cannot sustain itself across the time horizons required to achieve population-level impact.

The Program's five Governing Principles, published separately and reproduced in the Methodology Appendix of this manual (Part 5.1), are the authority that this Governance specification implements. The principles are: drive contamination reduction through consumer demand; safeguard consumers; safeguard brands that enter the program; ensure legal defensibility; and scale the program to become the de facto standard. No principle is subordinate to any other. Every operational rule in this Part 2 is evaluated against all five; conflicts among principles are surfaced and resolved rather than ignored.

Operational specification

1.2 Action Levels, Not Safety Thresholds

The HMTc Standards set category-specific concentration limits (action levels) for priority toxic heavy metals in finished seafood products. These

action levels are risk-management tools anchored to the best available occurrence data and regulatory benchmarks. They are expressly **not** safety thresholds. Exceeding an action level triggers sourcing, process, and quality control interventions under the Program, but does not by itself indicate that a product is unsafe or that a consumer has been harmed. This distinction is fundamental to the legal and scientific integrity of the Program and shall be reflected in all Program documentation, communications, and mark-usage guidelines.

1.3 ALARA Paradigm

The Program operates under an As Low As Reasonably Achievable (ALARA) paradigm. For metals where no safe exposure threshold is known in children (including lead and inorganic arsenic), the Program requires continuous, feasible reduction of concentrations over time. ALARA is not aspirational language; it is an enforceable obligation within this Program. Brands that achieve certification and then plateau without pursuing further reduction are not in compliance with the spirit or the requirements of this Policy. The ALARA obligation is operationalized through trend monitoring, CAPA requirements, Verification of Effectiveness protocols, and the Standards Ratchet Mechanism described in Part 2.10.

1.4 Concentration-Based Limits

All HMTc action levels are expressed as parts per billion (ppb, equivalent to micrograms per kilogram) on an as-sold basis. The Program does not use per-serving limits. This design prevents circumvention through serving-size manipulation and ensures that compliance is achieved through genuine reduction in contaminant concentrations rather than labeling strategies. Concentration-based limits provide a consistent, testable, and comparable basis for certification across all product categories.

1.5 Legal Compliance Disclaimer

Certification under this Program does not constitute legal advice and does not replace the Licensee's obligation to comply with all applicable laws and regulations in every jurisdiction where the product is sold. If a product's heavy metal content violates a regulatory limit or trigger level in any jurisdiction, that noncompliance is a hard-stop condition under this Policy and will trigger immediate corrective action regardless of the

product's certification status. The Program Operator will cooperate with regulatory authorities as required by law.

1.6 International Scope

HMTc applies anywhere seafood products are sold. The Program is operated by Paleo Certified, Inc., a Florida corporation doing business as The Paleo Foundation, with dispute resolution under California law (see Part 5.6). The Program applies operationally to products sold in any jurisdiction.

The 2026 Edition is calibrated against the global open literature aggregated on heavymetalindex.com. A brand selling in any jurisdiction is evaluated against the same per-category HMTc limits derived from the global corpus, with the lowest applicable regulatory cap in the brand's markets of sale serving as a hard upper bound (per the standards-setting principle in Part 5.1 and the regulatory escalation rule in Part 2.11).

Regional adaptability, roadmap for future Editions. Future Editions of the Standards will introduce region-specific calibration as the heavymetalindex.com corpus accumulates region-specific evidence at sufficient sample-size totals to support statistically defensible regional pooling. The Governing Principles are universal; their *calibration* may become regional in a future Edition. When that happens, the operational rule will be: the more stringent of (regional platform limit, global cleanest-achievable benchmark) governs, so that brands selling in higher-contamination regions face standards that incentivize sourcing from cleaner supply chains, the market-demand catalyst Governing Principle 1 names as the program's primary mechanism.

For the 2026 Edition, regional adaptability is named here as a structural commitment to future evolution, not as an operational feature. The honest disclosure is that the current Edition uses global calibration, and that the next Edition's regional-adaptability work will begin once Cat 2 and additional categories ship. A separate EU-domiciled corporate entity for heavymetalindex.com is on the roadmap; that entity will operate the corpus under EU Database Directive 96/9/EC protection once incorporated, complementing the U.S. copyright and trade-secret protection that the 2026 Edition relies on.

1.7 Program-Operator Continuity

The HMTc Program is operated by Paleo Certified, Inc., a Florida corporation doing business as The Paleo Foundation, with operational governance documented in the controlled-document registry. Brand licensees may rely on Program continuity as follows: the license agreement remains operative across changes in Program leadership; the methodology continues to evolve through the Standards Ratchet and the Governing Principles' revision cycle; and if Program operation transfers to another entity (by sale, merger, or governance reorganization), licensees receive written notice with a minimum 12-month grace period before any operational change affects their certification status. The Program Operator publishes annual operational continuity reporting that includes governance composition, financial sustainability, and any pending transition events.

Cross-references

- Governing Principles full text: see Part 5.1 or `docs/governance/hmtc-governing-principles.md`.
- Standards Ratchet operational specification: Part 2.10.
- Confidential Remediation Track (the structural answer to brands not yet at certification standards): Part 2.4.

2.2 Scope and Analyte Tiering

Narrative

Not all heavy metals present the same toxicological profile, and a defensible certification framework must reflect this. The HMTc Program classifies covered metals into two tiers based on their toxicological characteristics, with governance consequences flowing directly from the classification. Metals that are carcinogenic with no established safe threshold, neurotoxic with no established safe threshold, or bioaccumulative with long biological half-lives warrant zero tolerance for exceedance within the certification framework, these are Tier 1 metals. Metals with established tolerable daily intakes, shorter biological half-lives, or primarily acute rather than chronic toxicity profiles may be eligible for transitional allowances, these are Tier 2 metals.

The tiered approach means that every governance decision, every transitional allowance granted, every probation imposed, has a traceable rationale grounded in the toxicological literature. **If a regulatory body or plaintiff's attorney asks why a 130% nickel result received transitional certification while a 110% lead result triggered immediate probation, the answer traces directly to the comparative toxicological profiles of the two metals. That defensibility is the point.**

Before specifying the tier classification, this chapter establishes the most important framing the manual carries for brand legal teams and regulators: **even compliant products can carry meaningful exposure when single-product regulatory floors are the only benchmark.**

The single-product-floor framing

> Compliance with a single-product regulatory action level does not establish that aggregate or repeated exposure sits below a health-based reference value. FDA action levels and EU maximum levels are single-product risk-management triggers set against feasibility and occurrence data; they are not derived from, and do not guarantee compliance with, a health-based tolerable-intake benchmark for a frequent consumer of the category. A consumer eating a contaminated product at the regulatory ceiling on a regular basis can accumulate intake well above the relevant health-based reference dose even though every individual product is regulatory-compliant.
>
> This is why HMTc category limits sit below applicable regulatory action levels rather than at them. Repeated and cumulative exposure for a frequent consumer of the category, not single-product compliance, is the relevant exposure metric. The Program drives every product steadily and measurably closer to the lowest achievable levels of contamination.

This calculation is the answer to the question “why don’t HMTc limits just align with FDA action levels?” The answer is that compliance with single-product regulatory floors does not deliver compliance with health-based exposure reference values for a frequent consumer of the category. The HMTc Program calibrates limits so that intake at realistic consumption patterns sits well below health-based exposure benchmarks, not just below single-product regulatory action levels.

Covered product categories

This Policy applies to finished products in the seafood categories defined by the HMTc Seafood Standards: marine predatory fish (tuna, swordfish, shark, marlin); marine non-predatory fish (cod, herring, sardine, anchovy, mackerel); freshwater fish; shellfish (bivalves and crustaceans); and seaweed and kelp foods (nori, wakame, kombu, dulse). Canned forms are not a standalone category; a canned product routes to its matrix-native fish or shellfish chapter carrying a tin (Sn) flag. Five specific

product chapters appear in Part 3, each carrying its master limit table, per-analyte rationale, and product-specific toxicology callouts.

Covered analytes and metal classification

The Program evaluates products against action levels for **eight priority toxic heavy metals**. Each is classified into Tier 1 or Tier 2 based on its toxicological profile. Note that the **analytical testing panel** uses ten analytes (the eight priority metals plus speciated forms for arsenic, mercury, and chromium when triggered); this is described after the tier tables.

Tier 1 Metals (Zero Tolerance for Exceedance)

Tier 1 metals are those that are carcinogenic with no established safe threshold, neurotoxic with no established safe threshold, or bioaccumulative with long biological half-lives in children. **No transitional exceedance (Status B variance) is permitted for Tier 1 metals.** Any confirmed result exceeding 100% of the applicable limit for a Tier 1 metal triggers immediate escalation to Status C (Probation) or higher.

Metal	Form Evaluated	Rationale for Tier 1
Lead (Pb)	Total elemental lead	EFSA (2010)[8] concluded that "a threshold for the critical effects of lead in children's neurodevelopment could not be identified" (BMDL01 = 12 µg/L blood lead for one-point IQ decrement). JECFA (2011)[13] withdrew its PTWI on the grounds that "conventional tolerable intake values were no longer health-protective" for endpoints without an identifiable threshold. Full toxicology: see Part 5 Glossary and `wiki/metals/lead` on heavymetalindex.com.

Metal	Form Evaluated	Rationale for Tier 1
Cadmium (Cd)	Total elemental cadmium	Bioaccumulative with biological half-life of 10–30 years; renal cortex is the primary target organ; IARC[17] classifies cadmium and cadmium compounds as Group 1 (carcinogenic to humans). EFSA TWI 2.5 µg/kg bw/week[11] JECFA PTMI 25 µg/kg bw/month[15]. Full toxicology: `wiki/metals/cadmium` on heavymetalindex.com.
Arsenic (iAs)	Inorganic arsenic (speciation required for all seafood; total arsenic is arsenosugar/arsenobetaine-dominated and is not the regulated toxic species)	IARC classifies inorganic arsenic[9] as Group 1 (carcinogenic to humans) on the basis of lung, skin, and bladder cancer evidence. EFSA's 2024 reassessment[10] derived a cancer reference point of approximately 0.06 µg/kg bw/day and concluded that margins of exposure at current dietary intake levels remain low. Full toxicology: `wiki/metals/arsenic-inorganic` on heavymetalindex.com.
Mercury (MeHg)	Methylmercury (speciation required for fish-containing products; reflex for others)	Methylmercury crosses the blood-brain barrier and accumulates in developing neural tissue; prenatal and early postnatal exposure windows are most critical. EFSA TWI 1.3 µg/kg bw/week[12] EPA RfD 0.1 µg/kg bw/day[16]. Full toxicology:

Metal	Form Evaluated	Rationale for Tier 1
		`wiki/metals/mercury-methyl` on heavymetalindex.com.

Tier 2 Metals (Variance-Eligible for Transitional Allowance)

Tier 2 metals are those with established tolerable daily intakes, shorter biological half-lives, or primarily acute rather than chronic toxicity profiles. Transitional exceedances up to 150% of the applicable limit may qualify for Status B certification under the conditions specified in Part 2.3.

Metal	Form Evaluated	Rationale for Tier 2
Nickel (Ni)	Total elemental nickel	Chronic TDI established by EFSA (13 µg/kg bw/day); controllable via sourcing and process. Tier 2 reflects established benchmark rather than no-threshold framing.
Tin (Sn)	Total elemental tin	Acute GI endpoint only; sentinel for packaging integrity; very large margin to chronic-effect levels.
Aluminum (Al)	Total elemental aluminum	JECFA PTWI established (2 mg/kg bw/week); controllable via water treatment and process controls.
Chromium (Cr)	Total chromium (Cr(VI) speciation only if credible pathway exists)	Cr(III) is essential nutrient and is assumed in food; Cr(VI) is Group 1 IARC carcinogen by ingestion. Speciation upgrades the analyte to effectively Tier 1 treatment when triggered.

Analytical testing panel

The Program tests for ten analytes:

- **Total forms (8):** Pb, Cd, total As, Ni, total Hg, Al, total Cr, Sn.

- **Speciated forms (3):** iAs (inorganic arsenic, required for rice-containing products and as a reflex trigger when total As approaches or exceeds 50% of limit); MeHg (methylmercury, required for fish-containing products and as a reflex trigger when total Hg approaches or exceeds 50% of limit); Cr(VI) (hexavalent chromium, required only when a credible Cr(VI) formation pathway exists, such as certain water-treatment processes).

The 8-tier enforcement framework operates on the eight priority metals; speciated forms inherit their parent metal's tier. iAs and MeHg inherit Tier 1 (zero tolerance); Cr(VI) inherits effectively Tier 1 treatment when triggered. Total As, total Hg, and total Cr are tested routinely to support reflex speciation decisions but do not themselves carry separate enforcement tiers, the speciated result governs when speciation has been performed.

The speciation logic operationalizes a key toxicological principle: the **form** of a heavy metal determines its toxicological significance, not just its total concentration. Total arsenic measurements alone are not sufficient for seafood because organic arsenic species (arsenosugars and arsenobetaine, predominant in seafood) are toxicologically distinct from inorganic arsenic (the genotoxic carcinogen) and dominate the total-arsenic signal. Similarly, total mercury alone is not sufficient for predatory fish because methylmercury, the bioaccumulating neurotoxic species, crosses the blood-brain barrier in a way that inorganic mercury does not. Failure to perform required speciation renders the lot ineligible for certification (see Part 4 §9).

Technical Committee authority over metal classification

The Program Operator reserves the right, upon recommendation of the Technical Committee, to reclassify a metal between Tier 1 and Tier 2 based on new toxicological evidence, regulatory changes, or emerging risk data. Any reclassification will be communicated to all Licensees with a defined transition period before enforcement. The Technical Committee may also designate additional analytes for inclusion in the testing panel.

Cross-references

- Per-product master limit tables: Part 3 (each of five chapters).
- Speciation testing protocol: Part 4 §9.
- Decision rules under measurement uncertainty: Part 2.6.
- Toxicology callouts for each metal: Part 5 Glossary, with full reviews on heavymetalindex.com via the canonical wikilinks.

2.3 Certification Status Framework

Narrative

The central design challenge for a heavy metal certification program is balancing rigor with accessibility. A program with unachievable standards attracts no participants and protects no consumers. A program with loose standards lacks credibility and provides no meaningful differentiation. The staged-compliance model resolves this tension by setting rigorous standards, calibrated to deliver the 90/30 row-pass intent specified in the Governing Principles, and giving brands structured pathways to reach full compliance rather than requiring every threshold to be met on day one.

The five-status framework absorbs normal lot-to-lot analytical variability without converting it into a commercial catastrophe, while maintaining clear escalation triggers for genuine control failures. A brand whose typical lead level is 4 ppb against a 5 ppb limit will occasionally produce a lot at 6 or 7 ppb through normal variation, not through negligence; the staged system treats that variability as the engineering reality it is, while preserving zero-tolerance treatment for Tier 1 metals and for systemic control failures regardless of which metal is involved.

Program entry and baseline establishment

Three-Lot Baseline Requirement

All SKUs entering the HMTc Program must be tested across a minimum of three (3) production lots to establish a baseline profile before any certification status is assigned. These three lots shall be tested for the full heavy metals panel using the Program's required methodology. The

three-lot baseline serves to distinguish inherent product variability from anomalous results, providing a statistically meaningful foundation for status assignment. Results from all three lots are evaluated collectively to determine the initial certification status.

Initial Status Assignment

Following completion of the three-lot baseline, the Program Operator will assign an initial certification status based on the results:

- **Status A:** All three lots have MaxPoL at or below 100% for all analytes.
- **Status B:** One or more lots have MaxPoL above 100% but at or below 150%, and the exceedance is limited to Tier 2 (variance-eligible) metals only.
- **Status C:** One or more lots have MaxPoL above 150% (up to 200%), or any Tier 1 metal exceeds 100%, or other probationary triggers are met.
- **Not Eligible for Certification:** Any lot exceeds 200% MaxPoL, or results indicate a regulatory limit violation, or conditions preclude entry into the staged compliance framework.

Brands whose baseline does not qualify for Status A or B may enter the program under the **Confidential Remediation Track** described in Part 2.4 rather than being excluded from the program entirely. The Track is the structural answer to the chilling-effect dynamics analyzed in Part 1.2: brands not yet at certification standards need a pathway that builds an improvement record without creating public exposure risk.

Expedited Baseline via Prior-Data Credit

A brand entering the Program with a documented history of accredited third-party heavy-metal lot testing has, in substance, already begun its baseline. The Program recognizes this directly. Under Governance Policy §4.5, such a brand may credit up to two of the three baseline lots and complete the baseline with a single new full-panel HMTc-methodology lot, shortening time-to-status without compromising the statistical and data-integrity purposes the three-lot baseline exists to serve. The pathway is available only when the prior results are produced by laboratories accredited to ISO/IEC 17025 for the relevant methods and matrices, meeting the limit-of-quantitation requirement for each credited analyte; tested within the twelve months preceding application, each lot a distinct

production run; supplied in structured, machine-readable form with chain-of-custody documentation; and expressed in, or reliably convertible to, the row's native basis. A prior lot is credited only for the analytes it actually measured at or below the applicable HMTc limit of quantitation. Any analyte not covered by the credited prior testing receives no credit and is established by the completing lot, which is sampled under the Program's independent-sampling and chain-of-custody requirements and run across the full panel with speciation where required. For an analyte established by the completing lot alone, the next two routine surveillance lots serve as confirmatory baseline points, and the more conservative interpretation governs if they diverge; in high-risk categories under monthly surveillance, that confirmation completes within the first surveillance cycle. The pathway is not available to brands entering under the Confidential Remediation Track, and it does not reduce the per-analyte data requirements for any analyte lacking qualifying prior data.

The five statuses

Status A: Certified (Full Compliance)

Criteria: All tested analytes at or below 100% of their respective HMTc limits (MaxPoL ≤ 100%). No action levels exceeded; no special allowances needed.

Mark Usage: Permitted. Product may bear the HMTc Certification Mark without restriction.

Testing Frequency: Routine surveillance per the Lot Testing Schedule (Part 4). Frequency is risk-tiered by product category: high-risk categories (predatory fish such as tuna, swordfish, shark, and marlin, where methylmercury bioaccumulation is greatest) tested most frequently; lower-bioaccumulation categories (non-predatory fish, shellfish, and seaweed) tested on the standard surveillance cadence.

CAPA: Not required for passing lots. Preventive action is expected if trend analysis shows an upward trajectory approaching limits.

Ongoing Obligations: Maintain all controls and record-keeping. Continue periodic trend analysis. **Pursue ALARA improvement even while compliant**, Status A is not a plateau, and the Standards

Ratchet (Part 2.10) ensures that the bar tightens as the certified market improves.

Status B: Certified (Transitional Compliance)

Criteria: MaxPoL above 100% but at or below 150%, and the exceedance is limited to Tier 2 (variance-eligible) metals. No Tier 1 metal exceeds 100%. No other hard-stop conditions triggered.

Mark Usage: Permitted with conditions. The product may bear the HMTc Certification Mark, but certification is flagged as transitional in the Program registry.

Testing Frequency: Elevated surveillance per the Lot Testing Schedule. Monthly independent sampling at minimum. The Program Operator may require every-lot testing based on risk assessment.

CAPA: Mandatory. CAPA must be opened within 10 business days of status determination, with a detailed plan submitted within 30 days.

VOE (Verification of Effectiveness): Required. The CAPA must include a VOE plan demonstrating measurable reduction in heavy metal concentrations over subsequent lots.

Time Limit: Brands are given a default of 12 months to achieve Status A. The Program Operator may adjust this timeline based on the complexity of the required changes. Failure to demonstrate an improving trend will trigger escalation to Status C.

Escalation: If any lot tested during Status B exceeds the 150% cap, involves a Tier 1 metal exceedance, or the brand fails to show improvement, the SKU is immediately escalated to Status C.

Status C: Probation (Heightened Surveillance)

Criteria: Any of the following: MaxPoL between 150% and 200% for any analyte; repeated Status B outcomes without improvement; any Tier 1 metal exceeding 100% of its limit; systemic process control failure as determined by the Program Operator.

Mark Usage: Not permitted on new production. Only individual lots that test at Status A or B levels may be released with the Certification Mark, on a lot-by-lot approval basis.

Testing Frequency: Every lot, with mandatory hold-and-release. No product may ship until test results are received and evaluated.

CAPA: Accelerated. CAPA must be opened within 5 business days, with a full plan submitted within 15 business days. The CAPA must include a Probation Lot Release Plan.

Graduated probation severity. To ensure proportionate response, Status C is differentiated by severity:

- **Low Probation (MaxPoL 150–175%):** Every-lot testing with hold-and-release. Brand's own accredited laboratory sampling is permitted, subject to Program Operator audit rights. The Program Operator may conduct unannounced verification sampling.
- **High Probation (MaxPoL 175–200%):** Every-lot testing with hold-and-release, plus mandatory independent third-party sampling directed by the Program Operator. The third-party sampler collects samples independently of the brand's quality team and submits them directly to a Program-designated laboratory. This dual-track approach ensures data integrity under heightened risk conditions.

Exit criteria. To exit probation, the brand must produce a minimum of five (5) consecutive lots that individually test at Status A or Status B levels (the Program Operator may require more depending on severity), and complete all CAPA milestones with verified effectiveness. Both conditions must be satisfied before probation is lifted.

Status D: Suspension

Criteria: MaxPoL exceeds 200% for any analyte; any hard-stop condition (including regulatory limit violation or data integrity failure); failure to execute required testing, CAPA, or enforcement directives during probation.

Mark Usage: Prohibited. The Certification Mark must be removed from all product units and marketing materials immediately.

Required Actions: Stop-ship of affected product; removal of seal from marketing; market correction as directed; CAPA within 2 business days with containment within 48 hours.

Reinstatement: Requires completion of CAPA with VOE, consecutive compliant lots as determined by the Program Operator, restored data integrity, and formal Program Operator approval.

Status E: Revocation

Criteria: Persistent suspension without effective correction; severe or repeated nonconformance; willful misconduct or data fraud.

Mark Usage: Permanently prohibited. Certification is fully withdrawn. License agreement is terminated.

Re-entry: Only by full re-application to the Program at the Program Operator's discretion. Trust must be rebuilt from the ground up.

Cross-references

- Confidential Remediation Track (for brands whose baseline does not qualify for A or B): Part 2.4.
- Expedited Baseline via Prior-Data Credit: Governance Policy §4.5; Lot Testing Schedule §4.3 (Part 4).
- Lot Testing Schedule (testing frequency by Status × Risk Tier): Part 4.
- CAPA procedures: Part 2.8.
- Mark Usage Rules: Part 2.5.
- Standards Ratchet (how limits evolve over time): Part 2.10.

2.4 The Confidential Remediation Track

The structural problem this chapter solves

The commercial and legal landscape around heavy metals in seafood creates a paradox: **the brands most in need of a structured improvement program are the least likely to join one voluntarily if doing so creates public exposure.** Part 1.2 documented this dynamic at length. In a climate where consumer advocacy groups, state attorneys general, and plaintiffs' attorneys are actively scrutinizing the industry, any public acknowledgment that a brand's products contain elevated heavy metals, even in the context of a voluntary improvement program, could be weaponized in litigation or media coverage. The chilling effect is real, documented in the compliance-theory literature, and structurally hostile to public-health objectives.

The Confidential Remediation Track is the program's structural answer to this paradox. It is the single most important differentiator of HMTc from the exposé-driven enforcement environment that Part 1 attacks. It is the reason this chapter is hoisted to its own section in Part 2 rather than appearing as a subsection of program-entry rules. A brand legal team reading this manual to decide whether to recommend HMTc participation should read this chapter alongside Part 5.2 (the Program-Level Adversarial Read), together they answer the question "what does HMTc give my client that internal testing alone does not?"

What the Track is

Brands whose initial baseline results do not qualify for Status A or Status B may enter the Program under the Confidential Remediation Track,

provided that all tested lots meet applicable legal obligations for heavy metals in their markets of sale at the time of testing. (Regulatory limit violations remain a hard-stop condition under Part 2.11 regardless of certification status.)

Under the Confidential Remediation Track:

- **Enrollment status is confidential.** The brand's participation in HMTc is treated as confidential business information under the license agreement. The Program Operator will not unilaterally publicize a brand's enrollment, certification efforts, or test data.
- **Test data is confidential.** All baseline testing results, surveillance results, CAPA documentation, and remediation progress data are confidential business information. The Program Operator may use aggregated, anonymized data from the Track for supplier-intelligence purposes (per Part 2.9) and for ratchet-trigger analysis (per Part 2.10), but individual brand identities and SKU-level results are not disclosed.
- **Full Program resources are available.** Brands in the Track receive full access to the Program's remediation support, supplier intelligence, testing infrastructure, technical guidance, and the Program Operator's working relationships with accredited laboratories and remediation consultancies.
- **Mark usage is restricted until qualification.** The brand may not use the HMTc Certification Mark on any product, packaging, or marketing materials until at least one SKU achieves Status A or Status B. Confidentiality is the Track's structural feature; mark usage is the Program's signal of verified compliance. The two are deliberately decoupled until the brand earns the signal.
- **Quality improvement initiative framing.** The brand's participation in HMTc and all associated testing data constitute a quality improvement initiative undertaken in good faith to reduce contaminant levels. This framing is intended to support the brand's legal defensibility by establishing a documented record of proactive, third-party-verified improvement efforts.

The legal defensibility argument

A primary value proposition of HMTc enrollment, including the Confidential Remediation Track, is the creation of a **documented, time-stamped, third-party-verified record of continuous improvement.**

In the current litigation environment, the most legally exposed posture for a brand is the *absence* of any documented heavy metal reduction program. A brand that has no testing data, no improvement record, and no documented engagement with reduction efforts faces plaintiffs' counsel arguing that the brand "knew or should have known" about contamination and "did nothing." This framing is the basis of the neurodevelopmental-injury theories advanced in MDL 3101[24] and parallel state actions.

A brand enrolled in the Confidential Remediation Track has a different posture available. The brand can document, with third-party verification: when it became aware of its baseline contamination profile; what remediation actions it undertook; how those actions performed against predefined improvement targets; and whether its lots are trending toward Status A or B over time. **The legal narrative shifts from "the brand knew and did nothing" to "the brand knew, enrolled in an independent certification program, and has documented evidence of continuous improvement." That shift is the difference between exposure and defense.**

This shift does not eliminate litigation risk. It changes the brand's posture from passive to active. A plaintiff's expert challenging the brand's controls now faces a documented improvement record that begins before any inquiry was filed; a defense attorney has a contemporaneous, externally validated narrative of due diligence to anchor the defense. Defense counsel familiar with MDL-style litigation will recognize this as the structural difference between a brand that can be discovered into a corner and a brand that has a story to tell.

Discovery dynamics, honest disclosure

Track data is held by the Program Operator as a service provider to the Licensee under the License Agreement. In a discovery scenario, Track data is **potentially compellable** from the Program Operator under subpoena. The License Agreement specifies that the Program Operator

will notify the Licensee of any subpoena and cooperate with the Licensee's defense to the extent legally permissible.

The Track does not provide attorney-client privilege or work-product protection over the testing data. What it provides is contemporaneous, third-party-verified evidence of due diligence. That evidence is admissible *for* the brand's defense; it is also discoverable by plaintiffs. The structural value of the Track is that the evidence record exists, is timestamped, and was generated in good faith for quality-improvement purposes before any litigation was foreseeable. Brands and their counsel evaluate whether that evidentiary posture is preferable to the alternative posture (no documented program at all).

Brands evaluating the Track should consult counsel about how the Track interacts with their broader discovery posture. The Program Operator does not offer legal advice, and the Track is not a substitute for counsel-directed quality programs that may carry stronger privilege protections in specific litigation contexts. The Track is one element of a brand's overall legal-defensibility strategy, not a complete substitute for that strategy.

Why confidentiality is not a loophole

A regulator or consumer-advocacy reader may ask: doesn't the Confidential Remediation Track let brands hide contamination data from the public?

The answer is that the Track is the *structural* answer to the chilling-effect failure mode, not a workaround of consumer protection. **A brand that remains outside the program because it fears the consequences of joining is a brand whose products are not improving.** A brand that enters confidentially and begins reducing heavy metal levels is a brand whose products *are* getting safer, even if the public does not yet know. The relevant public-health question is which arrangement produces more contamination reduction in the food supply, and the compliance theory literature[43, 44, 45], summarized in Part 1.2, is clear that punitive-disclosure regimes produce less reduction than confidential-improvement regimes.

A brand that successfully completes the Track and qualifies for Status A or B then *can* display the mark publicly. Their improvement record

becomes a competitive asset rather than a liability. The Track does not let brands hide forever; it lets brands enter from a starting position that would otherwise exclude them, build the improvement record that justifies the mark, and then operate publicly when they have earned the right to do so.

A brand that enrolls in the Track, fails to improve, and exits without ever qualifying for the mark is in approximately the same position they would have occupied without the Track, except that they have a confidential record of attempted improvement that may still factor into a future legal defense. They have not gained the mark, and they have not gained immunity, but they have gained documented good faith. That is the proportionate outcome of the Track's design.

Aggregated data and public reporting

The Program Operator publishes annual aggregate statistics on Program participation and trend movement, including:

- Total number of SKUs enrolled across all statuses (including the Track, in aggregate).
- Per-analyte aggregate distribution shifts year-over-year (the data that feeds the Standards Ratchet trigger).
- Number of Confidential Remediation Track enrollees that transitioned to Status A or B in the reporting period.
- Number of CAPA-driven supplier intelligence alerts issued (anonymized).

These aggregate statistics are public. **Individual brand identities and SKU-level results are not.** The aggregate data demonstrates that the Program is achieving population-level contamination reduction without exposing individual participants to the litigation amplification cycle analyzed in Part 1.2.

This is the targeted-transparency model described by the National Academies in its 2011 report on FSIS establishment-specific data release: aggregate disclosure that serves a defined public-health purpose, structured so the disclosure is interpretable and proportionate, with confidentiality at the individual-participant level where confidentiality serves the goal of maximizing program participation. The Track is built on this principle.

Exit from the Track

A brand exits the Confidential Remediation Track in one of three ways:

1. **Successful qualification for Status A or B on at least one SKU.** The brand transitions to the standard certification framework; mark usage becomes available for the qualifying SKU; the brand's confidentiality protections remain for any SKU still in the Track.
2. **Voluntary withdrawal.** The brand may withdraw from the Program at any time. Their Track participation remains confidential per the license agreement; the Program Operator does not publicize past enrollment.
3. **Program Operator termination.** The Program Operator may terminate a brand's Track participation for data integrity violations, failure to engage with remediation activities, regulatory limit violations, or material breach of the license agreement. Terminated brands forfeit Track confidentiality protections only to the extent required by enforcement obligations or regulatory cooperation; the Program Operator does not publicize terminations as a punitive measure.

Cross-references

- Game-theoretic and naming-and-shaming analysis that motivates the Track: Part 1.2.
- Aggregate data feeds for the ratchet trigger: Part 2.10.
- Supplier intelligence sharing (uses anonymized Track data): Part 2.9.
- Program-level adversarial read on Track confidentiality: Part 5.2.
- License agreement summary (Track confidentiality is contractual): Part 5.6.

2.5 Mark Usage Rules and Restrictions

Narrative

The HMTc Certification Mark is the public signal that a product has been independently tested and meets defined action levels under an ongoing surveillance protocol. Mark usage rules are designed to ensure that the mark means what it says, that consumers, retailers, and regulators can rely on its presence as an indicator of verified compliance, and that brands cannot game the mark through arbitrage, status manipulation, or misleading marketing claims.

6.1 Eligibility

Only SKUs with a current Status A or Status B certification may bear the HMTc Certification Mark. Status C, D, and E products may not display the mark under any circumstances. Products in the Confidential Remediation Track may not display the mark. A brand must have at least one SKU at Status A or B before any use of the mark is permitted.

6.2 Prohibited Marketing Claims

Brands using the HMTc Certification Mark shall not make claims that imply absolute safety, absence of heavy metals, or that the certified product poses zero risk. Specifically prohibited claims include but are not limited to: “heavy metal free,” “no toxic metals,” “100% safe from heavy metals,” or any language suggesting that the HMTc mark indicates the absence of heavy metals or the elimination of risk.

The mark certifies that the product has been independently tested and meets defined action levels under an ongoing surveillance protocol. **It does not certify safety.** This distinction is fundamental to the program's legal and scientific integrity and traces directly to the action-levels-not-safety-thresholds framing in Part 2.1.

6.3 Anti-Arbitrage: Re-Entry After Dropping the Mark

If a brand removes the HMTc Certification Mark from a SKU during a remediation period, whether voluntarily or by enforcement, the brand cannot simply resume mark usage after producing one passing lot. Re-entry to mark usage requires:

- Three (3) consecutive passing lot results at Status A or B levels, and
- Closure of any open CAPA associated with the SKU.

This prevents status arbitrage, where a brand might drop the mark during bad periods (avoiding the documented exceedance) and re-enter when results improve by chance (claiming a clean certification record without the intervening problem appearing in the registry). The anti-arbitrage rule ensures that the mark's signal, *this product has demonstrated sustained compliance*, is not subvertible by selective mark display.

6.4 Brand Visibility Controls

Brands may opt out of publicly displaying their certified SKUs in any Program registry or consumer-facing platform. The decision to show or not show certified SKUs rests entirely with the brand. This capability is built into the HMTc platform, giving brands full control over the visibility of individual SKUs that have been approved for mark usage.

The Program Operator will not unilaterally publicize a brand's participation or certified SKUs without written consent, except as required by enforcement actions or regulatory obligations. This provision is consistent with the confidentiality principles that govern the Confidential Remediation Track (Part 2.4) and reflects the Program's broader posture: confidentiality is the default; public disclosure is the brand's choice.

Cross-references

- Status framework that determines mark eligibility: Part 2.3.
- Confidential Remediation Track (mark prohibition until qualification): Part 2.4.
- License agreement terms governing mark usage: Part 5.6.
- Trademark and certification mark IP protections: Part 5.7 (Colophon).

2.6 Decision Rules under Measurement Uncertainty

Narrative

The analytical chemistry of heavy metal testing at low parts-per-billion concentrations introduces inherent measurement uncertainty. A reported result of 11 ppb against a 10 ppb limit does not necessarily indicate that the true concentration exceeds the limit; the measurement could reflect analytical noise. Conversely, a result of 9 ppb does not guarantee the true concentration is below the limit. **Certification decision rules must account for this uncertainty to be both scientifically valid and legally defensible.**

The HMTc decision rule follows established metrological principles documented in international standards (EURACHEM/CITAC[34], ILAC-G8) and accredited laboratory practice under ISO/IEC 17025[33]. Under cross-examination by a plaintiff's expert or a hostile regulatory reviewer, a decision based on this framework is defensible because it is neither biased toward the brand nor toward the regulator: it is biased toward analytical accuracy.

7.1 Primary Decision Rule

Certification pass/fail decisions incorporate the laboratory's reported Measurement Uncertainty (MU) using a two-sided decision rule:

- **Pass:** (Result + MU) ≤ Limit. Even at the upper end of the uncertainty range, the concentration does not exceed the action level.
- **Fail:** (Result – MU) > Limit. Even at the lower end of the uncertainty range, the concentration clearly exceeds the limit.

- **Borderline:** Neither condition is met. The true value could be either below or above the limit given the uncertainty. Borderline results require confirmatory action before a status is assigned.

This protects brands from being penalized for measurement noise while protecting consumers from borderline results being treated as passes. The Program does not deliver a pass that the laboratory's uncertainty calls into question; it does not deliver a fail that the laboratory's uncertainty cannot support.

7.2 Borderline Result Resolution

No product will be certified on the basis of a borderline result alone. The Program Operator will require at least one of the following before finalizing certification status:

- Confirmatory re-analysis of the same sample extract;
- Confirmatory testing on a new sample from the same lot under strict chain-of-custody; or
- Lot hold-and-release pending resolution.

The approach is documented, and the more conservative interpretation governs in cases of ambiguity.

7.3 No Testing into Compliance

Repeated, ad hoc retesting to obtain a passing result is **strictly prohibited.** Any retesting must follow a pre-defined pathway approved by the Program Operator and be scientifically justified. Undocumented resampling, lab-shopping, or selective reporting of results is a data integrity violation subject to suspension or revocation (per Part 2.7).

This prohibition is structural rather than punitive. The program's evidence record must withstand discovery in litigation: a defense attorney pointing to a certified SKU's test history needs to be able to say that the record represents the program's actual analytical findings, not a curated subset chosen to support a pre-determined outcome. Anti-retesting language is the discipline that makes the evidence record admissible.

7.4 Default Rule When MU Is Not Reported

If a laboratory does not report Measurement Uncertainty, the measured result is evaluated directly against the limit without uncertainty adjustment. A result above the limit is treated as a failure; a result at or below the limit is treated as a pass. Laboratories that cannot report MU are not preferred for Program use, and the Program Operator may require transition to a laboratory that provides MU.

Cross-references

- Laboratory accreditation requirements (ISO/IEC 17025[33]): Part 2.7 and Part 4 §10.
- Data integrity violations and their consequences: Part 2.7.
- CAPA procedures when a fail is confirmed: Part 2.8.
- Chain-of-custody requirements: Part 4 §10.

2.7 Data Governance and Integrity

Narrative

A certification program's credibility ultimately rests on the integrity of its data. The Standards Ratchet (Part 2.10) operates on aggregated lot-testing results across all participating brands; the supplier-intelligence system (Part 2.9) operates on root-cause findings shared across the program; the legal defensibility argument (Parts 1.3, 2.4) rests on a documented improvement record that withstands discovery. Each of these depends on data being structured, auditable, and resistant to manipulation.

Data integrity is therefore not a procedural concern but a structural condition of the program. Falsification, manipulation, selective omission, or unauthorized alteration of test data is a hard-stop violation. This chapter specifies the data-governance regime that operationalizes the program's integrity requirements.

8.1 Structured Data Requirement

Brands must permit the Program Operator, in writing, to access the data from their laboratory tests. **Submission of results as PDF documents is not permitted.** All test data must be provided in a structured, machine-readable format that enables trend analysis, statistical evaluation, and population-level aggregation. The specific format requirements are defined by the Program Operator and communicated to all Licensees.

This requirement exists because the Program's ability to demonstrate aggregate improvement over time, which is both the strongest legal defense for individual brands and the strongest proof that the Program

achieves its public-health objectives, depends on structured, queryable data. Static PDFs cannot be aggregated, cannot be trend-analyzed, and cannot feed the ratchet trigger. The structured-data requirement is the architectural precondition for the Standards Ratchet to function.

8.2 Data Integrity as a Hard-Stop Condition

Falsification, manipulation, selective omission, or unauthorized alteration of test data is a hard-stop violation that triggers immediate suspension (Status D) and likely revocation (Status E). Data integrity violations include but are not limited to:

- Submitting falsified laboratory results.
- Failing to report required test results.
- Using non-accredited laboratories without authorization.
- Breaking chain-of-custody without documentation.
- Undisclosed retesting or lab-shopping (per the no-testing-into-compliance rule in Part 2.6).

The Program Operator will audit data integrity as part of routine oversight and may require random verification testing at designated laboratories.

The consequence calibration is intentional. Status D suspension is reserved for cases where remediation may eventually restore certification; Status E revocation terminates the license agreement and bars re-entry except by full re-application at the Program Operator's discretion. Data integrity violations are treated at the Status D-to-E severity level because, unlike contamination events that may reflect supply-chain or process variability, integrity violations reflect a deliberate choice by the brand to subvert the program's evidence record. The program cannot defend a certification mark whose evidence base is corrupted by participant falsification.

8.3 Laboratory Requirements

All testing used for certification decisions must be performed by laboratories accredited to ISO/IEC 17025[33] for the relevant analytical methods and food matrices. Analytical methods must be fit for purpose, meaning they can reliably quantify target heavy metals at concentrations at or

below the applicable HMTc category limits. The laboratory's Limit of Quantitation (LOQ) for each analyte must be at or below the HMTc action level. If an LOQ exceeds the Program limit, a non-detect result is not acceptable proof of compliance.

Laboratories that the Program Operator has reason to question, based on proficiency-test performance, audit findings, or pattern analysis of results across participating brands, may be removed from the list of Program-acceptable laboratories. Licensees using removed laboratories must transition to a Program-acceptable laboratory before their next surveillance cycle. The Program Operator maintains the current list of acceptable laboratories under document control.

Cross-references

- Decision rules under measurement uncertainty (MU reporting requirements): Part 2.6.
- Chain-of-custody requirements: Part 4 §10.
- Anti-arbitrage rules (related to data presentation manipulation): Part 2.5.
- Standards Ratchet (depends on structured data): Part 2.10.
- Supplier intelligence sharing (depends on structured CAPA root-cause data): Part 2.9.

2.8 Corrective and Preventive Action (CAPA)

Narrative

Corrective and Preventive Action is the operational mechanism that converts a status assignment into measurable improvement. CAPA is required whenever a SKU enters Status B, C, or D. The required elements, timelines, and verification protocol are specified below. Every CAPA must connect a documented root cause to documented corrective and preventive measures, and must be verified through follow-up testing before closure.

9.1 CAPA Initiation Timelines

Status	CAPA Initiation	Full Plan Due	Containment
Status B	10 business days	30 calendar days	As needed
Status C	5 business days	15 business days	Within 48 hours
Status D	2 business days	10 business days	Immediate

These timelines are maximum allowable periods; faster action is always expected. The Program Operator may require accelerated timelines based on the severity of the underlying exceedance or the risk profile of the affected SKU.

9.2 Required CAPA Elements

Every CAPA must include the following documented components:

- **Issue Identification:** Affected SKU, lot(s), analyte(s), numerical results, applicable limits, and calculated PoL/MaxPoL.

- **Impact Quantification:** Extent of exceedance, historical trend data, and context (outlier vs. systemic).
- **Immediate Containment:** Actions taken to prevent potentially noncompliant product from reaching consumers.
- **Root Cause Analysis:** Thorough investigation of ingredients, water, process, equipment, packaging, and formulation. **Structured methodology (5-Why, fishbone, fault tree) is required.** Unstructured root-cause assertions are not acceptable.
- **Corrective Actions:** Short-term fixes to bring current and upcoming lots into compliance.
- **Preventive Actions:** Long-term systemic changes to prevent recurrence (supplier qualification, process modification, formulation changes, enhanced specifications).
- **Verification of Effectiveness (VOE):** A defined plan specifying how many validation lots will be tested and what criteria constitute success. The CAPA cannot be closed until VOE confirms the problem is resolved.

The root-cause analysis is the source of supplier intelligence data shared anonymously across the program per Part 2.9. CAPAs that identify supplier, ingredient-source, or geographic-origin contributors generate anonymized alerts to other Licensees using the same supply-chain inputs.

9.3 CAPA Closure

A CAPA is closed only after the VOE demonstrates that the product has returned to controlled status and meets defined success criteria. The Program Operator reviews the evidence and formally approves closure. If criteria are not met, the CAPA is extended or additional actions are required.

Cross-references

- Status framework (when CAPA is required): Part 2.3.
- Lot Testing Schedule (testing frequency during open CAPA): Part 4.
- Supplier intelligence sharing (uses CAPA root-cause data): Part 2.9.
- Data governance (CAPA documentation requirements): Part 2.7.

2.9 Supplier Intelligence Sharing

Narrative

Individual brands testing their own finished products can identify contamination after it has already occurred. A certification program operating across multiple brands has the opportunity to detect contamination patterns at the supply chain level and alert affected participants *before* contamination reaches finished products. This is the upstream remediation pathway that Part 1.2 identified as structurally absent from the exposé-driven enforcement environment.

The Supplier Intelligence System is the operational mechanism that converts individual-brand CAPA root-cause analyses into program-wide collective intelligence. It is what makes HMTc more than a sum of its participants: each additional brand contributes data that benefits all other participants, creating a positive network effect that strengthens retention and attracts new entrants. From a public-health perspective, it accelerates the identification and remediation of contamination sources across the entire supply chain rather than within individual brands operating in isolation. From a competitive-strategy perspective, supplier intelligence is the durable moat that distinguishes mature HMTc certification from any single-brand internal-testing program.

The system

The Program Operator maintains an anonymized supplier intelligence system. When a Licensee's testing identifies a supplier or ingredient source with elevated heavy metal levels, the Program Operator may issue anonymized alerts to other Licensees using the same supplier or ingredient category, without identifying the reporting brand. The system operates as follows:

- **All supplier-related findings submitted through CAPA root cause analyses are evaluated for potential cross-brand relevance.** Findings that name a supplier, ingredient origin, geographic source, or processing pathway with elevated heavy metal levels are candidates for alerting.
- **Alerts are anonymized.** The identity of the reporting brand is never disclosed to recipients of the alert. The Program Operator preserves brand confidentiality even when issuing alerts that may indirectly benefit competitors.
- **Alerts identify the ingredient category, geographic origin (if known), and the nature of the concern.** Example: "Elevated arsenic levels observed in rice flour sourced from [geographic region X]. Affected brands report typical results 30-50% above category baseline. Suspected mechanism: [soil irrigation source or processing-water contamination]. Recommended action: independent verification testing of incoming rice flour lots."
- **Licensees receiving an alert are expected to evaluate their own supply chain for the flagged risk and take appropriate action.** Failure to respond to a Supplier Intelligence Alert is documented and may inform future status determinations.

Participation in the supplier intelligence system is a condition of the license agreement. **Brands benefit from collective intelligence, creating a powerful incentive to remain in the Program and a significant informational disadvantage for those outside it.**

Anonymization limits, honest disclosure

The Program Operator anonymizes alerts by removing brand identifiers. However, ingredient-category, geographic-origin, and timing metadata can correlate with a brand's publicly disclosed sourcing in ways that may identify the reporting brand to a sophisticated observer. For example, if a brand has publicly disclosed sourcing cocoa from a specific Ecuadorian region, and a supplier-intelligence alert is issued in the same quarter referencing "elevated cadmium in Ecuadorian high-altitude cocoa," correlated observers could infer the reporting brand.

The Program Operator mitigates the metadata-correlation risk through three operational controls:

- **Batching.** Alerts are batched so a single alert does not correspond to a single reporting brand's CAPA timeline. Multiple reporting brands' findings about the same supplier or region are combined into one alert where the protective intent is preserved.
- **Geographic generalization.** Geographic specificity is generalized (region rather than exact origin) where the alert's protective intent is preserved by the broader framing.
- **Discretionary hold.** Alerts may be held for a discretionary delay period when the reporting brand requests it, allowing the reporting brand's quarterly disclosures to settle before the alert publishes.

Brands relying on anonymization for confidentiality should treat anonymization as **strong but not absolute**, and should consider the metadata-correlation risk in their participation decision. The Program Operator's commitment is to apply the mitigations above as a matter of operational discipline; the Program Operator does not warrant that a sophisticated observer with full access to a brand's public sourcing disclosures could not infer the reporting brand from a specific alert.

Why anonymization is a design feature

A reader might ask: doesn't anonymization let problem suppliers escape accountability?

The answer is that anonymization serves the same structural function that the Confidential Remediation Track does for brands. A program in which CAPA root-cause findings *named* specific suppliers in public alerts would create the same chilling effect Part 1.2 analyzed: brands would underreport supplier issues, supply-chain remediation would slow, and the collective intelligence value would collapse. Anonymization is what makes the system work.

The Program Operator addresses upstream supplier accountability through bilateral channels (working with affected brands on supplier qualification, requalification, or replacement) rather than through public naming. This is the same logic that drives the staged-compliance framework in Part 2.3: structural responses calibrated to outcomes, not punitive responses calibrated to attribution.

A consumer-advocacy reader who believes that named-supplier disclosure would drive faster contamination reduction is invited to read the Part

1.2 analysis of why naming-and-shaming dynamics produce the opposite outcome in practice. The empirical evidence across naming-and-shaming literature, supply-chain economics, and food-safety regulation converges on the conclusion that confidential collective intelligence outperforms public attribution at reducing contamination.

Cross-references

- Game-theoretic and naming-and-shaming analysis that motivates anonymization: Part 1.2.
- CAPA root-cause requirements: Part 2.8.
- Reflex testing triggered by Supplier Intelligence Alerts: Part 4 §8.
- Standards Ratchet (uses aggregated improvement data): Part 2.10.

2.10 The Standards Ratchet Mechanism

Narrative

A static certification standard is inherently self-defeating. If standards remain unchanged while industry improves, they eventually certify the status quo rather than driving further progress. Conversely, if standards are perceived as arbitrary or subject to surprise changes, brands lose the planning certainty necessary to invest in reformulation and supply-chain improvements. The Standards Ratchet Mechanism resolves this through a formal commitment to tighten standards over time, moving in only one direction, on data-driven trigger conditions, with a **capped step size** that bounds the per-cycle impact on any participating brand.

The mechanism is what makes HMTc a forward-moving signal rather than a static gate. As the certified market improves, the standards tighten; brands that have already invested in reduction find that their leadership is rewarded by tighter limits that exclude less-invested competitors; the population-level contamination reduction that Governing Principle 1 requires is delivered cumulatively, year over year.

This chapter is the operational specification of that mechanism. Each sub-section corresponds to a clause that brand legal teams, regulators, and adversarial reviewers should be able to verify on its own terms. The full §12 specification appears below.

12.1 Review Cycle

The HMTc Seafood Standards shall be formally reviewed on a cycle of not more than three (3) years, or sooner if significant new toxicological data, regulatory changes, or aggregate Program data warrant revision.

The review is conducted by the Program Operator with input from the Technical Committee.

The annual operational review (referenced throughout §12.3 below) is distinct from the formal three-year revision cycle. The annual review checks ratchet-trigger conditions and publishes the forward forecast. The three-year revision cycle reviews the methodology, the Tier 1/Tier 2 classification, the panel composition, and any structural updates to the framework.

12.2 Directional Commitment

The Standards Ratchet operates under a directional commitment: **action levels may be tightened (lowered) but shall not be loosened (raised) unless the underlying science changes in a manner that justifies relaxation.** This commitment ensures that the Program continuously drives exposure reduction and that brands can plan their improvement trajectories with confidence that the goalposts will move in only one direction.

A science-based relaxation, for example, a future EFSA reassessment[10] that meaningfully revises a metal's reference dose upward, would require Technical Committee review, Program Operator approval, and a defined transition period before any limit relaxation took effect. The directional commitment is not absolute, but its scope is narrow and its conditions are documented.

12.3 Trigger Conditions

The Standards Ratchet for a given analyte fires when **both** of the following conditions hold for two consecutive annual reviews:

(a) The 80th percentile of the combined Status A + Status B trigger pool for the analyte sits at or below 80% of the current limit.

(b) At least 50% of SKUs in the trigger pool have lots at or below 60% of the current limit.

The first condition tests aggregate market shift. The second tests breadth, the ratchet does not fire on a single large brand pulling the percentile down. Both conditions must hold across two consecutive annual

reviews to confirm sustained improvement rather than a transient market event.

The two-condition design is intentional. Condition (a) alone would allow a single dominant brand to drive the trigger by aggressive reformulation across many SKUs; condition (b) alone would not capture the magnitude of market shift. Together, the two conditions require that the market has moved *and* that the movement is broad-based. The two-consecutive-cycle requirement protects against transient effects (a particularly clean harvest year, a temporary supplier change) that would otherwise distort the trigger.

Anti-gaming provisions

Trigger pool composition. The trigger pool combines Status A and Status B lot results. This prevents *withholding gaming*, in which a brand could keep its best-performing SKUs at Status B by delaying CAPA closure on marginal Tier 2 exceedances to keep strong data out of the Status A pool. Combining the two pools makes the trigger insensitive to status-classification gaming on the brand's side.

Single-Licensee cap. No single Licensee may contribute more than 25% of the SKUs in the trigger pool for a given analyte. When a Licensee's SKU count exceeds the cap, the Program Operator excludes that Licensee's SKUs above the cap from the trigger calculation (selected randomly to avoid Licensee selection bias) and recalculates against the capped pool. This prevents a single dominant Licensee from driving trigger firing through aggressive reformulation across many SKUs; the ratchet reflects industry-wide movement rather than one Licensee's individual progress.

Exit-and-re-enter prohibition. A Licensee that voluntarily withdraws from the Program and subsequently re-enrolls within 24 months is evaluated against the standard in effect at the time of re-entry, not the standard in effect at the time of original exit. Grandfathering protections under §12.5 apply only to continuous certification. This rule prevents the exit-and-re-enter gaming pathway, in which a brand could withdraw before a ratchet event to escape transition obligations and then re-enter once the new standard's transition period has elapsed. A Licensee that withdraws for documented commercial reasons (product discontinua-

tion, line restructuring, change of control) and re-enrolls after 24 months is treated as a new applicant per standard program-entry procedures.

12.4 Step Size and New Limit Calculation

When the trigger fires, the new limit equals the **maximum of**:

(a) The actual 80th percentile of the trigger pool for the analyte, or (b) 80% of the current limit.

The second clause is the **step cap**. **No brand faces more than a 20% reduction per ratchet event, regardless of how far the market has actually moved.** Residual tightening above 20% is held over and may trigger again at the next cycle. This converts the ratchet from a potential shock into a predictable, capped progression.

The step cap is the program's structural protection for participating brands. A brand that has invested in reduction expects to be rewarded by tighter standards that exclude competitors who have not invested; the brand does not expect to be punished by surprise tightening that exceeds its planning horizon. The 20% cap establishes a maximum per-cycle obligation that brands can model and budget against.

Worked example

Current Pb limit is 30 ppb. The 80th percentile of the Status A + Status B trigger pool is 15 ppb (50% of the limit). The breadth condition is met (≥50% of SKUs at or below 60% PoL = ≤18 ppb). The trigger fires.

New limit candidate (a): 15 ppb (the actual 80th percentile). New limit candidate (b): 80% of 30 ppb = 24 ppb (the step cap). New limit = max(15, 24) = **24 ppb**.

The step cap held; the market actually moved beyond the cap, but the program tightens only by the capped 20% increment.

At the next annual review, the 80th percentile of the trigger pool against the new 24 ppb limit is evaluated. If the market stays at the same absolute level (15 ppb), the 80th percentile is now 15/24 ≈ 62.5% of the new limit, below the 80% trigger condition. Both conditions hold. The trigger fires again.

New limit candidate (a): 15 ppb. New limit candidate (b): 80% of 24 ppb = 19.2 ppb. New limit = max(15, 19.2) = **19.2 ppb**.

The residual surfaces in the next cycle, but the cadence is determined by market behavior after the first tightening, not by mechanical compounding. A brand modeling the program's future obligations can plan against this worked example with confidence in the predictability of the trajectory.

12.5 Transition

When standards are tightened, a transition period of not less than 12 months and not more than 18 months will be provided between announcement and enforcement.

Status A certifications grandfather for the entire transition period. A Status A brand at certification renewal during the transition window keeps Status A even if their typical lots no longer meet the new limit, until their next annual review after enforcement begins. This protects brands who have invested in achieving Status A under the previous standard from being penalized by the standard's tightening before they have had time to respond.

Status B and Status C SKUs re-evaluate against the new limit at their next CAPA cycle. Status B and C cannot preferentially evaluate against the old standard during the transition window, they get the more stringent of (old standard, new standard), to prevent gaming of the transition window. This prevents a brand that is already in a remediation status from "freezing" their improvement trajectory at the old standard while the rest of the program tightens.

Status D and Status E products are out regardless.

12.6 Forward Forecast

At every annual review, the Program Operator publishes a forward forecast:

> "Current 80th percentile of trigger pool: X ppb. Current limit: Y ppb. If current improvement trends continue, the projected next-cycle limit is Z ppb, with expected announcement in N months and enforcement Q months after."

This converts the ratchet from a surprise into a planning parameter that brands can model years in advance. A brand considering certification can review the forward forecast for the relevant category before signing the license agreement and understand the next three-to-five-year obligation profile. A brand operating under the current standard can budget reformulation against the projected next-cycle limit.

The forward forecast is published in the annual Program Operator report and posted to the heavymetaltested.com Program Registry under document control. Forecasts are *projections*, not commitments, actual trigger firing depends on whether the two-condition test in §12.3 is satisfied at the time of the relevant annual review. Brands and regulators can rely on the forecast as the program's best current estimate of trajectory; the trigger conditions remain the operative test.

Forecast accountability

Forecasts are projections, not commitments. The Program Operator publishes an annual retrospective comparison of prior forecasts to actual ratchet outcomes in the transparency report. Material divergence between a forecast and its corresponding outcome, operationally defined as a forecasted limit that differs from the eventual actual limit by more than 25%, triggers a methodology review of the projection method.

This retrospective accountability ensures that the forecast is not a marketing artifact but a calibrated planning tool with documented predictive accuracy over time. Brands relying on forecasts for reformulation planning should treat them as the Program's best current estimate, not as guaranteed future limits, and can verify the Program Operator's track record in the published retrospectives.

12.7 Exit-and-Re-Enter Prohibition

A Licensee that voluntarily withdraws from the Program and subsequently re-enrolls within 24 months is evaluated against the standard in effect at the time of re-entry, not the standard in effect at the time of original exit. Grandfathering protections under §12.5 apply only to continuous certification.

This rule prevents the exit-and-re-enter gaming pathway, in which a brand could withdraw before a ratchet event to escape transition oblig-

ations and then re-enter once the new standard's transition period has elapsed. A Licensee that withdraws for documented commercial reasons (product discontinuation, line restructuring, change of control) and re-enrolls after 24 months is treated as a new applicant per standard program-entry procedures.

12.8 Why this rule survives each Governing Principle

The Standards Ratchet as specified above satisfies all five HMTc Governing Principles simultaneously:

- **Principle 1 (drive contamination down through consumer demand):** The ratchet moves whenever the market actually moves, up to 20% per cycle. The market cannot stall the ratchet by collectively holding still.
- **Principle 2 (safeguard and protect consumers):** Limits never loosen. Tightening is data-driven and cumulative.
- **Principle 3 (safeguard and protect brands who enter the program):** Capped step size, 12–18 month transition, grandfather period during transition, published forward forecast at every cycle. Brands know the direction, the cadence, and the maximum step size in advance.
- **Principle 4 (legal defensibility):** The rule is data-driven, two-cycle-confirmed, and transparently published with specific thresholds. Adversarial readers cannot attack it as arbitrary because every clause has a specified rationale.
- **Principle 5 (scale to ubiquity):** Predictable cadence brands can plan around. Brands considering certification can model the next 3–5 years of obligation before they sign.

A plaintiff's expert attacking the ratchet must attack one of the specific clauses, the 80th-percentile threshold, the breadth condition, the step cap, the transition period, the grandfather clause, and must explain why an alternative specification would deliver the same five-principle satisfaction. The Program-Level Adversarial Read in Part 5.2 addresses the categories of attack that the program anticipates.

Cross-references

- Governing Principles (the source authority): Part 5.1 or `docs/governance/hmtc-governing-principles.md`.
- Annual data inputs to the trigger: Part 2.7 (structured data requirement).
- Status framework that grandfathering protects: Part 2.3.
- Confidential Remediation Track (independent of ratchet for non-certified SKUs): Part 2.4.
- Methodology Reference (how the initial limits were set, which the ratchet evolves): Part 5.1.

2.11 Regulatory Escalation

Narrative

HMTc certification is not a substitute for regulatory compliance. The Program's action levels are intentionally set below applicable regulatory limits so that a brand passing HMTc certification is comfortably inside the regulatory envelope. But when a tested lot reveals a regulatory limit violation, distinct from an HMTc action level exceedance, the Program's response is not graduated remediation. It is hard-stop escalation.

This chapter specifies the regulatory-escalation rule and its consequences. The rule is short because the principle is simple: regulatory compliance takes absolute precedence over Program-internal status determinations.

Operational specification

If any tested lot reveals a result that violates an applicable regulatory limit (as distinguished from an HMTc action level) in any jurisdiction where the product is sold, the following actions are triggered **regardless of the product's HMTc certification status:**

- Immediate notification of the Licensee.
- Mandatory product hold (if not already shipped) or market correction (if distributed).
- CAPA initiation within 2 business days.
- The Program Operator will not shield any Licensee from regulatory obligations and will cooperate with regulatory investigations as required by law.

This escalation path is a hard-stop condition. It applies irrespective of whether the result would otherwise qualify for transitional certification under HMTc. The Program's action levels are set below regulatory limits

precisely to prevent this scenario, but when it occurs, **regulatory compliance takes absolute precedence.**

Why this rule is structural rather than discretionary

A regulator reading this chapter should understand that HMTc is not a shadow regulatory regime. Brands that violate FDA action levels[2], EU maximum levels under Regulation 2023/915[19] for toxic elements in fish and shellfish, or any other operative regulatory threshold cannot use HMTc enrollment to defer the regulator's response. The Program Operator's enforcement-cooperation obligation is contractually binding under the license agreement.

A brand legal team should understand the corollary: the Program is designed so that brands that pass HMTc certification are confidently inside the regulatory envelope. A brand that triggers a §2.11 escalation has not just exceeded an HMTc limit; they have produced a lot that, on the merits, was already a regulatory violation regardless of HMTc status. The escalation rule does not create a new violation; it acknowledges an existing one and ensures that Program-internal processes do not slow the brand's regulatory response.

The §2.11 rule is the operational expression of Governing Principle 4 (legal defensibility). A program whose internal status determinations conflicted with regulatory action would fail the principle on its first regulatory inquiry.

Cross-references

- Action levels not safety thresholds (the foundational framing): Part 2.1.
- Data governance (regulatory-cooperation obligations): Part 2.7.
- License agreement summary (regulatory-cooperation contractual language): Part 5.6.
- The Regulatory Trigger Level (RTL) definition: Part 5.3 Glossary.

2.12 Appeals, Reinstatement, and Document Control

Narrative

Every certification program requires three procedural mechanisms that operate independently of the operational rules: a way for a Licensee to challenge a status determination they believe is incorrect; a defined path back into the Program for a Licensee whose certification has been suspended or revoked; and a document-control regime that ensures the rules governing a decision are unambiguous at the time the decision is made.

This chapter specifies all three. The mechanisms are procedural rather than substantive, they govern how the Program responds to disagreement, recovery, and document evolution, not what the underlying standards or determinations are. Together, they ensure that brands are treated fairly within the Program's framework and that the rules in force at the time of any decision are knowable, traceable, and applied consistently.

14.1 Appeals Process, Two-Tier Structure

The appeals process is structured in two tiers to address the institutional conflict between the Program Operator's certification revenue and its role as initial appeal-reviewer. Tier 1 is internal to the Program Operator; Tier 2 is independent of the Program Operator. The two-tier structure is the program's structural answer to the conflict-of-interest concern that an internal-only appeals process operating within the Program Operator's authority cannot fully mitigate the institutional conflict between the Operator's certification revenue and the Operator's role as reviewer.

Tier 1: Initial appeal

A Licensee may appeal any certification status decision by submitting a written appeal to the Program Operator within 10 business days of notification. The appeal must include:

- The technical or factual basis for disagreement; and
- Supporting evidence (laboratory data, chain-of-custody records, methodology documentation, or analogous materials).

The Program Operator will review the appeal within 15 business days and may convene the Technical Committee for methodological or interpretive questions. Operational appeals (testing logistics, sampling disputes, laboratory selection) are reviewed by the Program Operator directly.

During Tier 1 review, the originally assigned status remains in effect unless the Program Operator grants temporary relief. Temporary relief is exceptional and is granted only when the originally assigned status would impose disproportionate harm relative to the strength of the appeal, for example, when chain-of-custody documentation suggests sample mishandling and the decision would otherwise impose Status C consequences pending resolution.

The standard for granting an appeal is whether the originally assigned status reflects the correct application of this Policy and the Lot Testing Schedule to the actual facts of the case.

Tier 2: Independent Appeals Committee

If the Licensee disagrees with the Tier 1 outcome, the Licensee may escalate to the independent Appeals Committee within 10 business days of the Tier 1 decision. The Appeals Committee is structurally independent of the Program Operator:

- **Composition.** Three external reviewers, none of whom are employed by or under contract to The Paleo Foundation: one toxicologist or food-safety scientist with demonstrated independence; one food-industry quality professional with no financial relationship to the Licensee under review or to the Program Operator; one legal advisor with regulatory-affairs background. Members are appointed for two-year terms with staggered rotation to preserve institutional memory while preventing entrenchment.

- **Caseload.** No member may sit on more than three appeals per calendar year. Members recuse from any appeal involving a Licensee with whom they have a current or past commercial relationship.
- **Authority.** The Appeals Committee reviews the Tier 1 decision on the merits and may affirm, reverse, or remand to the Program Operator for further investigation. **Reversals are binding on the Program Operator.** The Committee may also recommend methodology revisions where the appeal surfaces a generalizable issue.
- **Timeline.** The Committee renders its decision within 30 business days of escalation, with one 15-day extension permitted for cases requiring additional evidence or expert consultation.
- **Transparency.** The Program Operator publishes anonymized Appeals Committee outcomes in the annual transparency report, including the number of appeals heard, the disposition (affirm/reverse/remand), and the categories of issue raised. Individual brand identities and SKU-specific details are not disclosed.
- **Funding.** The Appeals Committee operates on a per-appeal fee structure paid by the Program Operator and a per-Licensee contribution from the licensing fee schedule. This funding arrangement ensures that the Committee is not financially dependent on case outcomes and that no individual appeal carries financial pressure for the reviewers.

Tier 2 review is binding. A Licensee that prevails at Tier 2 has the Program Operator's status determination reversed, with retroactive effect to the original determination date.

14.2 Reinstatement After Suspension or Revocation

Reinstatement requires:

- Completion of CAPA with verified effectiveness (per Part 2.8);
- A defined number of consecutive compliant lots (set case-by-case by the Program Operator); and
- Restored data integrity (per Part 2.7).

The Program Operator determines specific reinstatement requirements on a case-by-case basis. The specific number of compliant lots, the depth of CAPA review, the extent of independent verification, and the timeline are calibrated to the severity and nature of the original nonconformance.

Revocation (Status E) requires **full re-application to the Program** at the Program Operator's discretion. A brand whose license was revoked for data integrity violations or willful misconduct faces the highest reinstatement bar; the Program Operator's discretion to refuse re-application is broad. This is consistent with Part 2.3′s framing: Status E is the program's mechanism for terminating relationships that cannot be repaired through standard remediation.

14.3 Document Precedence

In the event of any discrepancy between documents within the HMTc Program (including between this Program Manual, the Governance Policy, and the Lot Testing Schedule), the Program Operator will issue a formal Technical Clarification.

Until such clarification is issued, the **more conservative interpretation governs.** "More conservative" means the interpretation that results in stricter limits, tighter testing requirements, or more rigorous CAPA obligations. This default ensures that ambiguity in the document set does not advantage a Licensee at the consumer's expense.

The Program Operator maintains the operative version of each controlled document under document control with version numbers, effective dates, and change logs. The version in effect at the time of a certification decision governs that decision. Subsequent revisions do not retroactively alter prior decisions.

All revisions are managed under document control procedures with transition periods for new requirements. Material changes, those that would alter eligibility, testing frequency, status criteria, or CAPA obligations, receive minimum transition periods consistent with the Standards Ratchet's 12-to-18-month rule in Part 2.10.

14.4 Conflict of Interest

Individuals involved in certification determinations shall be free from conflicts of interest with the Licensee under review. The Program Operator maintains impartiality and objectivity throughout the certification process. Appeal and special committee members shall be independent from the day-to-day business decisions of the affected Licensee.

The conflict-of-interest standard applies at the individual level (no individual reviewer with a financial interest in or personal relationship to the Licensee may participate in their certification or appeal proceedings) and at the institutional level (the Paleo Foundation discloses any institutional relationships that could create the appearance of conflict). The Standards Architect's dual role as Paleo Foundation CEO is disclosed publicly in the Colophon (Part 5.7) and addressed in the Program-Level Adversarial Read (Part 5.2) on the merits.

Closing of Part 2

Part 2 has now specified the HMTc Program in its operational form. A reader who has worked through chapters 2.0 through 2.12 can:

- Trace a SKU's lifecycle from inquiry to ratchet review via the program-flow diagram (2.0).
- Understand the program's foundational principles and international scope (2.1).
- Locate any analyte in the Tier 1/Tier 2 classification and understand its enforcement consequence (2.2).
- Read the five-status framework with its eligibility criteria, mark-usage permissions, testing frequencies, CAPA requirements, and escalation paths (2.3).
- Understand the Confidential Remediation Track and its legal-defensibility role (2.4).
- Apply mark-usage rules and anti-arbitrage provisions (2.5).
- Operate the two-sided measurement-uncertainty decision rule (2.6).
- Comply with structured-data and data-integrity requirements (2.7).
- Initiate and close CAPAs according to status-specific timelines (2.8).
- Receive and respond to supplier-intelligence alerts (2.9).
- Plan against the Standards Ratchet trigger conditions, step cap, transition rules, and forward forecast (2.10).
- Handle regulatory escalation as a hard-stop condition (2.11).
- Appeal status determinations, recover from suspension, and operate under document control (2.12).

Part 2.5, which follows, demonstrates how brands and QA teams use heavymetalindex.com, the public corpus introduced in Part 1.5, as an active operational tool. Part 3 publishes the per-product standards.

Part 2.5, Using heavymetalindex.com

Part 1.5 introduced heavymetalindex.com as the public curated corpus of the heavy-metals-in-food literature that HMTc's standards rest on. That chapter established the architectural separation between HMI and the certification program and explained why the separation produces durable epistemic authority. This chapter is the operational complement: it demonstrates how brand QA teams, brand legal teams, and certification auditors **use** the corpus day-to-day in the course of operating under HMTc.

The chapter has four sections. The first three describe how the corpus serves brand operations today; the fourth describes a roadmap capability, the brand-facing percentile-placement dashboard, that is on the program's near-term build path and that brands considering certification should expect to come online during their first license cycle.

This chapter exists for a strategic reason. The certification mark is the program's visible signal; the corpus is the program's value engine. A reader who finishes Part 2.5 should understand that HMTc is not just a certification mark backed by a methodology, but a certification mark backed by a continuously-maintained operational tool that brand QA teams lean on. That distinction is what separates HMTc from any single-point certification.

2.5.1 The diagnostic tool

When a brand's lot tests come back with one or more analytes elevated above their typical range, even when still within the HMTc limit, the brand's first operational question is: *why*? Random analytical noise, a seasonal supplier shift, a new ingredient lot, a process change that wasn't

documented as material, an upstream supply-chain event the brand has not yet been alerted to? The diagnostic tool is the relevant ingredient pages on heavymetalindex.com.

Each ingredient page on HMI carries a structured `contamination_profile` block summarizing what the literature says about typical and 95th-percentile concentrations for each priority metal in that commodity, with three additional structured layers:

- **Geographic variance**, typical concentration ranges by growing region, with sample sizes from contributing sources. A brand sourcing cocoa from Ecuador can see what the published literature reports for Cd in Ecuadorian cocoa specifically, distinguished from Cd in cocoa from Côte d'Ivoire, Ghana, or Indonesia.
- **Seasonal variance**, concentration ranges by harvest period and climate condition, where the literature reports them. The El Niño years 2015–2016 and 2023 are documented as drought-stress periods that elevate Cd uptake in Ecuadorian cocoa by roughly 40% above non-El-Niño baselines.
- **Processing effects**, how polishing, rinsing, parboiling, fermentation, drying, milling, and refining shift concentrations of which metals. Each processing-effect claim is anchored to a cited source.

Worked example

A brand producing canned albacore tuna sees their July 2026 lot test return MeHg meaningfully above the brand's typical baseline, though still within the HMTc methylmercury limit of 1000 ppb (as-sold). The result is within limit, but the jump prompts the question of why. Brand QA opens `heavymetalindex.com/ingredients/tuna`, navigates to the MeHg contamination_profile block, and finds:

- A typical and 95th-percentile methylmercury range across all reported tuna, with sample sizes from the contributing sources.
- A species and fishing-ground breakdown distinguishing higher-trophic predatory species and older, larger fish from younger, smaller fish and lower-trophic species, since methylmercury bioaccumulates with size and trophic level.
- Processing and product-form effects, where the literature reports them, distinguishing loin, drained canned, and other forms.

The brand cross-references their supplier records: their June 2026 tuna lot was sourced from a new supplier landing larger, older fish from a different fishing ground (replacing their typical supplier after a price shift). The diagnostic answer is clear, traceable to the literature, and actionable: source verification on the new supplier, possibly a return to the prior supplier or a shift toward smaller fish and lower-trophic species to lower the bioaccumulated methylmercury load.

This diagnostic conversation happened in twenty minutes using the corpus. It would have taken brand QA days of supplier outreach and ingredient testing to reach the same conclusion working from scratch.

2.5.2 The remediation playbook

Each ingredient page on HMI carries a Mitigation Options section ranked by impact magnitude. The categories are: sourcing levers (origin region, supplier specification, soil-Cd or paddy-iAs pre-screening); agronomic levers (soil amendments, water management, cultivar selection); processing levers (rinsing, polishing, fermentation, drying); formulation levers (substitute or dilute); testing and QC levers (lot-level vs. spot-check, internal vs. third-party); and packaging and storage levers where relevant.

Each lever entry names the magnitude of the effect on the metal in question and cites the source page that establishes the effect. A brand reading the tuna ingredient page's Mitigation Options finds, for methylmercury:

- **Sourcing level (highest impact):** shifting toward smaller, younger fish and lower-trophic species, and away from fishing grounds and species with higher bioaccumulated methylmercury, can substantially reduce typical MeHg (anchored to comparative studies cited on the page).
- **Species and size level:** specifying maximum fish size or age at procurement reduces supplier-batch variance, since methylmercury accumulates with size and trophic level.
- **Processing and product-form level:** product form and trimming can shift the measured methylmercury of the finished product, where the cited literature reports such effects.

- **Formulation level:** blending lower-MeHg fish with higher-MeHg fish reduces final-product MeHg proportionally (a 50/50 blend of high-MeHg and low-MeHg fish typically produces 50% of the high-MeHg value, allowing for the interactions documented in the cited sources).
- **Testing and QC level:** lot-by-lot incoming-fish testing with rejection thresholds at the 80% PoL internal warning level can prevent elevated lots from entering production.

Brand QA can read the cited sources for each lever, evaluate which fit their operational reality (sourcing changes have long lead times; processing changes have capital implications; formulation changes have R&D and labeling implications), and assemble a defensible reformulation plan. The remediation pathway becomes a reading exercise, not a research project.

For brands enrolled in the Confidential Remediation Track, this is the daily operational tool the Track's technical guidance points them to. The Track's value is not opacity but accelerated access to the same evidence base that successful Status A brands have already used to reduce their contamination profiles.

2.5.3 Percentile placement (roadmap)

A future capability on the HMTc roadmap is the brand-facing percentile-placement dashboard. The dashboard will allow a brand with a license to log into the heavymetaltested.com platform and see, in real time:

- Where their lot results sit relative to the pooled distribution of all certified Status A SKUs for the same product category and metal. A brand's lot at 12 ppb Pb in a category with a 30 ppb HMTc limit and a Status A median of 8 ppb sees themselves at "P67 of certified Status A pool", better than two-thirds of certified competitors, worse than one-third.
- Their trend over time, with the pool's distribution overlay. Is the brand improving faster than the certified market, slower, or at the same pace? Is the brand's percentile placement getting better or worse as the certified pool tightens around them?
- The projected next-cycle limit (per Part 2.10′s forward forecast) overlaid on their current lot trajectory. Will the brand's current trajectory

clear the projected next-cycle limit before enforcement, or will the brand need to accelerate reduction?

This is the operational expression of the Standards Ratchet's planning-tool framing. A brand that knows where they sit and where they are heading can budget reformulation against actual data rather than abstract limits. A brand that is at the cleaner end of the certified pool sees their leadership reflected in real time; a brand at the dirtier end of the pool sees the urgency of their reformulation runway.

The dashboard is a roadmap capability rather than a current operational feature. Brands considering certification should expect the dashboard to come online during their first license cycle; brands currently certified will receive access as the capability ships. The roadmap is named here so that brands evaluating HMTc understand the operational trajectory they are signing up for, not just the standards they are signing up against.

2.5.4 The pre-certification baseline

Before a brand signs an HMTc license agreement, the brand can use heavymetalindex.com to model their pre-certification baseline. The exercise is straightforward and takes one to two hours per product category.

The brand identifies the product-category page on HMI that matches their SKU (one of the five Category 6 seafood pages, or another category's page when those come online). The product-category page carries a Master Limit Table previewing what the HMTc limit would be for each priority analyte, the applicable regulatory cap, and the basis note. The brand cross-references this against their internal testing data, if any, or against the relevant ingredient pages' contamination_profile blocks if no internal data exists.

This pre-certification exercise tells the brand:

- Whether their current production is likely to qualify for Status A immediately (most metals at or below the HMTc limit), Status B (one or more Tier 2 metals between 100% and 150% PoL), or only the Confidential Remediation Track.
- Which metals are closest to the limit, the analytes most likely to drive a status determination.

- Which ingredients in their formulation are the likely contributors, sourcing lever candidates.
- What the rough magnitude of reformulation effort would be, small adjustments versus full sourcing-chain restructuring.

A brand whose pre-certification baseline suggests Status A or B can sign the license with reasonable confidence about their initial status. A brand whose pre-certification baseline suggests Status C-or-worse can sign the license with the understanding that the Confidential Remediation Track is their entry path, and can begin the technical-guidance conversations with the Program Operator before formal enrollment. Either way, the brand enters the program with their eyes open and their reformulation strategy informed by the same evidence base the Program Operator is using to set their limits.

This is the closing argument for the chapter and for the corpus's role in the Program. The HMTc certification mark is the visible signal. The HMI corpus is the operational tool that brand QA, brand legal, and brand management use to plan, diagnose, remediate, and verify the certification effort. The two work together; neither alone would be sufficient. Part 3 publishes the standards that the corpus supports; Part 4 specifies the testing schedule that operationalizes the surveillance; Part 5 documents the methodology that the corpus feeds. The bound volume in front of the reader is the program's specification; heavymetalindex.com is where the program lives.

Cross-references

- The corpus's structure and the wiki/HMTc firewall: Part 1.5.
- The product-category pages corresponding to Part 3′s per-product chapters: `wiki/products/*` on heavymetalindex.com.
- The ingredient pages with contamination profiles, geographic variance, seasonal variance, and Mitigation Options: `wiki/ingredients/*` on heavymetalindex.com.
- The metal pages with toxicology, exposure routes, and regulatory limits: `wiki/metals/*` on heavymetalindex.com (the toxicology callouts in Part 2.2 reference these directly).
- The Standards Ratchet forward forecast: Part 2.10.
- The Confidential Remediation Track: Part 2.4.

Part 3 — The Standards

The HMTc Seafood standards, one chapter per product row. Each Master Limit Table is the set of published thresholds for that row, in its native basis. Canned seafood is not a standalone row: per the canned/tin rule, canned forms route to their matrix-native fish or shellfish chapter carrying a tin (Sn) flag handled by the food-contact cans platform.

3.1 Marine Fish, Predatory (tuna, swordfish, shark, marlin)

One of 5 per-product chapters in Part 3, self-contained as a licensing exhibit. The methodology is in Part 5.1; the certification status framework in Part 2.3; the Standards Ratchet in Part 2.10. Held/withdrawn analytes (e.g. total arsenic in fish and seaweed, which is arsenosugar/arsenobetaine-dominated — inorganic arsenic is the toxic and regulated analyte) are shown as "adjudication pending" and carry no published threshold; the published values are validated by the Publication QC Gate (no value exceeds a regulatory ceiling).

Product class: Fish — marine, predatory (tuna, swordfish, shark, king mackerel) **Subcategory role:** whole_row_platform **Native basis for published thresholds:** as_sold **Methodology:** HMTc-10 analyte panel; per-analyte standards derived from pooled occurrence distribution at P97 (clean / independent rows) or P45 (dirty rows), capped at the lowest applicable regulatory ceiling. See "Methodology in one page" below. **Date generated:** 2026-06-17 **Author:** Karen Pendergrass, Standards Architect, Paleo Foundation **Status:** Draft for stakeholder review. Not yet through the 9 pre-publication gates.

Who this briefing is for

The HMTc certification mark serves multiple audiences with different concerns. This briefing is written so each audience finds what they need in one document. Audience-tagged "read" callouts appear inline next to each per-analyte section.

Brand legal teams — Read the per-analyte Defensibility narrative and the Adversarial read section. The methodology section is your Daubert-resistance reference. Every numeric claim traces to a source page; the contribution logs document admission criteria.

Retailer quality and compliance — Read the Master Limit Table and the per-analyte Regulator read callouts. The certification mark on a product means the brand has tested at or below every value in the Master Limit Table, in the product's native basis. The regulatory comparison rows show which government caps apply and how HMTc thresholds compare.

Regulators (FDA, EFSA, state AGs) — Read the Methodology section, the Sources Cited table, and the Anti-Circumvention argument. The methodology is open, reproducible, and follows the WHO/JECFA, EFSA, ATSDR, and Codex precedents cited. Numeric thresholds are derived from a pooled occurrence distribution of peer-reviewed and agency data, weighted by source N with a 50% relative-share cap.

Adversarial readers (plaintiff's experts, competing standards bodies) — Read the per-analyte Defensibility narrative, the Adversarial read section, the contribution logs, and the sensitivity analyses. Every value carries: the pool composition, the basis-conversion math, the regulatory cap comparison, the parametric sensitivity, and the rationale for any gap between literature and threshold.

Master Limit Table – Executive Summary

All ten HMTc analytes for this product class. Each row shows the published HMTc threshold (rounded up to the primary increment ladder), the precise pooled-engine value before rounding, the rationale tag, the contributing source count, the parametric-sensitivity band, and the lowest applicable regulatory ceiling. Brands meeting every value in this table on a like-for-like basis qualify for the HMTc certification mark for fish marine predatory.

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
Pb	**300 ppb** [as-sold]≈ 50 ppb [prepared]	3248.446 ppb (capped to 300 by reg cap)	regulatory-alignment	8	provisional	300 ppb (eu2023_pb_fish_muscle_300)
iAs	**200 ppb** [as-sold]≈ 20 ppb [prepared]	107.089 ppb	literature-baseline	2	stable	—
tAs	*adjudication pending* ≈ 10000 ppb [prepared]	41247.268 ppb	literature-baseline	3	provisional	—
Ni	**200 ppb** [as-sold]≈ 25 ppb [prepared]	146.675 ppb	literature-baseline	1	stable	—
Cd	**50 ppb** [as-sold]≈ 10 ppb [prepared]	202.015 ppb (capped to 50 by reg cap)	regulatory-alignment	6	stable	50 ppb (eu2023_cd_fish_muscle_50)
Al	**10000 ppb** [as-sold]≈ 2000 ppb [prepared]	11961.176 ppb	literature-baseline	1	stable	—
tHg	**1000 ppb** [as-sold]≈ 200 ppb	1920.921 ppb (capped to 1000	regulatory-alignment	16	stable	1000 ppb (eu2023_hg_predatory_

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
	[prepared]	by reg cap)				fish_ 1000)
MeHg	**1000 ppb** [as-sold]≈ 200 ppb [prepared]	829.001 ppb	literature-baseline	12	stable	1000 ppb (eu2023_ hg_predatory_ fish_ 1000)

How to read this table. "Published" values are rounded up to the HMTc primary increment ladder (1, 2, 3, 5, 10, 15, 20, 25, 50, 100, 200, 500, 1000, 1300, 1500, 2000, 5000, 10000 ppb) per the program's publication conventions; this gives brands an unambiguous bin to test against. The "precise" column shows the pooled-engine output before rounding so the methodology is transparent and re-derivable. Rationale tags map to CLAUDE.md Part 19′s four allowed tags: `literature-baseline` (pooled value binds), `regulatory-alignment` (regulatory ceiling binds), `feasibility-driven` (analytical method binds), and `precautionary` (manual override below pooled and regulatory).

Methodology – one page

Per-analyte standards. For each (product row, analyte) cell, every source loaded for that cell contributes to a pooled distribution via moment-matching pseudo-sample expansion. Sample-level data uses raw values; summary statistics (mean, median, max, lb/ub, reported percentiles) reconstruct under log-normal or Tobit assumptions documented per source. Pseudo-samples are weighted by source N and capped at 50% relative pool share to prevent any one survey from dominating. The pooled distribution is sorted and queried at the per-analyte target percentile:

- **Clean subcategory:** standard = pooled P97

- **Dirty subcategory:** standard = pooled P45
- **Independent (no within-pair partner):** standard = pooled P97 by default

Why P97 / P45, not P90 / P30 (the row-pass calibration). A brand certifies a product (one row, ten analytes simultaneously) — not one analyte at a time. Setting each per-analyte limit at the raw P90 of the clean distribution would deliver a joint row-pass rate close to 0.9^10 = 35% under analyte independence. Under realistic analyte correlation (0.6, since supply-chain hygiene drives multiple metals together), the joint rate sits around 50-70%. To deliver the row-level outcomes the Governing Principles require — approximately 90% of clean-subcategory brands and approximately 30% of dirty-subcategory brands pass row certification — per-analyte limits are uplifted to P97 (clean) and P45 (dirty). This is the calibration that delivers the program's stated objective. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening; per-analyte differentiation (e.g., Pb tighter than Ni) is deferred until HMTc has 6-12 months of lot-testing data to measure actual joint pass rates.

Basis. Each product row's standard is published in the native basis of the product form: powder rows in powder-as-placed-on-market, ready-to-feed liquid rows in as-consumed, dry cereal rows in dry-as-sold, puree rows in as-consumed. Brands ship and test in their product's native form; the certification mark matches that form. Sources reporting in non-native bases are converted into the row's native basis during pooling via documented factors (the 1:7 reconstitution ratio for powder ↔ prepared; within-class identity for as_sold ↔ dry_as_sold). The cross-basis equivalent of every published value is shown in the Master Limit Table for ease of reference by audiences operating in different bases.

Regulatory cap. The final published HMTc value never exceeds the lowest applicable loaded regulatory ceiling, expressed in the product's native basis. When the pooled P97/P45 exceeds the regulatory cap, the cap binds and the rationale tag is `regulatory-alignment`. When the pooled value is below the cap, the rationale is `literature-baseline` and the cap is shown for context only.

Sensitivity. Each pooled value is recomputed under three parametric assumptions: log-normal (headline), gamma (alternative), and nonparametric reconstruction. The spread across the three is reported as the

sensitivity band: stable (≤25%), provisional (25–50%), or needs_review (>50%). Confidence intervals are computed via meta-analytic bootstrap that resamples whole sources (not individual pseudo-samples) to capture between-source uncertainty.

Methodology authorities. The pooling approach follows WHO/JECFA pooled total-diet study methodology, the EFSA Scientific Opinions on lead (2010, 2024) and cadmium (2009, 2024) for log-normal contaminant occurrence reconstruction, ATSDR moment-matching guidance for summary-statistic-only sources, and Codex CAC/GL 50-2004 weighted pooling. Per-source contribution logs document every admission decision and every basis conversion.

Tooling and reproducibility. The pooling engine is `tools/evidence/build-hmtc-aggregate-distributions.mjs` (1170 lines of deterministic JavaScript, 37 unit tests). The briefing generator is `tools/standards/build-standards-briefing.mjs` (this file). Both are versioned in git and re-runnable. Every numeric claim in this briefing can be regenerated by checking out the commit named in the Provenance Footer and running `npm run evidence:pooling && npm run briefing -- --product=fish-marine-predatory`.

Per-Analyte Sections

Pb

Published HMTc threshold: 300 ppb [as-sold] · 300 ppb (pre-rounding) **Cross-basis: ≈ 50 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `regulatory-alignment` **Regulatory ceiling reference:** 300 ppb (eu2023_pb_fish_muscle_300) **Pool composition:** 8 sources contributing · 162 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [365.07, 4814.85] ppb Sensitivity: log-normal 2004.76 · gamma 3133.21 · nonparametric 2526.58 ppb · spread 36% (provisional)**

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_pb_fish_muscle_300). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the under-

lying pool is moot — the cap binds. The pooled P97 of 3248.45 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_pb_fish_muscle_300). A Daubert challenge on the value reduces to "is eu2023_pb_fish_muscle_300 a legitimate regulatory anchor?" — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 300 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_pb_fish_muscle_300 cap. The pooled occurrence distribution from 8 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Reconstruction	Pool share
bhoyroo2015-scombrids-heavy-metals-mauritius (2015) · MU	Detection of heavy metals bioaccumulation in scombrids for …	marine predatory fish muscle, wet weight	5	uniform_max_only	3.1%
depaiva2017-canned-tuna-metals-sao-paulo (2017) · BR	Cadmium, lead, tin, total mercury, and methylmercury in cann…	canned tuna drained product, wet weight/as sold	30	uniform_max_only	6.2%
lehel2023-yellowfin-tuna-indian-ocean (2023) · HU, LK, EU	Metal Load of Potentially Toxic Elements in Tuna (Thunnus al…	yellowfin tuna flesh, wet weight	40	lognormal_mean_max	24.7%

Source	Title	Basis	N used	Recon-struction	Pool share
lozano-bilbao2023-canary-tuna-metals (2023) · ES, EU	Risk Assessment and Characterization in Tuna Species of the …	tuna muscle, wet weight	15	lognormal_mean_max	9.3%
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	marine predatory fish whole weight/wet weight as reported	5	uniform_max_only	3.1%
ormaza-gonzalez2020-tuna-canned-eastern-pacific (2020) · EC, EU	Concentration of mercury, cadmium and lead concentrations in tuna product…	canned tuna and pre-cooked tuna loins, wet/drained weight	2572	lognormal_mean_max	50.0%
smolikova2016-heavy-metals-fish-czech (2016) · CZ, EU	Determination of heavy metals in fish products	fish muscle, wet/as reported	1	single_point	0.6%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	marine predatory fish muscle, as reported	5	uniform_max_only	3.1%

iAs

Published HMTc threshold: 200 ppb [as-sold] · 107.089 ppb (pre-rounding) **Cross-basis: ≈ 20 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling**

reference: no applicable cap loaded **Pool composition:** 2 sources contributing · 10 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [73.35, 145.72] ppb Sensitivity: log-normal 81.26 · gamma 103.97 · nonparametric 108.20 ppb · spread 25% (stable)**

Defensibility. Pooled P97 from 2 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 2 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 200 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 2 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
lehel2023-yellowfin-tuna-indian-ocean (2023) · HU, LK, EU	Metal Load of Potentially Toxic Elements in Tuna (Thunnus al…	yellowfin tuna flesh, wet weight	40	lognormal_mean_max	50.0%
macdonald2025-fss-uk-fishery-contaminants-review	Occurrence of chemical contaminants in wild-	marine predatory fish whole weight/wet	5	uniform_max_only	50.0%

Source	Title	Basis	N used	Recon-struction	Pool share
(2025) · GB, EU	caught fishery p…	weight as reported			

tAs

Published HMTc threshold: adjudication pending — held provisional (wrong-analyte-not-material) · 41247.268 ppb (pre-rounding) **Cross-basis: ≈ 10000 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 3 sources contributing · 20 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [1540.64, 77290.05] ppb Sensitivity: log-normal 69595.07 · gamma 44089.01 · nonparametric 69947.57 ppb · spread 37% (provisional)**

Defensibility. Pooled P97 from 3 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is provisional — log-normal, gamma, and nonparametric variants disagree by more than 25%, signaling heterogeneity that should be resolved before final publication. Likely sources: basis mixing, regional outliers, or sparse pseudo-sample contributions.

Brand legal read. The value is pooled P97 from 3 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 10000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 3 contributing sources across multiple jurisdictions. Sensitivity band provisional.

Source	Title	Basis	N used	Recon-struction	Pool share
lehel2023-yellowfin-tuna-in-dian-ocean (2023) · HU, LK, EU	Metal Load of Poten-tially Toxic Elements in Tuna (Thunnus al…	yellowfin tuna flesh, wet weight	40	lognormal_ mean_max	50.0%
macdonald2025-fss-uk-fish-ery-cont-aminants-review (2025) · GB, EU	Occurrence of chem-ical cont-aminants in wild-caught fishery p…	marine predatory fish whole weight/wet weight as reported	5	uniform_ max_only	25.0%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	marine predatory fish mus-cle, as re-ported	5	uniform_ max_only	25.0%

Ni

Published HMTc threshold: 200 ppb [as-sold] · 146.675 ppb (pre-rounding) **Cross-basis: ≈ 25 ppb [prepared] (÷7 (1:7 recon-stitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 1 sources contributing · 1 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [50.30, 867.35] ppb Sensitivity: log-normal 148.85 · gamma 148.70 · nonparametric 149.18 ppb · spread 0% (stable)**

Defensibility. Pooled P97 from 1 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 1 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 200 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 1 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
bhoyroo2015-scombrids-heavy-metals-mauritius (2015) · MU	Detection of heavy metals bio-accumulation in scombrids for ...	marine predatory fish muscle, wet weight	5	uniform_max_only	100.0%

Cd

Published HMTc threshold: 50 ppb [as-sold] · 50 ppb (pre-rounding) **Cross-basis: ≈ 10 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `regulatory-alignment` **Regulatory ceiling reference:** 50 ppb (eu2023_cd_fish_muscle_50) **Pool composition:** 6 sources contributing · 142 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [109.10, 649.12] ppb Sensitivity: log-normal 263.28 · gamma 198.77 · nonparametric 217.13 ppb · spread 25% (stable)**

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_cd_fish_muscle_50). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the underlying pool is moot — the cap binds. The pooled P97 of 202.01 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_cd_fish_muscle_50). A Daubert challenge on the value reduces to "is eu2023_cd_fish_muscle_50 a legitimate regulatory anchor?" — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 50 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_cd_fish_muscle_50 cap. The pooled occurrence distribution from 6 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Reconstruction	Pool share
depaiva2017-canned-tuna-metals-sao-paulo (2017) · BR	Cadmium, lead, tin, total mercury, and methylmercury in cann…	canned tuna drained product, wet weight/as sold	30	uniform_max_only	7.0%
lehel2023-yellowfin-tuna-indian-ocean (2023) · HU, LK, EU	Metal Load of Potentially Toxic Elements in Tuna (Thunnus al…	yellowfin tuna flesh, wet weight	40	lognormal_mean_max	28.2%
lozano-bilbao2023-canary-tuna-metals (2023) · ES, EU	Risk Assessment and Characterization in Tuna Species of the …	tuna muscle, wet weight	15	lognormal_mean_max	10.6%
ormaza-gonzalez2020-tuna-canned-eastern-	Lead, mercury, cadmium and lead con-	canned tuna and pre-cooked tuna loins,	2572	lognormal_mean_max	50.0%

Source	Title	Basis	N used	Recon-struction	Pool share
pacific (2020) · EC, EU	centrations in tuna product…	wet/ drained weight			
smolikova2016-heavy-metals-fish-czech (2016) · CZ, EU	Determination of heavy metals in fish products	fish muscle, wet/as reported	1	single_ point	0.7%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	marine predatory fish muscle, as reported	5	uniform_ max_only	3.5%

Al

Published HMTc threshold: 10000 ppb [as-sold] · 11961.176 ppb (pre-rounding) **Cross-basis: ≈ 2000 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 1 sources contributing · 1 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [9027.50, 42380.66] ppb Sensitivity: log-normal 15318.48 · gamma 12468.82 · nonparametric 16559.18 ppb · spread 25% (stable)**

Defensibility. Pooled P97 from 1 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 1 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international

precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 10000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 1 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
lozano-bilbao2023-canary-tuna-metals (2023) · ES, EU	Risk Assessment and Characterization in Tuna Species of the ...	tuna muscle, wet weight	15	lognormal_mean_max	100.0%

tHg

Published HMTc threshold: 1000 ppb [as-sold] · 1000 ppb (pre-rounding) **Cross-basis: ≈ 200 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `regulatory_alignment` **Regulatory ceiling reference:** 1000 ppb (eu2023_hg_predatory_fish_1000) **Pool composition:** 16 sources contributing · 1955 pseudo-samples in pool · 4 sample-level contributors **Bootstrap 95% CI: [1304.52, 2470.16] ppb Sensitivity: log-normal 1878.53 · gamma 1829.22 · nonparametric 1883.15 ppb · spread 3% (stable)**

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_hg_predatory_fish_1000). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the underlying pool is moot — the cap binds. The pooled P97 of 1920.92 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_hg_predatory_fish_1000). A Daubert challenge on the value reduces to "is eu2023_hg_predatory_fish_1000 a legitimate regulatory anchor?" — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 1000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_hg_predatory_fish_1000 cap. The pooled occurrence distribution from 16 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Reconstruction	Pool share
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web ...	unknown	11	lognormal_mean_median	0.6%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web ...	unknown	16	single_point	0.1%
brodziak-dopierala2023-mercury-fish-poland (2023) · PL, EU	Analysis of the Mercury Content in Fish for Human Consumptio...	tuna edible flesh, wet weight	5	uniform_max_only	0.3%
depaiva2017-canned-tuna-metals-sao-paulo (2017) · BR	Cadmium, lead, tin, total mercury, and methylmer-	canned tuna drained product, wet	30	lognormal_mean_max	1.5%

Source	Title	Basis	N used	Recon-struction	Pool share
	cury in cann…	weight/as sold			
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in Seafood P…	marine predatory fish, wet weight	522	percentile_interpola-tion	25.6%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in Seafood P…	marine predatory fish, wet weight	247	percentile_interpola-tion	12.6%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in Seafood P…	marine predatory fish, wet weight	450	percentile_interpola-tion	23.0%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in Seafood P…	marine predatory fish, wet weight	158	percentile_interpola-tion	8.1%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccu-mulation of methylmer-cury within the marine food web …	marine predatory fish mus-cle, wet weight	11	lognormal_mean_me-dian	0.6%
harding2018-bay-fundy-methylmercury	Bioaccu-mulation of methylmer-cury within	marine predatory fish mus-	5	uniform_max_only	0.3%

Source	Title	Basis	N used	Recon-struction	Pool share
(2018) · CA, US	the marine food web …	cle, wet weight			
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccu-mulation of methylmer-cury within the marine food web …	marine predatory fish mus-cle, wet weight	16	lognormal_mean_me-dian	0.8%
lehel2023-yellowfin-tuna-in-dian-ocean (2023) · HU, LK, EU	Metal Load of Poten-tially Toxic Elements in Tuna (Thunnus al…	yellowfin tuna flesh, wet weight	40	uniform_max_only	0.5%
macdonald2025-fss-uk-fish-ery-cont-aminants-review (2025) · GB, EU	Oc-currence of chem-ical cont-aminants in wild-caught fishery p…	marine predatory fish whole weight/wet weight as reported	5	uniform_max_only	0.3%
ormaza-gonzalez2020-tuna-canned-eastern-pacific (2020) · EC, EU	Lo-mercury, cad-mium and lead con-centrations in tuna product…	canned tuna and pre-cooked tuna loins, wet/drained weight	2572	lognormal_mean_max	25.6%
smolikova2016-heavy-metals-fish-czech (2016) · CZ, EU	De-termi-nation of heavy met-als in fish products	fish mus-cle, wet/as reported	1	single_point	0.1%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT,	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	marine predatory fish mus-cle, as re-ported	5	uniform_max_only	0.3%

Source	Title	Basis	N used	Recon-struction	Pool share
PK, PL, SA, SK, TR, VN					

MeHg

Published HMTc threshold: 1000 ppb [as-sold] · 829.001 ppb (pre-rounding) **Cross-basis: ≈ 200 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** 1000 ppb (eu2023_hg_predatory_fish_1000) **Pool composition:** 12 sources contributing · 84 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [510.35, 1212] ppb Sensitivity: log-normal 739.98 · gamma 909.76 · nonparametric 769.34 ppb · spread 19% (stable)**

Defensibility. Pooled P97 from 12 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination below the pooled value (cap = 1000 ppb). The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 12 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 1000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 12 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web ...	unknown	11	lognormal_mean_median	13.1%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web ...	unknown	16	single_point	1.2%
depaiva2017-canned-tuna-metals-sao-paulo (2017) · BR	Cadmium, lead, tin, total mercury, and methylmercury in cann...	canned tuna drained product, wet weight/as sold	30	lognormal_mean_max	35.7%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/seafood consumption compar...	fish meat/products, source MeHg occurrence basis	1	single_point	1.2%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/seafood consumption compar...	fish meat/products, source MeHg occurrence basis	1	single_point	1.2%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/seafood consumption compar...	fish meat/products, source MeHg occurrence basis	1	single_point	1.2%

Source	Title	Basis	N used	Recon-struction	Pool share
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	1.2%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	1.2%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web …	marine predatory fish muscle, wet weight	11	lognormal_ mean_median	13.1%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web …	marine predatory fish muscle, wet weight	5	uniform_ max_only	5.9%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web …	marine predatory fish muscle, wet weight	16	lognormal_ mean_median	19.1%
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	marine predatory fish whole weight/wet weight as reported	5	uniform_ max_only	5.9%

Adversarial Read – what a plaintiff's expert will attack, and the response

This section explicitly addresses the most likely attack vectors a plaintiff's expert, a competing standards body, or a journalist working an adversarial angle will use against this briefing. Each attack is named and answered.

Attack 1: "The pool is too small; the methodology cherry-picks." Response: The pool admits every source that contributes a usable statistic (sample-level data, percentile vector, mean, median, max, lb/ub mean, range). No source is excluded for being "summary-only." The contribution log per analyte lists every admission decision and every basis conversion. Per-source pool share is capped at 50% to prevent dominance by any single survey. The methodology is reproducible from the structured CSVs in `data/evidence/` and the engine in `tools/evidence/build-hmtc-aggregate-distributions.mjs`.

Attack 2: "Why P97 instead of P90? You're loosening the standard." Response: P97 per analyte is calibrated to deliver approximately 90% row-pass for the clean subcategory across the 10-analyte joint. Setting each analyte's limit at P90 of its own marginal would deliver a joint pass rate of 35% under independence and 50-70% under realistic correlation — well short of the Governing Principles' 90/30 row-pass target. P97 is the per-analyte percentile that delivers the row-level outcome the program promises. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening.

Attack 3: "Why is your dirty subcategory P45 instead of P10?" Response: P45 of the dirty subcategory's distribution, applied per-analyte, delivers approximately 30% joint row-pass for dirty-subcategory brands. This is the design intent stated in the Governing Principles v2.0: the dirty subcategory standard must give dirty-platform brands a realistic path to certification so the program drives reformulation rather than excluding the platform. A P10 dirty standard would deliver <1% joint row-pass and the program would have no commercial uptake on the dirty side.

Attack 4: "Your threshold matches the EU/FDA cap; HMTc adds no value over the regulator." Response: For Pb, Cd, tHg, the published threshold matches the binding regulatory cap because the pooled P97 sits above the cap. HMTc cannot defensibly publish a tighter

standard than the literature supports under its row-pass calibration. The value HMTc adds is row-level certification — a brand must meet every value in the Master Limit Table simultaneously on a like-for-like basis, with documented contribution logs and sensitivity analyses. Single-analyte regulatory caps do not deliver row-level certification.

Attack 5: "Your sensitivity analysis shows the methodology is unstable for some analytes." Response: The sensitivity bands are explicitly reported per analyte. 2 analytes (Pb, tAs) are flagged provisional — log-normal, gamma, and nonparametric variants disagree by more than 25%. This is transparent reporting, not a methodological flaw. Provisional values are published with the band stated and a commitment to resolve before final publication. The alternative — silently picking one parametric form — would be the actual methodological flaw.

Attack 7: "Where does basis conversion happen and what authority do you cite?" Response: The 1:7 powder ↔ prepared-for-feeding conversion is the conservative reconstitution ratio (1 g powder + 6 g water = 7 g prepared) standard across infant-formula regulatory frameworks (FDA Closer-to-Zero, EU 2023/915). Within-class conversions (as_sold ↔ dry_as_sold) pass through with factor 1. Every basis conversion is logged per source in the contribution log; unknown-conversion cases are flagged and not silently coerced.

Attack 8: "Show me one source you excluded." Response: No source contributing a usable statistic is excluded. Sources marked "no_usable_statistic" in the contribution log are LOD-censored entries with neither mean nor lb/ub recorded. Sources without a structured CSV row are documented in the routing audit and the gap report; those are extraction gaps, not exclusions, and they are tracked in Phase 1 backlog with target completion timelines.

Sources Cited

Every numeric claim in this briefing traces to one of the source pages below. Full source metadata lives at `wiki/sources/<source-id>.md` in the heavymetalindex.com repository.

Source ID	Title	Citation
`bhoyroo2015-scombrids-heavy-metals-mauritius` (B)	Detection of heavy metals bio-accumulation in scombrids for the determination of possible health hazard	Bhoyroo V; Soobratty N; Lalljee B · 2015 · African Journal of Food Science and Technology 6(4):098-107 · doi:10.14303/ajfst.2015.041
`brodziak-dopierala2023-mercury-fish-poland` (A)	Analysis of the Mercury Content in Fish for Human Consumption in Poland	Brodziak-Dopierala B; Fischer A · 2023 · Toxics · doi:10.3390/toxics11080717
`depaiva2017-canned-tuna-metals-sao-paulo` (A)	Cadmium, lead, tin, total mercury, and methylmercury in canned tuna commercialised in São Paulo, Brazil	Lima de Paiva E; Morgano MA; Milani RF · 2017 · Food Additives & Contaminants: Part B, Vol. 10, No. 3, pp 185–191·doi:10.1080/19393210.2017.1311379
`efsa-sc2015-fish-seafood-mehg-risk-benefit` (A)	Statement on the benefits of fish/seafood consumption compared to the risks of methylmercury in fish/seafood	EFSA Scientific Committee · 2015 · EFSA Journal 2015;13(1):3982, 36 pp. · doi:10.2903/j.efsa.2015.3982
`garofalo2025-seafood-cd-pb-hg-italy-10yr` (A)	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood Products: A Ten-Year Analysis	Garofalo L; Sala M; Focardi C; Pasqualetti P; Delfino D; D'Onofrio F; Droghei B; Pasquali F; Nicolini V; Galli FS; Scaramozzino P; Ubaldi A; Russo K; Neri B · 2025 · Foods 14(3):451 · doi:10.3390/foods14030451
`harding2018-bay-fundy-methylmercury` (A)	Bioaccumulation of methylmercury within the marine food web of the outer Bay of Fundy, Gulf of Maine	Harding G; Dalziel J; Vass P · 2018 · PLoS ONE · doi:10.1371/journal.pone.0197220
`lehel2023-yellowfin-tuna-indian-ocean` (A)	Metal Load of Potentially Toxic Elements in Tuna (Thunnus albacares) - Food Safety Aspects	Lehel J; Papp Z; Bartha A; Palotás P; Szabó R; Budai P; Süth M · 2023 · Foods · doi:10.3390/foods12163038
`lozano-bilbao2023-canary-tuna-metals` (A)	Risk Assessment and Characterization in Tuna	Lozano-Bilbao E; Delgado-Suárez I; Paz-

Source ID	Title	Citation
	Species of the Canary Islands According to Their Metal Content	Montelongo S; Hardisson A; Pascual-Fernández JJ; Rubio C; González Weller D; Gutiérrez ÁJ · 2023 · Foods · doi:10.3390/foods12071438
`macdonald2025-fss-uk-fishery-contaminants-review` (A)	Occurrence of chemical contaminants in wild-caught fishery products of relevance to Scottish and wider UK Fishing Waters: A Review	MacDonald S; Stead S; Rose M; Boss K; Campbell K · 2025 · Fera Science Ltd report to Food Standards Scotland (Report FR/002826) · doi:null
`ormaza-gonzalez2020-tuna-canned-eastern-pacific` (A)	Low mercury, cadmium and lead concentrations in tuna products from the eastern Pacific	Ormaza-Gonzalez FI; Ponce-Villao GE; Pin-Hidalgo GM · 2020 · Heliyon · doi:10.1016/j.heliyon.2020.e04576
`smolikova2016-heavy-metals-fish-czech` (B)	Determination of heavy metals in fish products	Smolikova V; Ridoskova A; Pelcova P · 2016 · MendelNet 2016: Proceedings of International PhD Students Conference, pp. 651-656 · doi:null
`tolkou2023-fish-heavy-metals-review` (B)	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury in Fish: Effects on the Sustainable and Healthy Development of Aquatic Life and Human Consumers	Tolkou AK; Toubanaki DK; Kyzas GZ · 2023 · Sustainability · doi:10.3390/su152316242

In addition to occurrence sources, this briefing cites:

- `docs/governance/hmtc-governing-principles.md` (v2.0, May 2026) — the five governing principles and the locked P97/P45 methodology
- `wiki/regulations/efsa-methylmercury-twi.md` — EFSA TWI 1.3 µg/kg-bw/week for MeHg
- `wiki/regulations/epa-iris-methylmercury-rfd.md` — EPA IRIS RfD 0.1 µg/kg-bw/day for MeHg

- `wiki/regulations/jecfa-methylmercury-ptwi.md` — JECFA PTWI 1.6 µg/kg-bw/week for MeHg
- WHO/JECFA pooled total-diet-study methodology
- EFSA Scientific Opinions on Pb (2010, 2024) and Cd (2009, 2024)
- ATSDR moment-matching guidance for summary-statistic sources
- Codex Alimentarius CAC/GL 50-2004 weighted pooling

Provenance Footer

This briefing is reproducible by:

1. Checking out commit `ab29385c5de7db8654be6c315d1657f93e89b1a4` in the heavymetalindex.com repository.
2. Running `npm run evidence:pooling` to regenerate `data/evidence/hmtc_aggregate_distributions.csv`.
3. Running `npm run briefing -- --product=fish-marine-predatory` to regenerate this document.

Field	Value
Generated at	2026-06-17T19:30:25.426Z
Generator	`tools/standards/build-standards-briefing.mjs`
Pooling engine	`tools/evidence/build-hmtc-aggregate-distributions.mjs`
Methodology version	HMTc-10 / P97-P45 per CLAUDE.md Part 19 / Governing Principles v2.0
Git commit	`ab29385c5de7db8654be6c315d1657f93e89b1a4`
Author	Karen Pendergrass, Standards Architect, Paleo Foundation
Status	Draft for stakeholder review

Disclosure: organizational role

Karen Pendergrass also serves as CEO of the Paleo Foundation, which operates the HMT&C certification program. This dual role is disclosed for transparency; the methodology in this briefing is reproducible from the open-source pooling engine and source data, independent of the author's organizational title.

Methodology implementation note

Methodology implementation: deterministic open-source pooling engine and document generator (this file's repository at the commit named in the Provenance Footer). Engine code and document templates were drafted with AI assistance (Claude, Anthropic) under the supervision and direction of the Standards Architect. All policy decisions, methodology choices, source-admission rules, basis conventions, and final values are the Standards Architect's; the AI assistance is a calculator-and-drafter, not a co-author.

End of briefing.

3.2 Marine Fish, Non-Predatory (cod, herring, sardine, anchovy, mackerel)

One of 5 per-product chapters in Part 3, self-contained as a licensing exhibit. The methodology is in Part 5.1; the certification status framework in Part 2.3; the Standards Ratchet in Part 2.10. Held/withdrawn analytes (e.g. total arsenic in fish and seaweed, which is arsenosugar/arsenobetaine-dominated — inorganic arsenic is the toxic and regulated analyte) are shown as "adjudication pending" and carry no published threshold; the published values are validated by the Publication QC Gate (no value exceeds a regulatory ceiling).

Product class: Fish — marine, non-predatory (sardines, anchovies, salmon, cod) **Subcategory role:** independent **Native basis for published thresholds:** as_sold **Methodology:** HMTc-10 analyte panel; per-analyte standards derived from pooled occurrence distribution at P97 (clean / independent rows) or P45 (dirty rows), capped at the lowest applicable regulatory ceiling. See "Methodology in one page" below. **Date generated:** 2026-06-17 **Author:** Karen Pendergrass, Standards Architect, Paleo Foundation **Status:** Draft for stakeholder review. Not yet through the 9 pre-publication gates.

Who this briefing is for

The HMTc certification mark serves multiple audiences with different concerns. This briefing is written so each audience finds what they need in one document. Audience-tagged "read" callouts appear inline next to each per-analyte section.

Brand legal teams — Read the per-analyte Defensibility narrative and the Adversarial read section. The methodology section is your Daubert-resistance reference. Every numeric claim traces to a source page; the contribution logs document admission criteria.

Retailer quality and compliance — Read the Master Limit Table and the per-analyte Regulator read callouts. The certification mark on a product means the brand has tested at or below every value in the Master Limit Table, in the product's native basis. The regulatory comparison rows show which government caps apply and how HMTc thresholds compare.

Regulators (FDA, EFSA, state AGs) — Read the Methodology section, the Sources Cited table, and the Anti-Circumvention argument. The methodology is open, reproducible, and follows the WHO/JECFA, EFSA, ATSDR, and Codex precedents cited. Numeric thresholds are derived from a pooled occurrence distribution of peer-reviewed and agency data, weighted by source N with a 50% relative-share cap.

Adversarial readers (plaintiff's experts, competing standards bodies) — Read the per-analyte Defensibility narrative, the Adversarial read section, the contribution logs, and the sensitivity analyses. Every value carries: the pool composition, the basis-conversion math, the regulatory cap comparison, the parametric sensitivity, and the rationale for any gap between literature and threshold.

Master Limit Table – Executive Summary

All ten HMTc analytes for this product class. Each row shows the published HMTc threshold (rounded up to the primary increment ladder), the precise pooled-engine value before rounding, the rationale tag, the contributing source count, the parametric-sensitivity band, and the lowest applicable regulatory ceiling. Brands meeting every value in this table on a like-for-like basis qualify for the HMTc certification mark for fish marine non predatory.

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
Pb	**300 ppb** [as-sold]≈ 50 ppb [prepared]	816.514 ppb (capped to 300 by reg cap)	regulatory-alignment	3	stable	300 ppb (eu2023_pb_fish_muscle_300)
iAs	**200 ppb** [as-sold]≈ 20 ppb [prepared]	122.099 ppb	literature-baseline	3	stable	—
tAs	*adjudication pending* ≈ 10000 ppb [prepared]	83013.966 ppb	literature-baseline	3	stable	—
Ni	**500 ppb** [as-sold]≈ 50 ppb [prepared]	324.294 ppb	literature-baseline	1	stable	—
Cd	**50 ppb** [as-sold]≈ 10 ppb [prepared]	277.083 ppb (capped to 50 by reg cap)	regulatory-alignment	3	provisional	50 ppb (eu2023_cd_fish_muscle_50)
tHg	**500 ppb** [as-sold]≈ 100 ppb [prepared]	454.197 ppb	literature-baseline	6	stable	—
MeHg	**200 ppb** [as-sold]≈ 25	174.690 ppb	literature-baseline	10	stable	—

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
	ppb [prepared]					

How to read this table. "Published" values are rounded up to the HMTc primary increment ladder (1, 2, 3, 5, 10, 15, 20, 25, 50, 100, 200, 500, 1000, 1300, 1500, 2000, 5000, 10000 ppb) per the program's publication conventions; this gives brands an unambiguous bin to test against. The "precise" column shows the pooled-engine output before rounding so the methodology is transparent and re-derivable. Rationale tags map to CLAUDE.md Part 19′s four allowed tags: `literature_baseline` (pooled value binds), `regulatory_alignment` (regulatory ceiling binds), `feasibility_driven` (analytical method binds), and `precautionary` (manual override below pooled and regulatory).

Methodology – one page

Per-analyte standards. For each (product row, analyte) cell, every source loaded for that cell contributes to a pooled distribution via moment-matching pseudo-sample expansion. Sample-level data uses raw values; summary statistics (mean, median, max, lb/ub, reported percentiles) reconstruct under log-normal or Tobit assumptions documented per source. Pseudo-samples are weighted by source N and capped at 50% relative pool share to prevent any one survey from dominating. The pooled distribution is sorted and queried at the per-analyte target percentile:

- **Clean subcategory:** standard = pooled P97
- **Dirty subcategory:** standard = pooled P45
- **Independent (no within-pair partner):** standard = pooled P97 by default

Why P97 / P45, not P90 / P30 (the row-pass calibration). A brand certifies a product (one row, ten analytes simultaneously) — not

one analyte at a time. Setting each per-analyte limit at the raw P90 of the clean distribution would deliver a joint row-pass rate close to 0.9^10 = 35% under analyte independence. Under realistic analyte correlation (0.6, since supply-chain hygiene drives multiple metals together), the joint rate sits around 50-70%. To deliver the row-level outcomes the Governing Principles require — approximately 90% of clean-subcategory brands and approximately 30% of dirty-subcategory brands pass row certification — per-analyte limits are uplifted to P97 (clean) and P45 (dirty). This is the calibration that delivers the program's stated objective. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening; per-analyte differentiation (e.g., Pb tighter than Ni) is deferred until HMTc has 6-12 months of lot-testing data to measure actual joint pass rates.

Basis. Each product row's standard is published in the native basis of the product form: powder rows in powder-as-placed-on-market, ready-to-feed liquid rows in as-consumed, dry cereal rows in dry-as-sold, puree rows in as-consumed. Brands ship and test in their product's native form; the certification mark matches that form. Sources reporting in non-native bases are converted into the row's native basis during pooling via documented factors (the 1:7 reconstitution ratio for powder ↔ prepared; within-class identity for as_sold ↔ dry_as_sold). The cross-basis equivalent of every published value is shown in the Master Limit Table for ease of reference by audiences operating in different bases.

Regulatory cap. The final published HMTc value never exceeds the lowest applicable loaded regulatory ceiling, expressed in the product's native basis. When the pooled P97/P45 exceeds the regulatory cap, the cap binds and the rationale tag is `regulatory_alignment`. When the pooled value is below the cap, the rationale is `literature_baseline` and the cap is shown for context only.

Sensitivity. Each pooled value is recomputed under three parametric assumptions: log-normal (headline), gamma (alternative), and nonparametric reconstruction. The spread across the three is reported as the sensitivity band: stable (≤25%), provisional (25–50%), or needs_review (>50%). Confidence intervals are computed via meta-analytic bootstrap that resamples whole sources (not individual pseudo-samples) to capture between-source uncertainty.

Methodology authorities. The pooling approach follows WHO/JECFA pooled total-diet study methodology, the EFSA Scientific Opinions on lead (2010, 2024) and cadmium (2009, 2024) for log-normal contaminant occurrence reconstruction, ATSDR moment-matching guidance for summary-statistic-only sources, and Codex CAC/GL 50-2004 weighted pooling. Per-source contribution logs document every admission decision and every basis conversion.

Tooling and reproducibility. The pooling engine is `tools/evidence/build-hmtc-aggregate-distributions.mjs` (1170 lines of deterministic JavaScript, 37 unit tests). The briefing generator is `tools/standards/build-standards-briefing.mjs` (this file). Both are versioned in git and re-runnable. Every numeric claim in this briefing can be regenerated by checking out the commit named in the Provenance Footer and running `npm run evidence:pooling && npm run briefing -- --product=fish-marine-non-predatory`.

Per-Analyte Sections

Pb

Published HMTc threshold: 300 ppb [as-sold] · 300 ppb (pre-rounding) **Cross-basis: ≈ 50 ppb [prepared] (÷7 (1:7 reconstitution))** **Rationale tag:** `regulatory-alignment` **Regulatory ceiling reference:** 300 ppb (eu2023_pb_fish_muscle_300) **Pool composition:** 3 sources contributing · 15 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [94.16, 907.43] ppb Sensitivity: log-normal 795.59 · gamma 843.71 · nonparametric 823.02 ppb · spread 6% (stable)**

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_pb_fish_muscle_300). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the underlying pool is moot — the cap binds. The pooled P97 of 816.51 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_pb_fish_muscle_300). A Daubert challenge on the value

reduces to “is eu2023_pb_fish_muscle_300 a legitimate regulatory anchor?” — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 300 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_pb_fish_muscle_300 cap. The pooled occurrence distribution from 3 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Reconstruction	Pool share
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	marine fish whole weight/wet weight as reported	5	uniform_max_only	33.3%
singhato2025-thailand-seafood-heavy-metals (2025) · TH	Risk Assessment of Toxic Heavy Metal Exposure in Selected Se…	marine fish fresh weight	5	uniform_max_only	33.3%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	marine fish muscle, as reported	5	uniform_max_only	33.3%

iAs

Published HMTc threshold: 200 ppb [as-sold] · 122.099 ppb (pre-rounding) **Cross-basis: ≈ 20 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 3 sources contributing · 8 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [30.52, 146.96] ppb Sensitivity: log-normal 127.47 · gamma 130.24 · nonparametric 144.46 ppb · spread 12% (stable)**

Defensibility. Pooled P97 from 3 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 3 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 200 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 3 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
munoz1999-seafood-inorganic-arsenic (1999) · ES	Rapid and quantitative release, separation and determination...	fish muscle, wet weight (as consumed)	2	lognormal_mean_max	25.0%

Source	Title	Basis	N used	Recon-struction	Pool share
norin1985-swedish-fish-arsenic-speciation (1985) · Sweden	Concentration of inorganic and total arsenic in fish from in…	fish muscle, wet weight (as consumed)	4	uniform_range_only	50.0%
park2005-korean-seafood-arsenic-speciation (2005) · KR	Speciation of Six Arsenic Compounds in Korean Seafood Sample…	fish muscle, wet weight (as consumed)	2	uniform_range_only	25.0%

tAs

Published HMTc threshold: adjudication pending — held provisional (wrong-analyte-not-material) · 83013.966 ppb (pre-rounding) **Cross-basis: ≈ 10000 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 3 sources contributing · 15 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [3274.66, 107462.96] ppb Sensitivity: log-normal 87006.27 · gamma 97962.96 · nonparametric 92465.67 ppb · spread 11% (stable)**

Defensibility. Pooled P97 from 3 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 3 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 10000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 3 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
macdonald2025-fss-uk-fish-ery-cont-aminants-review (2025) · GB, EU	Occurrence of chem-ical cont-aminants in wild-caught fishery p…	marine fish whole weight/wet weight as reported	5	uniform_ max_only	33.3%
singhato2025-thailand-seafood-heavy-met-als (2025) · TH	Risk As-sessment of Toxic Heavy Metal Ex-posure in Selected Se…	marine fish fresh weight	5	uniform_ max_only	33.3%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	marine fish muscle, as reported	5	uniform_ max_only	33.3%

Ni

Published HMTc threshold: 500 ppb [as-sold] · 324.294 ppb (pre-rounding) **Cross-basis: ≈ 50 ppb [prepared] (÷7 (1:7 recon-stitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 1 sources contributing · 1 pseudo-samples in pool · 0 sample-level contributors

Bootstrap 95% CI: [94.30, 703.17] ppb Sensitivity: log-normal 211.66 · gamma 259.90 · nonparametric 218.79 ppb · spread 19% (stable)

Defensibility. Pooled P97 from 1 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 1 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 500 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 1 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
yap-almutairi2022-nickel-seafood-malaysia (2022) · MY	Comparative Study of Potentially Toxic Nickel and Their Pote…	marine fish muscle, wet weight	57	lognormal_mean_max	100.0%

Cd

Published HMTc threshold: 50 ppb [as-sold] · 50 ppb (pre-rounding) **Cross-basis: ≈ 10 ppb [prepared] (÷7 (1:7 reconstitution))** **Rationale tag:** `regulatory-alignment` **Regulatory ceiling reference:** 50 ppb (eu2023_cd_fish_muscle_50) **Pool composition:** 3 sources

contributing · 15 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [96.61, 303.00] ppb Sensitivity: log-normal 192.92 · gamma 180.42 · nonparametric 301.77 ppb · spread 40% (provisional)**

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_cd_fish_muscle_50). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the underlying pool is moot — the cap binds. The pooled P97 of 277.08 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_cd_fish_muscle_50). A Daubert challenge on the value reduces to "is eu2023_cd_fish_muscle_50 a legitimate regulatory anchor?" — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 50 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_cd_fish_muscle_50 cap. The pooled occurrence distribution from 3 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Reconstruction	Pool share
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	marine fish whole weight/wet weight as reported	5	uniform_max_only	33.3%
singhato2025-thailand-seafood-heavy-metals (2025) · TH	Risk Assessment of Toxic Heavy Metal Exposure in	marine fish fresh weight	5	uniform_max_only	33.3%

Source	Title	Basis	N used	Recon-struction	Pool share
	Selected Se…				
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	marine fish muscle, as reported	5	uniform_ max_only	33.3%

tHg

Published HMTc threshold: 500 ppb [as-sold] · 454.197 ppb (pre-rounding) **Cross-basis: ≈ 100 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 6 sources contributing · 50 pseudo-samples in pool · 1 sample-level contributors **Bootstrap 95% CI: [145.93, 1016.73] ppb Sensitivity: log-normal 453.75 · gamma 454.72 · nonparametric 454.68 ppb · spread 0% (stable)**

Defensibility. Pooled P97 from 6 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 6 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 500 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table

simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 6 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
brodziak-dopierala2023-mercury-fish-poland (2023) · PL, EU	Analysis of the Mercury Content in Fish for Human Consumptio…	marine fish edible flesh, wet weight	5	uniform_max_only	10.0%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood P…	marine fish, wet weight	1142	percentile_interpolation	50.0%
helcom2017-core-indicator-metals-baltic (2017) · EU, DE, DK, EE, FI, LV, LT, PL, RU, SE	Metals (lead, cadmium and mercury) — HELCOM core indicator r…	fish muscle, wet weight	5	uniform_max_only	10.0%
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	marine fish whole weight/wet weight as reported	5	uniform_max_only	10.0%
singhato2025-thailand-seafood-heavy-metals (2025) · TH	Risk Assessment of Toxic Heavy Metal Exposure in	marine fish fresh weight	5	uniform_max_only	10.0%

Source	Title	Basis	N used	Recon-struction	Pool share
	Selected Se…				
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	marine fish muscle, as reported	5	uniform_ max_only	10.0%

MeHg

Published HMTc threshold: 200 ppb [as-sold] · 174.690 ppb (pre-rounding) **Cross-basis: ≈ 25 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 10 sources contributing · 10 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [98.51, 189] ppb Sensitivity: log-normal 174.69 · gamma 174.69 · nonparametric 174.69 ppb · spread 0% (stable)**

Defensibility. Pooled P97 from 10 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 10 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 200 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table

simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 10 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	10.0%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	10.0%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	10.0%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	10.0%
efsa-sc2015-fish-seafood-mehg-risk-	Statement on the benefits of fish/ seafood	fish meat/ products, source MeHg oc-	1	single_ point	10.0%

Source	Title	Basis	N used	Recon-struction	Pool share
benefit (2015) · EU	consump-tion com-par…	currence basis			
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consump-tion com-par…	fish meat/ products, source MeHg oc-currence basis	1	single_ point	10.0%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consump-tion com-par…	fish meat/ products, source MeHg oc-currence basis	1	single_ point	10.0%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consump-tion com-par…	fish meat/ products, source MeHg oc-currence basis	1	single_ point	10.0%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consump-tion com-par…	fish meat/ products, source MeHg oc-currence basis	1	single_ point	10.0%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consump-tion com-par…	fish meat/ products, source MeHg oc-currence basis	1	single_ point	10.0%

Adversarial Read – what a plaintiff's expert will attack, and the response

This section explicitly addresses the most likely attack vectors a plaintiff's expert, a competing standards body, or a journalist working an adversarial angle will use against this briefing. Each attack is named and answered.

Attack 1: "The pool is too small; the methodology cherry-picks." Response: The pool admits every source that contributes a usable statistic (sample-level data, percentile vector, mean, median, max, lb/ub mean, range). No source is excluded for being "summary-only." The contribution log per analyte lists every admission decision and every basis conversion. Per-source pool share is capped at 50% to prevent dominance by any single survey. The methodology is reproducible from the structured CSVs in `data/evidence/` and the engine in `tools/evidence/build-hmtc-aggregate-distributions.mjs`.

Attack 2: "Why P97 instead of P90? You're loosening the standard." Response: P97 per analyte is calibrated to deliver approximately 90% row-pass for the clean subcategory across the 10-analyte joint. Setting each analyte's limit at P90 of its own marginal would deliver a joint pass rate of 35% under independence and 50-70% under realistic correlation — well short of the Governing Principles' 90/30 row-pass target. P97 is the per-analyte percentile that delivers the row-level outcome the program promises. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening.

Attack 3: "Why is your dirty subcategory P45 instead of P10?" Response: P45 of the dirty subcategory's distribution, applied per-analyte, delivers approximately 30% joint row-pass for dirty-subcategory brands. This is the design intent stated in the Governing Principles v2.0: the dirty subcategory standard must give dirty-platform brands a realistic path to certification so the program drives reformulation rather than excluding the platform. A P10 dirty standard would deliver <1% joint row-pass and the program would have no commercial uptake on the dirty side.

Attack 4: "Your threshold matches the EU/FDA cap; HMTc adds no value over the regulator." Response: For Pb, Cd, the pub-

lished threshold matches the binding regulatory cap because the pooled P97 sits above the cap. HMTc cannot defensibly publish a tighter standard than the literature supports under its row-pass calibration. The value HMTc adds is row-level certification — a brand must meet every value in the Master Limit Table simultaneously on a like-for-like basis, with documented contribution logs and sensitivity analyses. Single-analyte regulatory caps do not deliver row-level certification.

Attack 5: "Your sensitivity analysis shows the methodology is unstable for some analytes." Response: The sensitivity bands are explicitly reported per analyte. 1 analytes (Cd) are flagged provisional — log-normal, gamma, and nonparametric variants disagree by more than 25%. This is transparent reporting, not a methodological flaw. Provisional values are published with the band stated and a commitment to resolve before final publication. The alternative — silently picking one parametric form — would be the actual methodological flaw.

Attack 7: "Where does basis conversion happen and what authority do you cite?" Response: The 1:7 powder ↔ prepared-for-feeding conversion is the conservative reconstitution ratio (1 g powder + 6 g water = 7 g prepared) standard across infant-formula regulatory frameworks (FDA Closer-to-Zero, EU 2023/915). Within-class conversions (as_sold ↔ dry_as_sold) pass through with factor 1. Every basis conversion is logged per source in the contribution log; unknown-conversion cases are flagged and not silently coerced.

Attack 8: "Show me one source you excluded." Response: No source contributing a usable statistic is excluded. Sources marked "no_usable_statistic" in the contribution log are LOD-censored entries with neither mean nor lb/ub recorded. Sources without a structured CSV row are documented in the routing audit and the gap report; those are extraction gaps, not exclusions, and they are tracked in Phase 1 backlog with target completion timelines.

Sources Cited

Every numeric claim in this briefing traces to one of the source pages below. Full source metadata lives at `wiki/sources/<source-id>.md` in the heavymetalindex.com repository.

Source ID	Title	Citation
`brodziak-dopierala2023-mercury-fish-poland` (A)	Analysis of the Mercury Content in Fish for Human Consumption in Poland	Brodziak-Dopierala B; Fischer A · 2023 · Toxics · doi:10.3390/toxics11080717
`efsa-sc2015-fish-seafood-mehg-risk-benefit` (A)	Statement on the benefits of fish/seafood consumption compared to the risks of methylmercury in fish/seafood	EFSA Scientific Committee · 2015 · EFSA Journal 2015;13(1):3982, 36 pp. · doi:10.2903/j.efsa.2015.3982
`garofalo2025-seafood-cd-pb-hg-italy-10yr` (A)	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood Products: A Ten-Year Analysis	Garofalo L; Sala M; Focardi C; Pasqualetti P; Delfino D; D'Onofrio F; Droghei B; Pasquali F; Nicolini V; Galli FS; Scaramozzino P; Ubaldi A; Russo K; Neri B · 2025 · Foods 14(3):451 · doi:10.3390/foods14030451
`helcom2017-core-indicator-metals-baltic` (A)	Metals (lead, cadmium and mercury) — HELCOM core indicator report	HELCOM; Zalewska T; Larsen MM; Fryer R; Danielsson S; Nyberg E · 2017 · HELCOM Core Indicator Report (HOLAS II component), ISSN 2343-2543 · doi:null
`macdonald2025-fss-uk-fishery-contaminants-review` (A)	Occurrence of chemical contaminants in wild-caught fishery products of relevance to Scottish and wider UK Fishing Waters: A Review	MacDonald S; Stead S; Rose M; Boss K; Campbell K · 2025 · Fera Science Ltd report to Food Standards Scotland (Report FR/002826) · doi:null
`munoz1999-seafood-inorganic-arsenic` (A)	Rapid and quantitative release, separation and determination of inorganic arsenic [As(III)+As(V)] in seafood products by microwave-assisted distillation and hydride generation atomic absorption spectrometry	Munoz O; Velez D; Cervera ML; Montoro R · 1999 · Journal of Analytical Atomic Spectrometry · doi:10.1039/a904999a

Source ID	Title	Citation
`norin1985-swedish-fish-arsenic-speciation` (B)	Concentration of inorganic and total arsenic in fish from industrially polluted water	Norin H; Vahter M; Christakopoulos A; Sandstrom M · 1985 · Chemosphere · doi:null
`park2005-korean-seafood-arsenic-speciation` (A)	Speciation of Six Arsenic Compounds in Korean Seafood Samples by HPLC-ICP-MS	Park KS; Kim JS; Lee HM; Pyo H; Kim ST; Lee KB · 2005 · Key Engineering Materials · doi:10.4028/www.scientific.net/KEM.277-279.431
`singhato2025-thailand-seafood-heavy-metals` (A)	Risk Assessment of Toxic Heavy Metal Exposure in Selected Seafood Species from Thailand	Alongkote Singhato; Narisa Rueangsri; Purimprat Thanaratsotornkun; Konpong Boonyingsathit; Piyanut Sridonpai; Nunnapus Laitip; Nattikarn Ornthai; Kunchit Judprasong · 2025 · Foods · doi:10.3390/foods14213725
`tolkou2023-fish-heavy-metals-review` (B)	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury in Fish: Effects on the Sustainable and Healthy Development of Aquatic Life and Human Consumers	Tolkou AK; Toubanaki DK; Kyzas GZ · 2023 · Sustainability · doi:10.3390/su152316242
`yap-almutairi2022-nickel-seafood-malaysia` (B)	Comparative Study of Potentially Toxic Nickel and Their Potential Human Health Risks in Seafood (Fish and Mollusks) from Peninsular Malaysia	“Yap; Chee Kong”; “Al-Mutairi; Khalid Awadh” · 2022 · Biology · doi:10.3390/biology11030376

In addition to occurrence sources, this briefing cites:

- `docs/governance/hmtc-governing-principles.md` (v2.0, May 2026) — the five governing principles and the locked P97/P45 methodology
- `wiki/regulations/efsa-methylmercury-twi.md` — EFSA TWI 1.3 µg/kg-bw/week for MeHg

- `wiki/regulations/epa-iris-methylmercury-rfd.md` — EPA IRIS RfD 0.1 µg/kg-bw/day for MeHg
- `wiki/regulations/jecfa-methylmercury-ptwi.md` — JECFA PTWI 1.6 µg/kg-bw/week for MeHg
- WHO/JECFA pooled total-diet-study methodology
- EFSA Scientific Opinions on Pb (2010, 2024) and Cd (2009, 2024)
- ATSDR moment-matching guidance for summary-statistic sources
- Codex Alimentarius CAC/GL 50-2004 weighted pooling

Provenance Footer

This briefing is reproducible by:

1. Checking out commit `ab29385c5de7db8654be6c315d1657f93e89b1a4` in the heavymetalindex.com repository.
2. Running `npm run evidence:pooling` to regenerate `data/evidence/hmtc_aggregate_distributions.csv`.
3. Running `npm run briefing -- --product=fish-marine-non-predatory` to regenerate this document.

Field	Value
Generated at	2026-06-17T19:30:25.591Z
Generator	`tools/standards/build-standards-briefing.mjs`
Pooling engine	`tools/evidence/build-hmtc-aggregate-distributions.mjs`
Methodology version	HMTc-10 / P97-P45 per CLAUDE.md Part 19 / Governing Principles v2.0
Git commit	`ab29385c5de7db8654be6c315d1657f93e89b1a4`
Author	Karen Pendergrass, Standards Architect, Paleo Foundation
Status	Draft for stakeholder review

Disclosure: organizational role

Karen Pendergrass also serves as CEO of the Paleo Foundation, which operates the HMT&C certification program. This dual role is disclosed for transparency; the methodology in this briefing is reproducible from the open-source pooling engine and source data, independent of the author's organizational title.

Methodology implementation note

Methodology implementation: deterministic open-source pooling engine and document generator (this file's repository at the commit named in the Provenance Footer). Engine code and document templates were drafted with AI assistance (Claude, Anthropic) under the supervision and direction of the Standards Architect. All policy decisions, methodology choices, source-admission rules, basis conventions, and final values are the Standards Architect's; the AI assistance is a calculator-and-drafter, not a co-author.

End of briefing.

3.3 Freshwater Fish

One of 5 per-product chapters in Part 3, self-contained as a licensing exhibit. The methodology is in Part 5.1; the certification status framework in Part 2.3; the Standards Ratchet in Part 2.10. Held/withdrawn analytes (e.g. total arsenic in fish and seaweed, which is arsenosugar/arsenobetaine-dominated — inorganic arsenic is the toxic and regulated analyte) are shown as "adjudication pending" and carry no published threshold; the published values are validated by the Publication QC Gate (no value exceeds a regulatory ceiling).

Product class: Fish — freshwater (tilapia, catfish, trout) **Subcategory role:** independent **Native basis for published thresholds:** as_sold **Methodology:** HMTc-10 analyte panel; per-analyte standards derived from pooled occurrence distribution at P97 (clean / independent rows) or P45 (dirty rows), capped at the lowest applicable regulatory ceiling. See "Methodology in one page" below. **Date generated:** 2026-06-17 **Author:** Karen Pendergrass, Standards Architect, Paleo Foundation **Status:** Draft for stakeholder review. Not yet through the 9 pre-publication gates.

Who this briefing is for

The HMTc certification mark serves multiple audiences with different concerns. This briefing is written so each audience finds what they need in one document. Audience-tagged "read" callouts appear inline next to each per-analyte section.

Brand legal teams — Read the per-analyte Defensibility narrative and the Adversarial read section. The methodology section is your Daubert-resistance reference. Every numeric claim traces to a source page; the contribution logs document admission criteria.

Retailer quality and compliance — Read the Master Limit Table and the per-analyte Regulator read callouts. The certification mark on a product means the brand has tested at or below every value in the Master Limit Table, in the product's native basis. The regulatory comparison rows show which government caps apply and how HMTc thresholds compare.

Regulators (FDA, EFSA, state AGs) — Read the Methodology section, the Sources Cited table, and the Anti-Circumvention argument. The methodology is open, reproducible, and follows the WHO/JECFA, EFSA, ATSDR, and Codex precedents cited. Numeric thresholds are derived from a pooled occurrence distribution of peer-reviewed and agency data, weighted by source N with a 50% relative-share cap.

Adversarial readers (plaintiff's experts, competing standards bodies) — Read the per-analyte Defensibility narrative, the Adversarial read section, the contribution logs, and the sensitivity analyses. Every value carries: the pool composition, the basis-conversion math, the regulatory cap comparison, the parametric sensitivity, and the rationale for any gap between literature and threshold.

Master Limit Table – Executive Summary

All ten HMTc analytes for this product class. Each row shows the published HMTc threshold (rounded up to the primary increment ladder), the precise pooled-engine value before rounding, the rationale tag, the contributing source count, the parametric-sensitivity band, and the lowest applicable regulatory ceiling. Brands meeting every value in this table on a like-for-like basis qualify for the HMTc certification mark for fish freshwater.

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
Pb	**300 ppb** [as-	1231.037 ppb	regulatory-	12	stable	300 ppb (eu2023_

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
	sold]≈ 50 ppb [prepared]	(capped to 300 by reg cap)	alignment			pb_fish_muscle_300)
iAs	**500 ppb** [as-sold]≈ 100 ppb [prepared]	359.471 ppb	literature-baseline	2	stable	—
tAs	*adjudication pending* ≈ 500 ppb [prepared]	2024.892 ppb	literature-baseline	8	stable	—
Cd	**50 ppb** [as-sold]≈ 10 ppb [prepared]	127.524 ppb (capped to 50 by reg cap)	regulatory-alignment	11	provisional	50 ppb (eu2023_cd_fish_muscle_50)
tHg	**1000 ppb** [as-sold]≈ 100 ppb [prepared]	678.374 ppb	literature-baseline	11	stable	—
MeHg	**500 ppb** [as-sold]≈ 100 ppb [prepared]	359.650 ppb	literature-baseline	6	stable	—

How to read this table. "Published" values are rounded up to the HMTc primary increment ladder (1, 2, 3, 5, 10, 15, 20, 25, 50, 100, 200, 500, 1000, 1300, 1500, 2000, 5000, 10000 ppb) per the program's publication conventions; this gives brands an unambiguous bin to test

against. The "precise" column shows the pooled-engine output before rounding so the methodology is transparent and re-derivable. Rationale tags map to CLAUDE.md Part 19's four allowed tags: `literature-baseline` (pooled value binds), `regulatory-alignment` (regulatory ceiling binds), `feasibility-driven` (analytical method binds), and `precautionary` (manual override below pooled and regulatory).

Methodology – one page

Per-analyte standards. For each (product row, analyte) cell, every source loaded for that cell contributes to a pooled distribution via moment-matching pseudo-sample expansion. Sample-level data uses raw values; summary statistics (mean, median, max, lb/ub, reported percentiles) reconstruct under log-normal or Tobit assumptions documented per source. Pseudo-samples are weighted by source N and capped at 50% relative pool share to prevent any one survey from dominating. The pooled distribution is sorted and queried at the per-analyte target percentile:

- **Clean subcategory:** standard = pooled P97
- **Dirty subcategory:** standard = pooled P45
- **Independent (no within-pair partner):** standard = pooled P97 by default

Why P97 / P45, not P90 / P30 (the row-pass calibration). A brand certifies a product (one row, ten analytes simultaneously) — not one analyte at a time. Setting each per-analyte limit at the raw P90 of the clean distribution would deliver a joint row-pass rate close to 0.9^10 = 35% under analyte independence. Under realistic analyte correlation (0.6, since supply-chain hygiene drives multiple metals together), the joint rate sits around 50-70%. To deliver the row-level outcomes the Governing Principles require — approximately 90% of clean-subcategory brands and approximately 30% of dirty-subcategory brands pass row certification — per-analyte limits are uplifted to P97 (clean) and P45 (dirty). This is the calibration that delivers the program's stated objective. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening; per-analyte differentiation (e.g., Pb tighter than Ni) is deferred until

HMTc has 6-12 months of lot-testing data to measure actual joint pass rates.

Basis. Each product row's standard is published in the native basis of the product form: powder rows in powder-as-placed-on-market, ready-to-feed liquid rows in as-consumed, dry cereal rows in dry-as-sold, puree rows in as-consumed. Brands ship and test in their product's native form; the certification mark matches that form. Sources reporting in non-native bases are converted into the row's native basis during pooling via documented factors (the 1:7 reconstitution ratio for powder ↔ prepared; within-class identity for as_sold ↔ dry_as_sold). The cross-basis equivalent of every published value is shown in the Master Limit Table for ease of reference by audiences operating in different bases.

Regulatory cap. The final published HMTc value never exceeds the lowest applicable loaded regulatory ceiling, expressed in the product's native basis. When the pooled P97/P45 exceeds the regulatory cap, the cap binds and the rationale tag is `regulatory-alignment`. When the pooled value is below the cap, the rationale is `literature-baseline` and the cap is shown for context only.

Sensitivity. Each pooled value is recomputed under three parametric assumptions: log-normal (headline), gamma (alternative), and nonparametric reconstruction. The spread across the three is reported as the sensitivity band: stable (≤25%), provisional (25–50%), or needs_review (>50%). Confidence intervals are computed via meta-analytic bootstrap that resamples whole sources (not individual pseudo-samples) to capture between-source uncertainty.

Methodology authorities. The pooling approach follows WHO/JECFA pooled total-diet study methodology, the EFSA Scientific Opinions on lead (2010, 2024) and cadmium (2009, 2024) for log-normal contaminant occurrence reconstruction, ATSDR moment-matching guidance for summary-statistic-only sources, and Codex CAC/GL 50-2004 weighted pooling. Per-source contribution logs document every admission decision and every basis conversion.

Tooling and reproducibility. The pooling engine is `tools/evidence/build-hmtc-aggregate-distributions.mjs` (1170 lines of deterministic JavaScript, 37 unit tests). The briefing generator is `tools/standards/build-standards-briefing.mjs` (this file). Both are ver-

sioned in git and re-runnable. Every numeric claim in this briefing can be regenerated by checking out the commit named in the Provenance Footer and running `npm run evidence:pooling && npm run briefing -- --product=fish-freshwater`.

Per-Analyte Sections

Pb

Published HMTc threshold: 300 ppb [as-sold] · 300 ppb (pre-rounding) **Cross-basis: ≈ 50 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `regulatory-alignment` **Regulatory ceiling reference:** 300 ppb (eu2023_pb_fish_muscle_300) **Pool composition:** 12 sources contributing · 206 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [671.20, 2800.55] ppb Sensitivity: log-normal 1612.58 · gamma 1430.01 · nonparametric 1334.74 ppb · spread 17% (stable)**

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_pb_fish_muscle_300). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the underlying pool is moot — the cap binds. The pooled P97 of 1231.04 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_pb_fish_muscle_300). A Daubert challenge on the value reduces to "is eu2023_pb_fish_muscle_300 a legitimate regulatory anchor?" — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 300 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_pb_fish_muscle_300 cap. The pooled occurrence distribution from 12 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Recon-struction	Pool share
lehel2024-rainbow-trout-aquaculture-italy (2024) · IT, HU, EU, WHO	Possible Metal Burden of Potentially Toxic Elements in Rainb…	unknown	40	lognormal_mean_max	19.4%
lehel2024-rainbow-trout-aquaculture-italy (2024) · IT, HU, EU, WHO	Possible Metal Burden of Potentially Toxic Elements in Rainb…	unknown	40	uniform_max_only	4.9%
lehel2024-rainbow-trout-aquaculture-italy (2024) · IT, HU, EU, WHO	Possible Metal Burden of Potentially Toxic Elements in Rainb…	unknown	40	lognormal_mean_max	19.4%
lehel2024-rainbow-trout-aquaculture-italy (2024) · IT, HU, EU, WHO	Possible Metal Burden of Potentially Toxic Elements in Rainb…	rainbow trout flesh, wet weight	40	lognormal_mean_max	19.4%
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	freshwater fish, wet whole-weight/ muscle as reported	16	uniform_max_only	4.9%
maikanov2020-fish-products-shuchinsk-burabay-	Quality and safety of fish products in the	freshwater fish muscle, wet/as-reported	44	uniform_max_only	4.9%

Source	Title	Basis	N used	Recon-struction	Pool share
metals (2020) · KZ	Shuchinsk-Burabay…				
majlesi2019-farmed-rainbow-trout-yasuj-iran (2019) · IR, WHO, US, EU	Heavy metal content in farmed rainbow trout in relation to a…	rainbow trout muscle, dry weight	30	lognormal_mean_max	14.6%
zhuzhassarova2024-fish-seafood-central-asia-review (2024) · KZ, KG, TJ, TM, UZ	Fish and Seafood Safety: Human Exposure to Toxic Metals from…	freshwater fish tissue, wet/as-reported	5	uniform_max_only	2.4%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood P…	freshwater fish, wet weight	57	uniform_max_only	4.9%
majlesi2017-khersan-river-fish-iran (2017) · IR, WHO, EU, US	The Concentration of Mercury, Cadmium and Lead in Muscular T…	freshwater fish muscle, dry/ambiguous basis	5	uniform_max_only	2.4%
subhanullah2024-panjkora-fish-metals (2024) · PK	The detrimental effects of heavy metals on tributaries exert…	fish muscle, dry/wet ambiguous basis	1	single_point	0.5%
tolkou2023-fish-heavy-	Detection of Arsenic,	freshwater fish mus-	5	uniform_max_only	2.4%

Source	Title	Basis	N used	Recon-struction	Pool share
metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Chromium, Cadmium, Lead, and Mercury i…	cle, as re-ported			

iAs

Published HMTc threshold: 500 ppb [as-sold] · 359.471 ppb (pre-rounding) **Cross-basis: ≈ 100 ppb [prepared] (÷7 (1:7 recon-stitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 2 sources contributing · 20 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [111.10, 406.18] ppb Sensitivity: log-normal 401.45 · gamma 327.17 · nonparametric 365.50 ppb · spread 19% (stable)**

Defensibility. Pooled P97 from 2 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 2 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 500 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 2 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
hoy2023-arsenic-speciation-freshwater-fish (2023) · CA, CN, FR, TH, TR, DE	Arsenic speciation in fresh-water fish: challenges and re-sear…	freshwa-ter fish dry weight	14	uniform_ max_only	50.0%
lehel2024-rainbow-trout-aquacul-ture-italy (2024) · IT, HU, EU, WHO	Possible Metal Bur-den of Po-tentially Toxic El-ements in Rainb…	rainbow trout flesh, wet weight	40	lognormal_ mean_max	50.0%

tAs

Published HMTc threshold: adjudication pending — held provisional (wrong-analyte-not-material) · 2024.892 ppb (pre-rounding) **Cross-basis: ≈ 500 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 8 sources contributing · 81 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [1132.65, 2518.93] ppb Sensitivity: log-normal 2252.28 · gamma 2195.34 · nonparametric 2218.91 ppb · spread 3% (stable)**

Defensibility. Pooled P97 from 8 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 8 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international

precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 5000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 8 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
hoy2023-arsenic-speciation-freshwater-fish (2023) · CA, CN, FR, TH, TR, DE	Arsenic speciation in fresh-water fish: challenges and re-sear…	freshwa-ter fish wet weight	5	uniform_ max_only	6.2%
hoy2023-arsenic-speciation-freshwater-fish (2023) · CA, CN, FR, TH, TR, DE	Arsenic speciation in fresh-water fish: challenges and re-sear…	freshwa-ter fish dry weight	14	uniform_ max_only	12.3%
lehel2024-rainbow-trout-aquacul-ture-italy (2024) · IT, HU, EU, WHO	Possible Metal Bur-den of Po-tentially Toxic El-ements in Rainb…	rainbow trout flesh, wet weight	40	lognormal_ mean_max	49.4%
macdonald2025-fss-uk-fish-ery-cont-aminants-review	Occurrence of chem-ical cont-aminants in wild-	freshwa-ter fish, wet whole-weight/ muscle as reported	5	uniform_ max_only	6.2%

Source	Title	Basis	N used	Recon-struction	Pool share
(2025) · GB, EU	caught fishery p…				
maikanov2020-fish-products-shuchinsk-burabay-metals (2020) · KZ	Quality and safety of fish products in the Shuchinsk-Burabay…	freshwater fish muscle, wet/as-reported	44	uniform_max_only	12.3%
zhuzhassarova2024-fish-seafood-central-asia-review (2024) · KZ, KG, TJ, TM, UZ	Fish and Seafood Safety: Human Exposure to Toxic Metals from…	freshwater fish tissue, wet/as-reported	5	uniform_max_only	6.2%
subhanullah2024-panjkora-fish-metals (2024) · PK	The detrimental effects of heavy metals on tributaries exert…	fish muscle, dry/wet ambiguous basis	1	single_point	1.2%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	freshwater fish muscle, as reported	5	uniform_max_only	6.2%

Cd

Published HMTc threshold: 50 ppb [as-sold] · 50 ppb (pre-rounding) **Cross-basis: ≈ 10 ppb [prepared] (÷7 (1:7 reconstitution))** **Rationale tag:** `regulatory-alignment` **Regulatory ceiling reference:** 50 ppb (eu2023_cd_fish_muscle_50) **Pool composition:** 11 sources contributing · 196 pseudo-samples in pool · 0 sample-level contributors

Bootstrap 95% CI: [49.39, 1760.32] ppb Sensitivity: log-normal 220.62 · gamma 143.82 · nonparametric 238.08 ppb · spread 40% (provisional)

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_cd_fish_muscle_50). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the underlying pool is moot — the cap binds. The pooled P97 of 127.52 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_cd_fish_muscle_50). A Daubert challenge on the value reduces to "is eu2023_cd_fish_muscle_50 a legitimate regulatory anchor?" — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 50 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_cd_fish_muscle_50 cap. The pooled occurrence distribution from 11 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Reconstruction	Pool share
lehel2024-rainbow-trout-aquaculture-italy (2024) · IT, HU, EU, WHO	Possible Metal Burden of Potentially Toxic Elements in Rainb…	unknown	40	lognormal_mean_max	20.4%
lehel2024-rainbow-trout-aquaculture-italy (2024) ·	Possible Metal Burden of Potentially Toxic Elements in Rainb…	unknown	40	lognormal_mean_max	20.4%

Source	Title	Basis	N used	Recon-struction	Pool share
IT, HU, EU, WHO					
lehel2024-rainbow-trout-aquaculture-italy (2024) · IT, HU, EU, WHO	Possible Metal Burden of Potentially Toxic Elements in Rainb…	rainbow trout flesh, wet weight	40	lognormal_mean_max	20.4%
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	freshwater fish, wet whole-weight/ muscle as reported	16	uniform_max_only	5.1%
maikanov2020-fish-products-shuchinsk-burabay-metals (2020) · KZ	Quality and safety of fish products in the Shuchinsk-Burabay…	freshwater fish muscle, wet/as-reported	44	uniform_max_only	5.1%
majlesi2019-farmed-rainbow-trout-yasuj-iran (2019) · IR, WHO, US, EU	Heavy metal content in farmed rainbow trout in relation to a…	rainbow trout muscle, dry weight	30	lognormal_mean_max	15.3%
zhuzhassarova2024-fish-seafood-central-asia-review (2024) · KZ, KG, TJ, TM, UZ	Fish and Seafood Safety: Human Exposure to Toxic Metals from…	freshwater fish tissue, wet/as-reported	5	uniform_max_only	2.5%
garofalo2025-seafood-cd-pb-hg-	Monitoring of Cadmium,	freshwater fish, wet weight	53	uniform_max_only	5.1%

Source	Title	Basis	N used	Recon-struction	Pool share
italy-10yr (2025) · IT, EU	Lead, and Mercury Levels in Seafood P…				
majlesi2017-khersan-river-fish-iran (2017) · IR, WHO, EU, US	The Concentration of Mercury, Cadmium and Lead in Muscular T…	freshwater fish muscle, dry/ambiguous basis	5	uniform_max_only	2.5%
subhanullah2024-panjkora-fish-metals (2024) · PK	The detrimental effects of heavy metals on tributaries exert…	fish muscle, dry/wet ambiguous basis	1	single_point	0.5%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	freshwater fish muscle, as reported	5	uniform_max_only	2.5%

tHg

Published HMTc threshold: 1000 ppb [as-sold] · 678.374 ppb (pre-rounding) **Cross-basis: ≈ 100 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 11 sources contributing · 1407 pseudo-samples in pool · 1 sample-level contributors **Bootstrap 95% CI: [346.14, 1063.23] ppb Sensitivity: log-normal 802.96 · gamma 736.28 · nonparametric 651.35 ppb · spread 19% (stable)**

Defensibility. Pooled P97 from 11 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across lognormal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 11 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 1000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 11 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
brodziak-dopiera2023-mercury-fish-poland (2023) · PL, EU	Analysis of the Mercury Content in Fish for Human Consumptio…	freshwater fish edible muscle, wet weight	30	lognormal_mean_median	2.1%
eaglessmith2016-freshwater-fish-mercury-western-us-canada (2016) · US, CA	Spatial and temporal patterns of mercury concentrations in f…	skinless boneless fillet standardized wet weight	96310	lognormal_mean_max	35.5%
epa1999-mercury-update-fish-	Mercury Update: Impact on	freshwater whole-fish composite, wet weight	315	lognormal_mean_max	22.4%

Source	Title	Basis	N used	Recon-struction	Pool share
advisories (1999) · US	Fish Advi-sories				
epa1999-mercury-update-fish-advisories (1999) · US	Mercury Update: Impact on Fish Advi-sories	freshwater fish tissue, wet weight	374	lognormal_mean_me-dian	26.6%
macdonald2025-fss-uk-fish-ery-cont-aminants-review (2025) · GB, EU	Occurrence of chem-ical cont-aminants in wild-caught fishery p…	freshwa-ter fish, wet whole-weight/ muscle as reported	16	uniform_max_only	0.7%
maikanov2020-fish-prod-ucts-shuchinsk-burabay-metals (2020) · KZ	Qual-ity and safety of fish prod-ucts in the Shuchinsk-Burabay…	freshwater fish mus-cle, wet/as-reported	44	uniform_max_only	0.7%
majlesi2019-farmed-rainbow-trout-yasuj-iran (2019) · IR, WHO, US, EU	Heavy metal con-tent in farmed rainbow trout in re-lation to a…	rainbow trout mus-cle, dry weight	30	lognormal_mean_max	2.1%
zhuzhassarova2024-fish-seafood-central-asia-review (2024) · KZ, KG, TJ, TM, UZ	Fish and Seafood Safety: Hu-man Ex-posure to Toxic Met-als from…	freshwater fish tissue, wet/as-re-ported	5	uniform_max_only	0.4%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in	freshwater fish, wet weight	123	percentile_interpola-tion	8.7%

Source	Title	Basis	N used	Recon-struction	Pool share
	Seafood P…				
majlesi2017-khersan-river-fish-iran (2017) · IR, WHO, EU, US	The Concentration of Mercury, Cadmium and Lead in Muscular T…	freshwater fish muscle, dry/ ambiguous basis	5	uniform_ max_only	0.4%
tolkou2023-fish-heavy-metals-review (2023) · AR, BD, CN, ES, GN, IN, IT, PK, PL, SA, SK, TR, VN	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury i…	freshwater fish muscle, as reported	5	uniform_ max_only	0.4%

MeHg

Published HMTc threshold: 500 ppb [as-sold] · 359.650 ppb (pre-rounding) **Cross-basis: ≈ 100 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 6 sources contributing · 6 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [86.55, 466.97] ppb Sensitivity: log-normal 374.81 · gamma 359.65 · nonparametric 383.64 ppb · spread 6% (stable)**

Defensibility. Pooled P97 from 6 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 6 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international

precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 500 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 6 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	16.7%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	16.7%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consumption compar…	fish meat/ products, source MeHg occurrence basis	1	single_ point	16.7%
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/ seafood consump-	fish meat/ products, source MeHg occurrence basis	1	single_ point	16.7%

Source	Title	Basis	N used	Recon-struction	Pool share
	tion compar…				
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/seafood consumption compar…	fish meat/products, source MeHg occurrence basis	1	single_point	16.7%
macdonald2025fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	trout/freshwater fish, wet weight	1	lognormal_mean_only	16.7%

Adversarial Read – what a plaintiff's expert will attack, and the response

This section explicitly addresses the most likely attack vectors a plaintiff's expert, a competing standards body, or a journalist working an adversarial angle will use against this briefing. Each attack is named and answered.

Attack 1: "The pool is too small; the methodology cherry-picks." Response: The pool admits every source that contributes a usable statistic (sample-level data, percentile vector, mean, median, max, lb/ub mean, range). No source is excluded for being "summary-only." The contribution log per analyte lists every admission decision and every basis conversion. Per-source pool share is capped at 50% to prevent dominance by any single survey. The methodology is reproducible from the structured CSVs in `data/evidence/` and the engine in `tools/evidence/build-hmtc-aggregate-distributions.mjs`.

Attack 2: "Why P97 instead of P90? You're loosening the standard." Response: P97 per analyte is calibrated to deliver approxi-

mately 90% row-pass for the clean subcategory across the 10-analyte joint. Setting each analyte's limit at P90 of its own marginal would deliver a joint pass rate of 35% under independence and 50-70% under realistic correlation — well short of the Governing Principles' 90/30 row-pass target. P97 is the per-analyte percentile that delivers the row-level outcome the program promises. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening.

Attack 3: "Why is your dirty subcategory P45 instead of P10?" Response: P45 of the dirty subcategory's distribution, applied per-analyte, delivers approximately 30% joint row-pass for dirty-subcategory brands. This is the design intent stated in the Governing Principles v2.0: the dirty subcategory standard must give dirty-platform brands a realistic path to certification so the program drives reformulation rather than excluding the platform. A P10 dirty standard would deliver <1% joint row-pass and the program would have no commercial uptake on the dirty side.

Attack 4: "Your threshold matches the EU/FDA cap; HMTc adds no value over the regulator." Response: For Pb, Cd, the published threshold matches the binding regulatory cap because the pooled P97 sits above the cap. HMTc cannot defensibly publish a tighter standard than the literature supports under its row-pass calibration. The value HMTc adds is row-level certification — a brand must meet every value in the Master Limit Table simultaneously on a like-for-like basis, with documented contribution logs and sensitivity analyses. Single-analyte regulatory caps do not deliver row-level certification.

Attack 5: "Your sensitivity analysis shows the methodology is unstable for some analytes." Response: The sensitivity bands are explicitly reported per analyte. 1 analytes (Cd) are flagged provisional — log-normal, gamma, and nonparametric variants disagree by more than 25%. This is transparent reporting, not a methodological flaw. Provisional values are published with the band stated and a commitment to resolve before final publication. The alternative — silently picking one parametric form — would be the actual methodological flaw.

Attack 7: "Where does basis conversion happen and what authority do you cite?" Response: The 1:7 powder ↔ prepared-for-feeding conversion is the conservative reconstitution ratio (1 g powder + 6 g water = 7 g prepared) standard across infant-formula regulatory frameworks (FDA Closer-to-Zero, EU 2023/915). Within-class conversions

(as_sold ↔ dry_as_sold) pass through with factor 1. Every basis conversion is logged per source in the contribution log; unknown-conversion cases are flagged and not silently coerced.

Attack 8: "Show me one source you excluded." Response: No source contributing a usable statistic is excluded. Sources marked "no_usable_statistic" in the contribution log are LOD-censored entries with neither mean nor lb/ub recorded. Sources without a structured CSV row are documented in the routing audit and the gap report; those are extraction gaps, not exclusions, and they are tracked in Phase 1 backlog with target completion timelines.

Sources Cited

Every numeric claim in this briefing traces to one of the source pages below. Full source metadata lives at `wiki/sources/<source-id>.md` in the heavymetalindex.com repository.

Source ID	Title	Citation
`brodziak-dopierala2023-mercury-fish-poland` (A)	Analysis of the Mercury Content in Fish for Human Consumption in Poland	Brodziak-Dopierala B; Fischer A · 2023 · Toxics · doi:10.3390/toxics11080717
`eaglessmith2016-freshwater-fish-mercury-western-us-canada` (A)	Spatial and temporal patterns of mercury concentrations in freshwater fish across the Western United States and Canada	Eagles-Smith CA; Ackerman JT; Willacker JJ; Tate MT; Lutz MA; Fleck JA; Stewart AR; Wiener JG; Evers DC; Lepak JM; Davis JA; Flanagan Pritz C · 2016 · Science of The Total Environment (2016, in press 2015) · doi:10.1016/j.scitotenv.2016.03.229
`efsa-sc2015-fish-seafood-mehg-risk-benefit` (A)	Statement on the benefits of fish/seafood consumption compared to the risks of methylmercury in fish/seafood	EFSA Scientific Committee · 2015 · EFSA Journal 2015;13(1):3982, 36 pp. · doi:10.2903/j.efsa.2015.3982
`epa1999-mercury-update-fish-advisories` (A)	Mercury Update: Impact on Fish Advisories	US EPA Office of Water · 1999 · US EPA Fact

Source ID	Title	Citation
		Sheet EPA-823-F-99-016 · doi:null
`garofalo2025-seafood-cd-pb-hg-italy-10yr` (A)	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood Products: A Ten-Year Analysis	Garofalo L; Sala M; Focardi C; Pasqualetti P; Delfino D; D'Onofrio F; Droghei B; Pasquali F; Nicolini V; Galli FS; Scaramozzino P; Ubaldi A; Russo K; Neri B · 2025 · Foods 14(3):451 · doi:10.3390/foods14030451
`hoy2023-arsenic-speciation-freshwater-fish` (A)	Arsenic speciation in freshwater fish: challenges and research needs	Hoy KS; Davydiuk T; Chen X; Lau C; Schofield JRM; Lu X; Graydon JA; Mitchell R; Reichert M; Le XC · 2023 · Food Quality and Safety · doi:10.1093/fqsafe/fyad032
`lehel2024-rainbow-trout-aquaculture-italy` (A)	Possible Metal Burden of Potentially Toxic Elements in Rainbow Trout (Oncorhynchus mykiss) on Aquaculture Farm	Lehel J; Plachy M; Palotás P; Bartha A; Budai P · 2024 · Fishes · doi:10.3390/fishes9070252
`macdonald2025-fss-uk-fishery-contaminants-review` (A)	Occurrence of chemical contaminants in wild-caught fishery products of relevance to Scottish and wider UK Fishing Waters: A Review	MacDonald S; Stead S; Rose M; Boss K; Campbell K · 2025 · Fera Science Ltd report to Food Standards Scotland (Report FR/002826) · doi:null
`maikanov2020-fish-products-shuchinsk-burabay-metals` (B)	Quality and safety of fish products in the Shuchinsk-Burabay Resort Zone	Maikanov BS; Auteleyeva LT; Ismagulova GT; Wisniewski J; Belkot Z; Anusz K · 2020 · Medycyna Weterynaryjna 76(10):585-588 · doi:10.21521/mw.6462
`majlesi2017-khersan-river-fish-iran` (B)	The Concentration of Mercury, Cadmium and Lead in Muscular Tissue of Fishes in Khersan River	Majlesi M; Khazaei Y; Berizi E; Sharifpour E · 2017 · International Journal of Nutrition Sciences · doi:null

Source ID	Title	Citation
`majlesi2019-farmed-rainbow-trout-yasuj-iran` (A)	Heavy metal content in farmed rainbow trout in relation to aquaculture area and feed pellets	Majlesi M; Malekzadeh J; Berizi E; Akbartabar Toori M · 2019 · Foods and Raw Materials · doi:10.21603/2308-40
`subhanullah2024-panjkora-fish-metals` (A)	The detrimental effects of heavy metals on tributaries exert pressure on water quality, Crossocheilus diplocheilus, and the well-being of human health	Subhanullah M; Hassan N; Ali S; Saleh IA; Ilyas M; Rawan B; Ullah W; Iqbal B; Okla MK; Alaraidh IA; Fahad S · 2024 · Scientific Reports · doi:10.1038/s41598-024-53340-5
`tolkou2023-fish-heavy-metals-review` (B)	Detection of Arsenic, Chromium, Cadmium, Lead, and Mercury in Fish: Effects on the Sustainable and Healthy Development of Aquatic Life and Human Consumers	Tolkou AK; Toubanaki DK; Kyzas GZ · 2023 · Sustainability · doi:10.3390/su152316242
`zhuzhassarova2024-fish-seafood-central-asia-review` (B)	Fish and Seafood Safety: Human Exposure to Toxic Metals from the Aquatic Environment and Fish in Central Asia	Zhuzhassarova G; Azarbayjani F; Zamaratskaia G · 2024 · International Journal of Molecular Sciences · doi:10.3390/ijms25031590

In addition to occurrence sources, this briefing cites:

- `docs/governance/hmtc-governing-principles.md` (v2.0, May 2026) — the five governing principles and the locked P97/P45 methodology
- `wiki/regulations/efsa-methylmercury-twi.md` — EFSA TWI 1.3 µg/kg-bw/week for MeHg
- `wiki/regulations/epa-iris-methylmercury-rfd.md` — EPA IRIS RfD 0.1 µg/kg-bw/day for MeHg
- `wiki/regulations/jecfa-methylmercury-ptwi.md` — JECFA PTWI 1.6 µg/kg-bw/week for MeHg
- WHO/JECFA pooled total-diet-study methodology
- EFSA Scientific Opinions on Pb (2010, 2024) and Cd (2009, 2024)
- ATSDR moment-matching guidance for summary-statistic sources
- Codex Alimentarius CAC/GL 50-2004 weighted pooling

Provenance Footer

This briefing is reproducible by:

1. Checking out commit `ab29385c5de7db8654be6c315d1657f93e89b1a4` in the heavymetalindex.com repository.
2. Running `npm run evidence:pooling` to regenerate `data/evidence/hmtc_aggregate_distributions.csv`.
3. Running `npm run briefing -- --product=fish-freshwater` to regenerate this document.

Field	Value
Generated at	2026-06-17T19:30:25.508Z
Generator	`tools/standards/build-standards-briefing.mjs`
Pooling engine	`tools/evidence/build-hmtc-aggregate-distributions.mjs`
Methodology version	HMTc-10 / P97-P45 per CLAUDE.md Part 19 / Governing Principles v2.0
Git commit	`ab29385c5de7db8654be6c315d1657f93e89b1a4`
Author	Karen Pendergrass, Standards Architect, Paleo Foundation
Status	Draft for stakeholder review

Disclosure: organizational role

Karen Pendergrass also serves as CEO of the Paleo Foundation, which operates the HMT&C certification program. This dual role is disclosed for transparency; the methodology in this briefing is reproducible from the open-source pooling engine and source data, independent of the author's organizational title.

Methodology implementation note

Methodology implementation: deterministic open-source pooling engine and document generator (this file's repository at the commit named in the Provenance Footer). Engine code and document templates were drafted with AI assistance (Claude, Anthropic) under the supervision and direction of the Standards Architect. All policy decisions, methodology choices, source-admission rules, basis conventions, and final values are

the Standards Architect's; the AI assistance is a calculator-and-drafter, not a co-author.

End of briefing.

3.4 Shellfish (Bivalves and Crustaceans)

One of 5 per-product chapters in Part 3, self-contained as a licensing exhibit. The methodology is in Part 5.1; the certification status framework in Part 2.3; the Standards Ratchet in Part 2.10. Held/withdrawn analytes (e.g. total arsenic in fish and seaweed, which is arsenosugar/arsenobetaine-dominated — inorganic arsenic is the toxic and regulated analyte) are shown as "adjudication pending" and carry no published threshold; the published values are validated by the Publication QC Gate (no value exceeds a regulatory ceiling).

Product class: Shellfish (shrimp, crab, lobster, clams, oysters, mussels) **Subcategory role:** whole_row_platform **Native basis for published thresholds:** as_sold **Methodology:** HMTc-10 analyte panel; per-analyte standards derived from pooled occurrence distribution at P97 (clean / independent rows) or P45 (dirty rows), capped at the lowest applicable regulatory ceiling. See "Methodology in one page" below. **Date generated:** 2026-06-18 **Author:** Karen Pendergrass, Standards Architect, Paleo Foundation **Status:** Draft for stakeholder review. Not yet through the 9 pre-publication gates.

Who this briefing is for

The HMTc certification mark serves multiple audiences with different concerns. This briefing is written so each audience finds what they need in one document. Audience-tagged "read" callouts appear inline next to each per-analyte section.

Brand legal teams — Read the per-analyte Defensibility narrative and the Adversarial read section. The methodology section is your Daubert-resistance reference. Every numeric claim traces to a source page; the contribution logs document admission criteria.

Retailer quality and compliance — Read the Master Limit Table and the per-analyte Regulator read callouts. The certification mark on a product means the brand has tested at or below every value in the Master Limit Table, in the product's native basis. The regulatory comparison rows show which government caps apply and how HMTc thresholds compare.

Regulators (FDA, EFSA, state AGs) — Read the Methodology section, the Sources Cited table, and the Anti-Circumvention argument. The methodology is open, reproducible, and follows the WHO/JECFA, EFSA, ATSDR, and Codex precedents cited. Numeric thresholds are derived from a pooled occurrence distribution of peer-reviewed and agency data, weighted by source N with a 50% relative-share cap.

Adversarial readers (plaintiff's experts, competing standards bodies) — Read the per-analyte Defensibility narrative, the Adversarial read section, the contribution logs, and the sensitivity analyses. Every value carries: the pool composition, the basis-conversion math, the regulatory cap comparison, the parametric sensitivity, and the rationale for any gap between literature and threshold.

Master Limit Table – Executive Summary

All ten HMTc analytes for this product class. Each row shows the published HMTc threshold (rounded up to the primary increment ladder), the precise pooled-engine value before rounding, the rationale tag, the contributing source count, the parametric-sensitivity band, and the lowest applicable regulatory ceiling. Brands meeting every value in this table on a like-for-like basis qualify for the HMTc certification mark for shellfish.

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
Pb	**1500 ppb** [as-sold]≈ 500 ppb [prepared]	3043.451 ppb (capped to 1500 by reg cap)	regulatory-alignment	18	stable	1500 ppb (eu2023_pb_bivalve_1500)
iAs	**200 ppb** [as-sold]≈ 20 ppb [prepared]	109.487 ppb	literature-baseline	4	needs_review	—
tAs	*adjudication pending* ≈ 5000 ppb [prepared]	33023.140 ppb	literature-baseline	6	needs_review	—
Ni	**10000 ppb** [as-sold]≈ 2000 ppb [prepared]	12189.015 ppb	literature-baseline	5	provisional	—
Cd	**1000 ppb** [as-sold]≈ 200 ppb [prepared]	5152.922 ppb (capped to 1000 by reg cap)	regulatory-alignment	20	stable	1000 ppb (eu2023_cd_bivalve_1000)
Al	*data gap*	—	no-data	0	no_data	—
tHg	**500 ppb** [as-sold]≈ 100 ppb [prepared]	376.492 ppb	literature-baseline	15	stable	500 ppb (eu2023_hg_crustaceans_molluscs_500)

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
MeHg	**200 ppb** [as-sold]≈ 25 ppb [prepared]	140.720 ppb	literature-baseline	4	stable	—

How to read this table. "Published" values are rounded up to the HMTc primary increment ladder (1, 2, 3, 5, 10, 15, 20, 25, 50, 100, 200, 500, 1000, 1300, 1500, 2000, 5000, 10000 ppb) per the program's publication conventions; this gives brands an unambiguous bin to test against. The "precise" column shows the pooled-engine output before rounding so the methodology is transparent and re-derivable. Rationale tags map to CLAUDE.md Part 19′s four allowed tags: `literature-baseline` (pooled value binds), `regulatory-alignment` (regulatory ceiling binds), `feasibility-driven` (analytical method binds), and `precautionary` (manual override below pooled and regulatory).

Methodology – one page

Per-analyte standards. For each (product row, analyte) cell, every source loaded for that cell contributes to a pooled distribution via moment-matching pseudo-sample expansion. Sample-level data uses raw values; summary statistics (mean, median, max, lb/ub, reported percentiles) reconstruct under log-normal or Tobit assumptions documented per source. Pseudo-samples are weighted by source N and capped at 50% relative pool share to prevent any one survey from dominating. The pooled distribution is sorted and queried at the per-analyte target percentile:

- **Clean subcategory:** standard = pooled P97
- **Dirty subcategory:** standard = pooled P45
- **Independent (no within-pair partner):** standard = pooled P97 by default

Why P97 / P45, not P90 / P30 (the row-pass calibration). A brand certifies a product (one row, ten analytes simultaneously) — not one analyte at a time. Setting each per-analyte limit at the raw P90 of the clean distribution would deliver a joint row-pass rate close to 0.9^10 = 35% under analyte independence. Under realistic analyte correlation (0.6, since supply-chain hygiene drives multiple metals together), the joint rate sits around 50-70%. To deliver the row-level outcomes the Governing Principles require — approximately 90% of clean-subcategory brands and approximately 30% of dirty-subcategory brands pass row certification — per-analyte limits are uplifted to P97 (clean) and P45 (dirty). This is the calibration that delivers the program's stated objective. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening; per-analyte differentiation (e.g., Pb tighter than Ni) is deferred until HMTc has 6-12 months of lot-testing data to measure actual joint pass rates.

Basis. Each product row's standard is published in the native basis of the product form: powder rows in powder-as-placed-on-market, ready-to-feed liquid rows in as-consumed, dry cereal rows in dry-as-sold, puree rows in as-consumed. Brands ship and test in their product's native form; the certification mark matches that form. Sources reporting in non-native bases are converted into the row's native basis during pooling via documented factors (the 1:7 reconstitution ratio for powder ↔ prepared; within-class identity for as_sold ↔ dry_as_sold). The cross-basis equivalent of every published value is shown in the Master Limit Table for ease of reference by audiences operating in different bases.

Regulatory cap. The final published HMTc value never exceeds the lowest applicable loaded regulatory ceiling, expressed in the product's native basis. When the pooled P97/P45 exceeds the regulatory cap, the cap binds and the rationale tag is `regulatory_alignment`. When the pooled value is below the cap, the rationale is `literature_baseline` and the cap is shown for context only.

Sensitivity. Each pooled value is recomputed under three parametric assumptions: log-normal (headline), gamma (alternative), and nonparametric reconstruction. The spread across the three is reported as the sensitivity band: stable (≤25%), provisional (25–50%), or needs_review (>50%). Confidence intervals are computed via meta-analytic bootstrap

that resamples whole sources (not individual pseudo-samples) to capture between-source uncertainty.

Methodology authorities. The pooling approach follows WHO/JECFA pooled total-diet study methodology, the EFSA Scientific Opinions on lead (2010, 2024) and cadmium (2009, 2024) for log-normal contaminant occurrence reconstruction, ATSDR moment-matching guidance for summary-statistic-only sources, and Codex CAC/GL 50-2004 weighted pooling. Per-source contribution logs document every admission decision and every basis conversion.

Tooling and reproducibility. The pooling engine is `tools/evidence/build-hmtc-aggregate-distributions.mjs` (1170 lines of deterministic JavaScript, 37 unit tests). The briefing generator is `tools/standards/build-standards-briefing.mjs` (this file). Both are versioned in git and re-runnable. Every numeric claim in this briefing can be regenerated by checking out the commit named in the Provenance Footer and running `npm run evidence:pooling && npm run briefing -- --product=shellfish`.

Per-Analyte Sections

Pb

Published HMTc threshold: 1500 ppb [as-sold] · 1500 ppb (pre-rounding) **Cross-basis: ≈ 500 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `regulatory-alignment` **Regulatory ceiling reference:** 1500 ppb (eu2023_pb_bivalve_1500) **Pool composition:** 18 sources contributing · 774 pseudo-samples in pool · 1 sample-level contributors **Bootstrap 95% CI: [771.10, 6306.83] ppb Sensitivity: log-normal 3130.83 · gamma 3032.83 · nonparametric 3057.21 ppb · spread 3% (stable)**

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_pb_bivalve_1500). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the underlying pool is moot — the cap binds. The pooled P97 of 3043.45 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_pb_bivalve_1500). A Daubert challenge on the value reduces to "is eu2023_pb_bivalve_1500 a legitimate regulatory anchor?" — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 1500 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_pb_bivalve_1500 cap. The pooled occurrence distribution from 18 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Reconstruction	Pool share
chouuthe1993-cadmium-american-lobster-belledune (1993) · CA	Cadmium in American lobster (Homarus americanus) from the ar…	cooked lobster tail/claw/knuckle meat, wet weight	5	uniform_max_only	0.7%
fera2015-uk-tds-metals (2015) · GB	Total Diet Study of metals and other elements in food	shellfish composite, as consumed/as prepared	1	single_point	0.1%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood P…	bivalve molluscs, wet weight	897	percentile_interpolation	50.0%
garofalo2025-seafood-cd-pb-hg-italy-10yr	Monitoring of Cadmium, Lead, and	crustaceans, wet weight	38	lognormal_mean_max	4.9%

Source	Title	Basis	N used	Recon-struction	Pool share
(2025) · IT, EU	Mercury Levels in Seafood P…				
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in Seafood P…	cephalo-pod mol-luscs, wet weight	650	uniform_ max_only	1.3%
gongora-gomez2017-oysters-gulf-cal-ifornia-metals (2017) · MX	Heavy-metal con-tents in oysters (Cras-sostrea gi-gas) culti-vat…	oyster soft tissue, dry weight	1	single_ point	0.1%
gongora-gomez2018-pen-shell-seasonal-metals (2018) · MX	Seasonal levels of heavy met-als in soft tissue and muscle of…	pen shell soft tissue, wet weight	1	single_ point	0.1%
helcom2017-core-indi-cator-met-als-baltic (2017) · EU, DE, DK, EE, FI, LV, LT, PL, RU, SE	Metals (lead, cad-mium and mercury) — HEL-COM core indicator r…	mussel soft body, wet weight	5	uniform_ max_only	0.7%
macdonald2023-fss-uk-fish-ery-cont-aminants-review (2025) · GB, EU	Occurrence of chem-ical cont-aminants in wild-caught fishery p…	mussel whole weight/wet weight	5	uniform_ max_only	0.7%

Source	Title	Basis	N used	Recon-struction	Pool share
nielsen1975-nz-molluscs-heavy-metals (1975) · NZ	Heavy metal levels in New Zealand molluscs	mollusc whole soft tissue, wet weight	5	uniform_max_only	0.7%
onyegeme-okerenta2023-shellfish-nigeria (2023) · NG	Potential Toxic elements in shellfish from three rivers in N…	shellfish tissue, dry weight	5	uniform_max_only	0.7%
rajeshkumar2018-taihu-lake-fish-oyster-metals (2018) · CN	Studies on seasonal pollution of heavy metals in water, sedi…	oyster whole body, wet weight	4	lognormal_mean_only	0.5%
raju2021-bivalves-cagayan-philippines (2021) · PH	Heavy Metal Determination of Bivalves in Cagayan Valley, Phi…	bivalve soft tissue, dry weight	5	uniform_max_only	0.7%
romero-esteves2020-bivalves-santa-rosa-ecuador (2020) · EC, EU	An Overview of Cadmium, Chromium, and Lead Content in Bivalv…	bivalve soft tissue, fresh weight	50	uniform_max_only	1.3%
sepulveda2020-callista-clam-metals (2020) · MX	Trace metals in two wild populations of the squalid callista…	clam tissue, dry weight	1	single_point	0.1%

Source	Title	Basis	N used	Reconstruction	Pool share
shue2012-bivalves-tapeng-taiwan (2012) · TW	Seasonal variations of heavy metals in the bivalve's mollusk...	bivalve soft tissue, dry weight	72	lognormal_mean_max	9.3%
singhato2025-thailand-seafood-heavy-metals (2025) · TH	Risk Assessment of Toxic Heavy Metal Exposure in Selected Se...	shellfish/crustacean/cephalopod edible tissue, fresh weight	5	uniform_max_only	0.7%
valencia2021-canacao-bay-oyster-pb-cd (2021) · PH	Heavy metal accumulation and risk assessment of lead and cad...	oyster soft tissue, fresh weight	214	lognormal_mean_max	27.7%

iAs

Published HMTc threshold: 200 ppb [as-sold] · 109.487 ppb (pre-rounding) **Cross-basis: ≈ 20 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 4 sources contributing · 34 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [27.13, 221.49] ppb Sensitivity: log-normal 69.42 · gamma 162.83 · nonparametric 88.98 ppb · spread 57% (needs_review)**

Defensibility. Pooled P97 from 4 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 4 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 200 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 4 contributing sources across multiple jurisdictions. Sensitivity band needs_review.

Source	Title	Basis	N used	Recon-struction	Pool share
fera2015-uk-tds-metals (2015) · GB	Total Diet Study of metals and other elements in food	shellfish composite, as consumed/as prepared	1	single_point	2.9%
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	crab/lobster shellfish subset, whole weight/wet weight	5	uniform_max_only	14.7%
munoz1999-seafood-inorganic-arsenic (1999) · ES	Rapid and quantitative release, separation and determination…	shellfish soft tissue, wet weight	11	lognormal_mean_median	32.4%
schoof2007-seafood-arsenic-speciation (2007) ·	Variation of total and speciated arsenic in commonly consume…	seafood soft tissue, wet weight	120	lognormal_mean_max	50.0%

Source	Title	Basis	N used	Recon-struction	Pool share
US, ES, NO, TH					

tAs

Published HMTc threshold: adjudication pending — held provisional (wrong-analyte-not-material) · 33023.140 ppb (pre-rounding) **Cross-basis: ≈ 5000 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 6 sources contributing · 20 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [2153.06, 75145.79] ppb Sensitivity: log-normal 24468.63 · gamma 13579.50 · nonparametric 27635.79 ppb · spread 51% (needs_review)**

Defensibility. Pooled P97 from 6 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 6 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 10000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 6 contributing sources across multiple jurisdictions. Sensitivity band needs_review.

Source	Title	Basis	N used	Recon-struction	Pool share
chou-uthe1993-cadmium-american-lobster-belledune (1993) · CA	Cadmium in Ameri-can lobster (Homarus ameri-canus) from the ar…	cooked lobster tail/claw/ knuckle meat, wet weight	5	uniform_ max_only	25.0%
fera2015-uk-tds-metals (2015) · GB	Total Diet Study of metals and other ele-ments in food	shellfish compos-ite, as con-sumed/as prepared	1	single_ point	5.0%
gongora-gomez2017-oysters-gulf-cal-ifornia-metals (2017) · MX	Heavy-metal con-tents in oysters (Cras-sostrea gi-gas) culti-vat…	oyster soft tissue, dry weight	1	single_ point	5.0%
macdonald2025-fss-uk-fish-ery-cont-aminants-review (2025) · GB, EU	Occurrence of chem-ical cont-aminants in wild-caught fishery p…	mussel whole weight/wet weight	5	uniform_ max_only	25.0%
onyegeme-oke2023-shellfish-nigeria (2023) · NG	Potential Toxic ele-ments in shellfish from three rivers in N…	shellfish tissue, dry weight	5	uniform_ max_only	25.0%
singhato2025-thailand-seafood-heavy-met-als (2025) · TH	Risk As-sessment of Toxic Heavy Metal Ex-posure in Selected Se…	shell-fish/crus-tacean/ cephalo-pod ed-ible tis-sue, fresh weight	3	lognormal_ mean_only	15.0%

Ni

Published HMTc threshold: 10000 ppb [as-sold] · 12189.015 ppb (pre-rounding) **Cross-basis: ≈ 2000 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 5 sources contributing · 102 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [6973.54, 17107.47] ppb Sensitivity: log-normal 13165.89 · gamma 9309.15 · nonparametric 11084.56 ppb · spread 29% (provisional)**

Defensibility. Pooled P97 from 5 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is provisional — log-normal, gamma, and nonparametric variants disagree by more than 25%, signaling heterogeneity that should be resolved before final publication. Likely sources: basis mixing, regional outliers, or sparse pseudo-sample contributions.

Brand legal read. The value is pooled P97 from 5 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 10000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 5 contributing sources across multiple jurisdictions. Sensitivity band provisional.

Source	Title	Basis	N used	Reconstruction	Pool share
de-anda-monitoring-baja-red-crab-el-	Spatial and Temporal Variabil-	whole adult red	5	uniform_max_only	4.9%

Source	Title	Basis	N used	Recon-struction	Pool share
ements (2023) · MX	ity of Trace and Macro Elements…	crab, dry weight			
gongora-gomez2017-oysters-gulf-california-metals (2017) · MX	Heavy-metal contents in oysters (Crassostrea gigas) cultivat…	oyster soft tissue, dry weight	1	single_point	1.0%
onyegeme-okerenta2023-shellfish-nigeria (2023) · NG	Potential Toxic elements in shellfish from three rivers in N…	shellfish tissue, dry weight	5	uniform_max_only	4.9%
shue2012-bivalves-tapeng-taiwan (2012) · TW	Seasonal variations of heavy metals in the bivalve's mollusk…	bivalve soft tissue, dry weight	72	lognormal_mean_max	50.0%
yap-almutairi2022-nickel-seafood-malaysia (2022) · MY	Comparative Study of Potentially Toxic Nickel and Their Pote…	mollusc soft tissue, wet weight	40	lognormal_mean_max	39.2%

Cd

Published HMTc threshold: 1000 ppb [as-sold] · 1000 ppb (pre-rounding) **Cross-basis: ≈ 200 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `regulatory-alignment` **Regulatory ceiling reference:** 1000 ppb (eu2023_cd_bivalve_1000) **Pool composition:** 20 sources contributing · 2165 pseudo-samples in pool · 3 sample-level contributors **Bootstrap 95% CI: [803.94, 8521.40] ppb Sensitivity: log-normal 5370.16 · gamma 5374.83 · nonparametric 5625.30 ppb · spread 5% (stable)**

Defensibility. The published value matches the lowest applicable regulatory ceiling (eu2023_cd_bivalve_1000). This is the strongest defense vector: HMTc certifies at the binding government limit, regardless of where the pooled literature value falls. Any attack on the underlying pool is moot — the cap binds. The pooled P97 of 5152.92 ppb sits above the cap, confirming the cap is the protective constraint.

Brand legal read. This threshold matches the binding government cap (eu2023_cd_bivalve_1000). A Daubert challenge on the value reduces to "is eu2023_cd_bivalve_1000 a legitimate regulatory anchor?" — and the answer is yes, it is published, finalized, and enforced.

Retailer compliance read. A certified product tests at or below 1000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. HMTc threshold = eu2023_cd_bivalve_1000 cap. The pooled occurrence distribution from 20 sources confirms the cap is the protective constraint; the cap binds at a value below where the pooled P97 falls.

Source	Title	Basis	N used	Reconstruction	Pool share
chouuthe1993-cadmium-american-lobster-belledune (1993) · CA	Cadmium in American lobster (Homarus americanus) from the ar…	cooked lobster tail/claw/knuckle meat, wet weight	5	uniform_max_only	0.2%
de-anda-montanez2023-baja-red-crab-elements (2023) · MX	Spatial and Temporal Variability of Trace and Macro Elements…	whole adult red crab, dry weight	5	uniform_max_only	0.2%
fera2015-uk-tds-metals (2015) · GB	Total Diet Study of metals and other ele-	shellfish composite, as consumed/as prepared	1	single_point	0.1%

Source	Title	Basis	N used	Recon-struction	Pool share
	ments in food				
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in Seafood P…	cephalo-pod mol-luscs, wet weight	1500	percentile_ interpola-tion	23.1%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in Seafood P…	bivalve molluscs, wet weight	904	percentile_ interpola-tion	23.1%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitor-ing of Cad-mium, Lead, and Mercury Levels in Seafood P…	crus-taceans, wet weight	424	percentile_ interpola-tion	19.6%
gongora-gomez2017-oysters-gulf-cal-ifornia-metals (2017) · MX	Heavy-metal con-tents in oysters (Cras-sostrea gi-gas) culti-vat…	oyster soft tissue, dry weight	1	single_ point	0.1%
gongora-gomez2018-pen-shell-seasonal-metals (2018) · MX	Seasonal levels of heavy met-als in soft tissue and muscle of…	pen shell soft tissue, wet weight	1	single_ point	0.1%
helcom2017-core-indi-cator-met-	Metals (lead, cad-mium and	mussel soft body, wet weight	5	uniform_ max_only	0.2%

Source	Title	Basis	N used	Reconstruction	Pool share
als-baltic (2017) · EU, DE, DK, EE, FI, LV, LT, PL, RU, SE	mercury) — HELCOM core indicator r…				
lordan2022-brown-crab-cadmium-review (2022) · IE, EU, NO, GB, PT, CN, HK, US, KR	Cadmium: A Focus on the Brown Crab (Cancer pagurus) Industry…	brown crab white meat, wet weight	5	uniform_max_only	0.2%
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	brown crabmeat and products, wet weight	399	lognormal_mean_median	18.4%
nielsen1975-nz-molluscs-heavy-metals (1975) · NZ	Heavy metal levels in New Zealand molluscs	mollusc whole soft tissue, wet weight	5	uniform_max_only	0.2%
onyegeme-okerenta2023-shellfish-nigeria (2023) · NG	Potential Toxic elements in shellfish from three rivers in N…	shellfish tissue, dry weight	5	uniform_max_only	0.2%
rajeshkumar2018-taihu-lake-fish-oyster-metals (2018) · CN	Studies on seasonal pollution of heavy metals in water, sedi…	oyster whole body, wet weight	4	lognormal_mean_only	0.2%

Source	Title	Basis	N used	Recon-struction	Pool share
raju2021-bivalves-cagayan-philippines (2021) · PH	Heavy Metal Determination of Bivalves in Cagayan Valley, Phi…	bivalve soft tissue, dry weight	5	uniform_max_only	0.2%
romero-esteves2020-bivalves-santa-rosa-ecuador (2020) · EC, EU	An Overview of Cadmium, Chromium, and Lead Content in Bivalv…	bivalve soft tissue, fresh weight	50	uniform_max_only	0.5%
sepulveda2020-callista-clam-metals (2020) · MX	Trace metals in two wild populations of the squalid callista…	clam tissue, dry weight	1	single_point	0.1%
shue2012-bivalves-tapeng-taiwan (2012) · TW	Seasonal variations of heavy metals in the bivalve's mollusk…	bivalve soft tissue, dry weight	72	lognormal_mean_max	3.3%
singhato2025-thailand-seafood-heavy-metals (2025) · TH	Risk Assessment of Toxic Heavy Metal Exposure in Selected Se…	shellfish/crustacean/cephalopod edible tissue, fresh weight	3	lognormal_mean_max	0.1%
valencia2021-canacao-bay-oyster-pb-cd (2021) · PH	Heavy metal accumulation and risk assessment of lead and cad…	oyster soft tissue, fresh weight	214	lognormal_mean_max	9.9%

Al

Published HMTc threshold: *data gap* **Rationale tag:** `no-data` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 0 sources contributing · 0 pseudo-samples in pool · 0 sample-level contributors **no_data**

Defensibility. This analyte has no defensible threshold today. The pool is empty; no read-across is available; no regulatory cap applies. The HMTc certification mark on this product does NOT cover this analyte until evidence is loaded. This is the honest position; we explicitly do not pretend coverage we cannot defend.

Brand legal read. The value is pooled P97 from 0 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below no published value (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 0 contributing sources across multiple jurisdictions. Sensitivity band no_data.

tHg

Published HMTc threshold: 500 ppb [as-sold] · 376.492 ppb (pre-rounding) **Cross-basis: ≈ 100 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** 500 ppb (eu2023_hg_crustaceans_molluscs_500) **Pool composition:** 15 sources contributing · 1260 pseudo-samples in pool · 1 sample-level contributors **Bootstrap 95% CI: [133.06, 817.96] ppb Sensitivity: log-normal 392.21 · gamma 403.85 · nonparametric 372.65 ppb · spread 8% (stable)**

Defensibility. Pooled P97 from 15 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination below the pooled value (cap = 500 ppb). The pooled value

derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 15 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 500 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 15 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
bjerregaard2023-mercury-mussels-denmark (2023) · DK, EU	Is Search of Mercury Lost from Sediments in a Previously Con…	dry_as_sold	137	lognormal_mean_only	10.9%
bjerregaard2023-mercury-mussels-denmark (2023) · DK, EU	Is Search of Mercury Lost from Sediments in a Previously Con…	blue mussel soft tissue, dry weight	20	lognormal_mean_max	1.6%
chouuthe1993-cadmium-american-lobster-belledune (1993) · CA	Cadmium in American lobster (Homarus americanus)	cooked lobster tail/claw/knuckle meat, wet weight	5	uniform_max_only	0.4%

Source	Title	Basis	N used	Recon-struction	Pool share
	from the ar…				
fera2015-uk-tds-metals (2015) · GB	Total Diet Study of metals and other elements in food	shellfish composite, as consumed/as prepared	1	single_point	0.1%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood P…	crustaceans, wet weight	60	percentile_interpolation	4.8%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood P…	bivalve molluscs, wet weight	898	lognormal_mean_max	39.7%
garofalo2025-seafood-cd-pb-hg-italy-10yr (2025) · IT, EU	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood P…	cephalopod molluscs, wet weight	687	lognormal_mean_max	39.7%
gongora-gomez2017-oysters-gulf-california-metals (2017) · MX	Heavy-metal contents in oysters (Crassostrea gigas) cultivat…	oyster soft tissue, dry weight	1	single_point	0.1%
gongora-gomez2018-pen-shell-seasonal-	Seasonal levels of heavy metals in soft	pen shell soft tissue, dry weight	1	single_point	0.1%

Source	Title	Basis	N used	Reconstruction	Pool share
metals (2018) · MX	tissue and muscle of…				
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web …	blue mussel soft tissue, wet weight	12	lognormal_mean_median	0.9%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccumulation of methylmercury within the marine food web …	lobster tissue, wet weight	9	lognormal_mean_median	0.7%
macdonald2025-fss-uk-fishery-contaminants-review (2025) · GB, EU	Occurrence of chemical contaminants in wild-caught fishery p…	shellfish whole weight/wet weight	5	uniform_max_only	0.4%
nielsen1975-nz-molluscs-heavy-metals (1975) · NZ	Heavy metal levels in New Zealand molluscs	mollusc whole soft tissue, wet weight	1	single_point	0.1%
pawlaczyk2020-mercury-canned-fish-poland (2020) · PL	Risk of Mercury Ingestion from Canned Fish in Poland	canned octopus drained product, wet/as consumed	3	lognormal_mean_only	0.2%
singhato2025-thailand-seafood-heavy-metals (2025) · TH	Risk Assessment of Toxic Heavy Metal Exposure in Selected Se…	shellfish/crustacean/cephalopod edible tissue, fresh weight	5	uniform_max_only	0.4%

MeHg

Published HMTc threshold: 200 ppb [as-sold] · 140.720 ppb (pre-rounding) **Cross-basis: ≈ 25 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 4 sources contributing · 22 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [9.44, 302] ppb Sensitivity: log-normal 140.72 · gamma 140.72 · nonparametric 144.30 ppb · spread 3% (stable)**

Defensibility. Pooled P97 from 4 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 4 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 200 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 4 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/seafood consump-	lobster, fish/seafood occurrence input, wet weight/as reported	1	single_point	4.5%

Source	Title	Basis	N used	Recon-struction	Pool share
	tion com-par…				
efsa-sc2015-fish-seafood-mehg-risk-benefit (2015) · EU	Statement on the benefits of fish/seafood consump-tion com-par…	squid, fish/seafood oc-currence input, wet weight/as reported	1	single_point	4.5%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccu-mulation of methylmer-cury within the marine food web …	blue mus-sel soft tis-sue, wet weight	12	lognormal_mean_me-dian	50.0%
harding2018-bay-fundy-methylmercury (2018) · CA, US	Bioaccu-mulation of methylmer-cury within the marine food web …	lobster tis-sue, wet weight	9	lognormal_mean_me-dian	40.9%

Adversarial Read – what a plaintiff's expert will attack, and the response

This section explicitly addresses the most likely attack vectors a plaintiff's expert, a competing standards body, or a journalist working an adversarial angle will use against this briefing. Each attack is named and answered.

Attack 1: "The pool is too small; the methodology cherry-picks." Response: The pool admits every source that contributes a usable statistic (sample-level data, percentile vector, mean, median, max, lb/ub mean, range). No source is excluded for being "summary-only." The contribution log per analyte lists every admission decision and every basis conversion. Per-source pool share is capped at 50% to prevent dominance by any single survey. The methodology is repro-

ducible from the structured CSVs in `data/evidence/` and the engine in `tools/evidence/build-hmtc-aggregate-distributions.mjs`.

Attack 2: "Why P97 instead of P90? You're loosening the standard." Response: P97 per analyte is calibrated to deliver approximately 90% row-pass for the clean subcategory across the 10-analyte joint. Setting each analyte's limit at P90 of its own marginal would deliver a joint pass rate of 35% under independence and 50-70% under realistic correlation — well short of the Governing Principles' 90/30 row-pass target. P97 is the per-analyte percentile that delivers the row-level outcome the program promises. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening.

Attack 3: "Why is your dirty subcategory P45 instead of P10?" Response: P45 of the dirty subcategory's distribution, applied per-analyte, delivers approximately 30% joint row-pass for dirty-subcategory brands. This is the design intent stated in the Governing Principles v2.0: the dirty subcategory standard must give dirty-platform brands a realistic path to certification so the program drives reformulation rather than excluding the platform. A P10 dirty standard would deliver <1% joint row-pass and the program would have no commercial uptake on the dirty side.

Attack 4: "Your threshold matches the EU/FDA cap; HMTc adds no value over the regulator." Response: For Pb, Cd, the published threshold matches the binding regulatory cap because the pooled P97 sits above the cap. HMTc cannot defensibly publish a tighter standard than the literature supports under its row-pass calibration. The value HMTc adds is row-level certification — a brand must meet every value in the Master Limit Table simultaneously on a like-for-like basis, with documented contribution logs and sensitivity analyses. Single-analyte regulatory caps do not deliver row-level certification.

Attack 5: "Your sensitivity analysis shows the methodology is unstable for some analytes." Response: The sensitivity bands are explicitly reported per analyte. 1 analytes (Ni) are flagged provisional — log-normal, gamma, and nonparametric variants disagree by more than 25%. This is transparent reporting, not a methodological flaw. Provisional values are published with the band stated and a commitment to resolve before final publication. The alternative — silently picking one parametric form — would be the actual methodological flaw.

Attack 7: "Where does basis conversion happen and what authority do you cite?" Response: The 1:7 powder ↔ prepared-for-feeding conversion is the conservative reconstitution ratio (1 g powder + 6 g water = 7 g prepared) standard across infant-formula regulatory frameworks (FDA Closer-to-Zero, EU 2023/915). Within-class conversions (as_sold ↔ dry_as_sold) pass through with factor 1. Every basis conversion is logged per source in the contribution log; unknown-conversion cases are flagged and not silently coerced.

Attack 8: "Show me one source you excluded." Response: No source contributing a usable statistic is excluded. Sources marked "no_usable_statistic" in the contribution log are LOD-censored entries with neither mean nor lb/ub recorded. Sources without a structured CSV row are documented in the routing audit and the gap report; those are extraction gaps, not exclusions, and they are tracked in Phase 1 backlog with target completion timelines.

Sources Cited

Every numeric claim in this briefing traces to one of the source pages below. Full source metadata lives at `wiki/sources/<source-id>.md` in the heavymetalindex.com repository.

Source ID	Title	Citation
`bjerregaard2023-mercury-mussels-denmark` (A)	In Search of Mercury Lost from Sediments in a Previously Contaminated Coastal Area, Harboøre Tange, Denmark	Bjerregaard P; Jensen CL; Juhl AVR; Markussen AJR; Poulsen SR · 2023 · Bulletin of Environmental Contamination and Toxicology · doi:null
`chou-uthe1993-cadmium-american-lobster-belledune` (B)	Cadmium in American lobster (Homarus americanus) from the area of Belledune Harbour, New Brunswick, Canada: 1980-1992 results	Chou CL; Uthe JF · 1993 · Canadian Technical Report of Fisheries and Aquatic Sciences No. 1916 · doi:null
`de-anda-montanez2023-baja-red-crab-elements` (A)	Spatial and Temporal Variability of Trace and Macro Elements in the	Juan Antonio De Anda-Montanez; Tania Zenteno-Savin; Eduardo F.

Source ID	Title	Citation
	Red Crab Pleuroncodes planipes in the Pacific Coast of the Baja California Peninsula, Mexico	Balart; Baudilio Acosta-Vargas; Ramon Gaxiola-Robles; Lia Celina Mendez-Rodriguez · 2023 · Animals · doi:10.3390/ani13050822
`efsa-sc2015-fish-seafood-mehg-risk-benefit` (A)	Statement on the benefits of fish/seafood consumption compared to the risks of methylmercury in fish/seafood	EFSA Scientific Committee · 2015 · EFSA Journal 2015;13(1):3982, 36 pp. · doi:10.2903/j.efsa.2015.3982
`fera2015-uk-tds-metals` (A)	Total Diet Study of metals and other elements in food	Baxter M; Brereton N · 2015 · Food and Environment Research Agency report for the UK Food Standards Agency, Fera report 15/06, project FS102081 · doi:null
`garofalo2025-seafood-cd-pb-hg-italy-10yr` (A)	Monitoring of Cadmium, Lead, and Mercury Levels in Seafood Products: A Ten-Year Analysis	Garofalo L; Sala M; Focardi C; Pasqualetti P; Delfino D; D'Onofrio F; Droghei B; Pasquali F; Nicolini V; Galli FS; Scaramozzino P; Ubaldi A; Russo K; Neri B · 2025 · Foods 14(3):451 · doi:10.3390/foods14030451
`gongora-gomez2017-oysters-gulf-california-metals` (B)	Heavy-metal contents in oysters (Crassostrea gigas) cultivated on the southeastern coast of the Gulf of California, Mexico	Gongora-Gomez Andres Martin; Garcia-Ulloa Manuel; Munoz-Sevilla Norma Patricia; Dominguez-Orozco Ana Laura; Villanueva-Fonseca Brenda Paulina; Hernandez-Sepulveda Juan Antonio; Ortega Izaguirre Rogelio · 2017 · Hidrobiologica · doi:10.24275/uam/izt/dcbs/hidro/2017v27n2/garcia

Source ID	Title	Citation
`gongora-gomez2018-pen-shell-seasonal-metals` (B)	Seasonal levels of heavy metals in soft tissue and muscle of the pen shell Atrina maura from a farm in the southeastern coast of the Gulf of California, Mexico	Gongora-Gomez Andres Martin; Dominguez-Orozco Ana Laura; Villanueva-Fonseca Brenda Paulina; Munoz-Sevilla Norma Patricia; Garcia-Ulloa Manuel · 2018 · Revista Internacional de Contaminacion Ambiental · doi:10.20937/rica.2018.34.01.05
`harding2018-bay-fundy-methylmercury` (A)	Bioaccumulation of methylmercury within the marine food web of the outer Bay of Fundy, Gulf of Maine	Harding G; Dalziel J; Vass P · 2018 · PLoS ONE · doi:10.1371/journal.pone.0197220
`helcom2017-core-indicator-metals-baltic` (A)	Metals (lead, cadmium and mercury) — HELCOM core indicator report	HELCOM; Zalewska T; Larsen MM; Fryer R; Danielsson S; Nyberg E · 2017 · HELCOM Core Indicator Report (HOLAS II component), ISSN 2343-2543 · doi:null
`lordan2022-brown-crab-cadmium-review` (A)	Cadmium: A Focus on the Brown Crab (Cancer pagurus) Industry and Potential Human Health Risks	Ronan Lordan; Ioannis Zabetakis · 2022 · Toxics · doi:10.3390/toxics10100591
`macdonald2025-fss-uk-fishery-contaminants-review` (A)	Occurrence of chemical contaminants in wild-caught fishery products of relevance to Scottish and wider UK Fishing Waters: A Review	MacDonald S; Stead S; Rose M; Boss K; Campbell K · 2025 · Fera Science Ltd report to Food Standards Scotland (Report FR/002826) · doi:null
`munoz1999-seafood-inorganic-arsenic` (A)	Rapid and quantitative release, separation and determination of inorganic arsenic [As(III)+As(V)] in seafood products by microwave-assisted distillation and hydride generation atomic absorption spectrometry	Munoz O; Velez D; Cervera ML; Montoro R · 1999 · Journal of Analytical Atomic Spectrometry · doi:10.1039/a904999a

Source ID	Title	Citation
`nielsen1975-nz-molluscs-heavy-metals` (B)	Heavy metal levels in New Zealand molluscs	Nielsen SA; Nathan A · 1975 · New Zealand Journal of Marine and Freshwater Research 9(4), 467-481·doi:10.1080/00288330.1975.951
`onyegeme-okerenta2023-shellfish-nigeria` (A)	Potential Toxic elements in shellfish from three rivers in Niger Delta, Nigeria: bioaccumulation, dietary intake, and human health risk assessment	Onyegeme-Okerenta BM; West LO · 2023 · Environmental Analysis, Health and Toxicology · doi:10.5620/eaht.2023015
`pawlaczyk2020-mercury-canned-fish-poland` (A)	Risk of Mercury Ingestion from Canned Fish in Poland	Pawlaczyk A; Przerywacz A; Gajek M; Szynkowska-Jozwik MI · 2020 · Molecules · doi:10.3390/molecules25245884
`rajeshkumar2018-taihu-lake-fish-oyster-metals` (A)	Studies on seasonal pollution of heavy metals in water, sediment, fish and oyster from the Meiliang Bay of Taihu Lake in China	Rajeshkumar S; Liu Y; Zhang X; Ravikumar B; Bai G; Li X · 2018 · Chemosphere · doi:10.1016/j.chemosphere.2017.10.078
`raju2021-bivalves-cagayan-philippines` (B)	Heavy Metal Determination of Bivalves in Cagayan Valley, Philippines	Raju J; Johnson J; Jeevan A; Capili JT · 2021 · Scholars Academic Journal of Biosciences 9(10):256-258 · doi:10.36347/sajb.2021.v09i10.001
`romero-estevez2020-bivalves-santa-rosa-ecuador` (B)	An Overview of Cadmium, Chromium, and Lead Content in Bivalves Consumed by the Community of Santa Rosa Island (Ecuador) and Its Health Risk Assessment	Romero-Estévez D; Yánez-Jácome GS; Dazzini Langdon M; Simbaña-Farinango K; Rebolledo Monsalve E; Durán Cobo G; Navarrete H · 2020 · Frontiers in Environmental Science 8:134 · doi:10.3389/fenvs.2020.00134
`schoof2007-seafood-arsenic-speciation` (B)	Variation of total and speciated arsenic in commonly consumed fish and seafood	Schoof RA; Yager JW · 2007 · Human and Ecological Risk Assessment·doi:10.1080/108

Source ID	Title	Citation
`sepulveda2020-callista-clam-metals` (B)	Trace metals in two wild populations of the squalid callista clam (Megapitaria squalida) in the southeastern Gulf of California, Mexico	Sepulveda Carlos Humberto; Gongora-Gomez Andres Martin; Perez-Alvarez Sandra; Rodriguez-Gonzalez Hervey; Munoz-Sevilla Norma Patricia; Villanueva-Fonseca Brenda Paulina; Hernandez-Sepulveda Juan Antonio; Garcia-Ulloa Gomez Manuel · 2020 · Revista Internacional de Contaminacion Ambiental · doi:10.20937/rica.53565
`shue2012-bivalves-tapeng-taiwan` (B)	Seasonal variations of heavy metals in the bivalve's mollusks from Ta-Peng Bay lagoon in southwestern Taiwan	Shue MF; Chen JJ; Tseng LH; Lai WL · 2012 · Advanced Materials Research · doi:10.4028/www.scientific.net/AMR.599.100
`singhato2025-thailand-seafood-heavy-metals` (A)	Risk Assessment of Toxic Heavy Metal Exposure in Selected Seafood Species from Thailand	Alongkote Singhato; Narisa Rueangsri; Purimprat Thanaratsotornkun; Konpong Boonyingsathit; Piyanut Sridonpai; Nunnapus Laitip; Nattikarn Ornthai; Kunchit Judprasong · 2025 · Foods · doi:10.3390/foods14213725
`valencia2021-canacao-bay-oyster-pb-cd` (B)	Heavy metal accumulation and risk assessment of lead and cadmium in cultured oysters (Crassostrea iredalei) of Cañacao Bay, Philippines	Valencia HB; Caballar EJM; Dioneda SCC; Gomez JAE; Obanan SP · 2021 · Sustinere: Journal of Environment and Sustainability 5(2), 64-78 · doi:10.22515/sustinerejes.v5i2.159
`yap-almutairi2022-nickel-seafood-malaysia` (B)	Comparative Study of Potentially Toxic Nickel and Their Potential Human Health Risks in Seafood (Fish and Mol-	"Yap; Chee Kong"; "Al-Mutairi; Khalid Awadh" · 2022 · Biology · doi:10.3390/biology11030376

Source ID	Title	Citation
	lusks) from Peninsular Malaysia	

In addition to occurrence sources, this briefing cites:

- `docs/governance/hmtc-governing-principles.md` (v2.0, May 2026) — the five governing principles and the locked P97/P45 methodology
- `wiki/regulations/efsa-methylmercury-twi.md` — EFSA TWI 1.3 µg/kg-bw/week for MeHg
- `wiki/regulations/epa-iris-methylmercury-rfd.md` — EPA IRIS RfD 0.1 µg/kg-bw/day for MeHg
- `wiki/regulations/jecfa-methylmercury-ptwi.md` — JECFA PTWI 1.6 µg/kg-bw/week for MeHg
- WHO/JECFA pooled total-diet-study methodology
- EFSA Scientific Opinions on Pb (2010, 2024) and Cd (2009, 2024)
- ATSDR moment-matching guidance for summary-statistic sources
- Codex Alimentarius CAC/GL 50-2004 weighted pooling

Provenance Footer

This briefing is reproducible by:

1. Checking out commit `e51473a013e92827fd88e3716dd7713928984779` in the heavymetalindex.com repository.
2. Running `npm run evidence:pooling` to regenerate `data/evidence/hmtc_aggregate_distributions.csv`.
3. Running `npm run briefing -- --product=shellfish` to regenerate this document.

Field	Value
Generated at	2026-06-18T13:34:12.002Z
Generator	`tools/standards/build-standards-briefing.mjs`
Pooling engine	`tools/evidence/build-hmtc-aggregate-distributions.mjs`
Methodology version	HMTc-10 / P97-P45 per CLAUDE.md Part 19 / Governing Principles v2.0
Git commit	`e51473a013e92827fd88e3716dd7713928984779`

Field	Value
Author	Karen Pendergrass, Standards Architect, Paleo Foundation
Status	Draft for stakeholder review

Disclosure: organizational role

Karen Pendergrass also serves as CEO of the Paleo Foundation, which operates the HMT&C certification program. This dual role is disclosed for transparency; the methodology in this briefing is reproducible from the open-source pooling engine and source data, independent of the author's organizational title.

Methodology implementation note

Methodology implementation: deterministic open-source pooling engine and document generator (this file's repository at the commit named in the Provenance Footer). Engine code and document templates were drafted with AI assistance (Claude, Anthropic) under the supervision and direction of the Standards Architect. All policy decisions, methodology choices, source-admission rules, basis conventions, and final values are the Standards Architect's; the AI assistance is a calculator-and-drafter, not a co-author.

End of briefing.

3.5 Seaweed and Kelp Foods (nori, wakame, kombu, dulse)

One of 5 per-product chapters in Part 3, self-contained as a licensing exhibit. The methodology is in Part 5.1; the certification status framework in Part 2.3; the Standards Ratchet in Part 2.10. Held/withdrawn analytes (e.g. total arsenic in fish and seaweed, which is arsenosugar/arsenobetaine-dominated — inorganic arsenic is the toxic and regulated analyte) are shown as "adjudication pending" and carry no published threshold; the published values are validated by the Publication QC Gate (no value exceeds a regulatory ceiling).

Product class: Seaweed/kelp foods (nori, wakame, kombu, dulse — as food products) **Subcategory role:** whole_row_platform **Native basis for published thresholds:** as_sold **Methodology:** HMTc-10 analyte panel; per-analyte standards derived from pooled occurrence distribution at P97 (clean / independent rows) or P45 (dirty rows), capped at the lowest applicable regulatory ceiling. See "Methodology in one page" below. **Date generated:** 2026-06-18 **Author:** Karen Pendergrass, Standards Architect, Paleo Foundation **Status:** Draft for stakeholder review. Not yet through the 9 pre-publication gates.

Who this briefing is for

The HMTc certification mark serves multiple audiences with different concerns. This briefing is written so each audience finds what they need in one document. Audience-tagged "read" callouts appear inline next to each per-analyte section.

Brand legal teams — Read the per-analyte Defensibility narrative and the Adversarial read section. The methodology section is your Daubert-resistance reference. Every numeric claim traces to a source page; the contribution logs document admission criteria.

Retailer quality and compliance — Read the Master Limit Table and the per-analyte Regulator read callouts. The certification mark on a product means the brand has tested at or below every value in the Master Limit Table, in the product's native basis. The regulatory comparison rows show which government caps apply and how HMTc thresholds compare.

Regulators (FDA, EFSA, state AGs) — Read the Methodology section, the Sources Cited table, and the Anti-Circumvention argument. The methodology is open, reproducible, and follows the WHO/JECFA, EFSA, ATSDR, and Codex precedents cited. Numeric thresholds are derived from a pooled occurrence distribution of peer-reviewed and agency data, weighted by source N with a 50% relative-share cap.

Adversarial readers (plaintiff's experts, competing standards bodies) — Read the per-analyte Defensibility narrative, the Adversarial read section, the contribution logs, and the sensitivity analyses. Every value carries: the pool composition, the basis-conversion math, the regulatory cap comparison, the parametric sensitivity, and the rationale for any gap between literature and threshold.

Master Limit Table – Executive Summary

All ten HMTc analytes for this product class. Each row shows the published HMTc threshold (rounded up to the primary increment ladder), the precise pooled-engine value before rounding, the rationale tag, the contributing source count, the parametric-sensitivity band, and the lowest applicable regulatory ceiling. Brands meeting every value in this table on a like-for-like basis qualify for the HMTc certification mark for seaweed kelp foods.

Analyte	HMTc Threshold (published)	Pooled value (precise, pre-rounding)	Rationale	Sources	Sensitivity	Reg cap
Pb	**10000 ppb** [as-sold]≈ 2000 ppb [prepared]	13020.408 ppb	literature-baseline	4	needs_review	—
iAs	**2000 ppb** [as-sold]≈ 500 ppb [prepared]	1829.114 ppb	literature-baseline	4	stable	—
tAs	*adjudication pending* ≈ 1000 ppb [prepared]	5516.476 ppb	literature-baseline	3	stable	—
Ni	*data gap*	—	no-data	0	no_data	—
Cd	**2000 ppb** [as-sold]≈ 500 ppb [prepared]	1874.087 ppb	literature-baseline	5	stable	—
tHg	**100 ppb** [as-sold]≈ 10 ppb [prepared]	62.916 ppb	literature-baseline	1	stable	—

How to read this table. "Published" values are rounded up to the HMTc primary increment ladder (1, 2, 3, 5, 10, 15, 20, 25, 50, 100, 200, 500, 1000, 1300, 1500, 2000, 5000, 10000 ppb) per the program's publication conventions; this gives brands an unambiguous bin to test against. The "precise" column shows the pooled-engine output before rounding so the methodology is transparent and re-derivable. Rationale

tags map to CLAUDE.md Part 19′s four allowed tags: `literature-baseline` (pooled value binds), `regulatory-alignment` (regulatory ceiling binds), `feasibility-driven` (analytical method binds), and `precautionary` (manual override below pooled and regulatory).

Methodology – one page

Per-analyte standards. For each (product row, analyte) cell, every source loaded for that cell contributes to a pooled distribution via moment-matching pseudo-sample expansion. Sample-level data uses raw values; summary statistics (mean, median, max, lb/ub, reported percentiles) reconstruct under log-normal or Tobit assumptions documented per source. Pseudo-samples are weighted by source N and capped at 50% relative pool share to prevent any one survey from dominating. The pooled distribution is sorted and queried at the per-analyte target percentile:

- **Clean subcategory:** standard = pooled P97
- **Dirty subcategory:** standard = pooled P45
- **Independent (no within-pair partner):** standard = pooled P97 by default

Why P97 / P45, not P90 / P30 (the row-pass calibration). A brand certifies a product (one row, ten analytes simultaneously) — not one analyte at a time. Setting each per-analyte limit at the raw P90 of the clean distribution would deliver a joint row-pass rate close to 0.9^10 = 35% under analyte independence. Under realistic analyte correlation (0.6, since supply-chain hygiene drives multiple metals together), the joint rate sits around 50-70%. To deliver the row-level outcomes the Governing Principles require — approximately 90% of clean-subcategory brands and approximately 30% of dirty-subcategory brands pass row certification — per-analyte limits are uplifted to P97 (clean) and P45 (dirty). This is the calibration that delivers the program's stated objective. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening; per-analyte differentiation (e.g., Pb tighter than Ni) is deferred until HMTc has 6-12 months of lot-testing data to measure actual joint pass rates.

Basis. Each product row's standard is published in the native basis of the product form: powder rows in powder-as-placed-on-market, ready-to-feed liquid rows in as-consumed, dry cereal rows in dry-as-sold, puree rows in as-consumed. Brands ship and test in their product's native form; the certification mark matches that form. Sources reporting in non-native bases are converted into the row's native basis during pooling via documented factors (the 1:7 reconstitution ratio for powder ↔ prepared; within-class identity for as_sold ↔ dry_as_sold). The cross-basis equivalent of every published value is shown in the Master Limit Table for ease of reference by audiences operating in different bases.

Regulatory cap. The final published HMTc value never exceeds the lowest applicable loaded regulatory ceiling, expressed in the product's native basis. When the pooled P97/P45 exceeds the regulatory cap, the cap binds and the rationale tag is `regulatory-alignment`. When the pooled value is below the cap, the rationale is `literature-baseline` and the cap is shown for context only.

Sensitivity. Each pooled value is recomputed under three parametric assumptions: log-normal (headline), gamma (alternative), and nonparametric reconstruction. The spread across the three is reported as the sensitivity band: stable (≤25%), provisional (25–50%), or needs_review (>50%). Confidence intervals are computed via meta-analytic bootstrap that resamples whole sources (not individual pseudo-samples) to capture between-source uncertainty.

Methodology authorities. The pooling approach follows WHO/JECFA pooled total-diet study methodology, the EFSA Scientific Opinions on lead (2010, 2024) and cadmium (2009, 2024) for log-normal contaminant occurrence reconstruction, ATSDR moment-matching guidance for summary-statistic-only sources, and Codex CAC/GL 50-2004 weighted pooling. Per-source contribution logs document every admission decision and every basis conversion.

Tooling and reproducibility. The pooling engine is `tools/evidence/build-hmtc-aggregate-distributions.mjs` (1170 lines of deterministic JavaScript, 37 unit tests). The briefing generator is `tools/standards/build-standards-briefing.mjs` (this file). Both are versioned in git and re-runnable. Every numeric claim in this briefing can be regenerated by checking out

the commit named in the Provenance Footer and running `npm run evidence:pooling && npm run briefing -- --product=seaweed-kelp-foods`.

Per-Analyte Sections

Pb

Published HMTc threshold: 10000 ppb [as-sold] · 13020.408 ppb (pre-rounding) **Cross-basis: ≈ 2000 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 4 sources contributing · 20 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [289.28, 23122.38] ppb Sensitivity: log-normal 5321.44 · gamma 11182.69 · nonparametric 9037.16 ppb · spread 52% (needs_review)**

Defensibility. Pooled P97 from 4 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 4 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 10000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 4 contributing sources across multiple jurisdictions. Sensitivity band needs_review.

Source	Title	Basis	N used	Recon-struction	Pool share
aziz2021-gracilaria-seaweed-bangladesh (2021) · BD	Poten-tial of Gracilaria tenuistip-itata var. liui grown in Nu…	dried Gracilaria, dry weight	3	lognormal_mean_only	15.0%
martinleon2021-edible-sea-weeds-ca-nary-met-als-nitrates (2021) · ES, EU, JP, KR	Human Exposure to Toxic Metals (Cd, Pb, Hg) and Nitrates (NO…	edible al-gae, dry weight	4	lognormal_mean_only	20.0%
rakib2021-macroalgae-bangladesh-metals (2021) · BD	Macroal-gae in bio-monitoring of metal pollution in the Bay of…	macroal-gae, dry weight	10	lognormal_mean_max	50.0%
razi-parjikolaei2016-red-macroal-gae-met-als (2016) · DK	Valuable Biomole-cules from Nine North Atlantic Red Macroalga…	red macroal-gae, dry matter	3	lognormal_mean_only	15.0%

iAs

Published HMTc threshold: 2000 ppb [as-sold] · 1829.114 ppb (pre-rounding) **Cross-basis: ≈ 500 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 4 sources contributing · 52 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [519.08, 1954.83] ppb Sensitivity: log-normal 1889.06 · gamma 1690.20 · nonparametric 1766.73 ppb · spread 11% (stable)**

Defensibility. Pooled P97 from 4 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching

methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 4 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 2000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 4 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
fsanz2012-seaweed-inorganic-arsenic-au (2012) · AU, NZ	Survey of Inorganic Arsenic in Seaweed and Seaweed-Containin…	dried seaweed, dry weight	37	uniform_range_only	38.5%
almela2002-algae-food-arsenic-metals (2002) · ES	Heavy Metal, Total Arsenic, and Inorganic Arsenic Contents o…	algae food product, dry weight	15	uniform_range_only	28.8%
llorente-miranda2011-commercial-seaweed-arsenic (2011) · ES	Determination of Water-Soluble Arsenic Compounds in Commerci…	dried seaweed, dry weight	6	uniform_range_only	11.5%

Source	Title	Basis	N used	Recon-struction	Pool share
bralatei2017-seaweed-inor-ganic-ar-senic-field-method (2017) · GB, JP, CN, KR, IE	A field de-ployable method for a rapid screening analysis of …	commer-cial dried seaweed, dry weight	11	uniform_ range_only	21.1%

tAs

Published HMTc threshold: adjudication pending — held provisional (wrong-analyte-not-material) · 5516.476 ppb (pre-rounding) **Cross-basis: ≈ 1000 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 3 sources contributing · 12 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [25.27, 18925.60] ppb Sensitivity: log-normal 6862.09 · gamma 7375.82 · nonparametric 6872.76 ppb · spread 7% (stable)**

Defensibility. Pooled P97 from 3 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 3 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 10000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 3 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Recon-struction	Pool share
aziz2021-gracilaria-seaweed-bangladesh (2021) · BD	Poten-tial of Gracilaria tenuistip-itata var. liui grown in Nu…	dried Gracilaria, dry weight	3	lognormal_mean_only	25.0%
rakib2021-macroalgae-bangladesh-metals (2021) · BD	Macroal-gae in bio-monitoring of metal pollution in the Bay of…	macroal-gae, dry weight	10	lognormal_mean_max	50.0%
razi-parjikolaei2016-red-macroal-gae-met-als (2016) · DK	Valuable Biomole-cules from Nine North Atlantic Red Macroalga…	red macroal-gae, dry matter	3	lognormal_mean_only	25.0%

Ni

Published HMTc threshold: *data gap* **Rationale tag:** `no-data` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 0 sources contributing · 0 pseudo-samples in pool · 0 sample-level contributors **no_data**

Defensibility. This analyte has no defensible threshold today. The pool is empty; no read-across is available; no regulatory cap applies. The HMTc certification mark on this product does NOT cover this analyte until evidence is loaded. This is the honest position; we explicitly do not pretend coverage we cannot defend.

Brand legal read. The value is pooled P97 from 0 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international

precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below no published value (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 0 contributing sources across multiple jurisdictions. Sensitivity band no_data.

Cd

Published HMTc threshold: 2000 ppb [as-sold] · 1874.087 ppb (pre-rounding) **Cross-basis: ≈ 500 ppb [prepared] (÷7 (1:7 reconstitution)) Rationale tag:** `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 5 sources contributing · 54 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [870.68, 3475.73] ppb Sensitivity: log-normal 1845.42 · gamma 2074.72 · nonparametric 2370.78 ppb · spread 22% (stable)**

Defensibility. Pooled P97 from 5 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 5 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 2000 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 5 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
aziz2021-gracilaria-seaweed-bangladesh (2021) · BD	Potential of Gracilaria tenuistipitata var. liui grown in Nu…	dried Gracilaria, dry weight	3	lognormal_mean_only	5.6%
martinleon2021-edible-seaweeds-canary-metals-nitrates (2021) · ES, EU, JP, KR	Human Exposure to Toxic Metals (Cd, Pb, Hg) and Nitrates (NO…	edible algae, dry weight	3	lognormal_mean_only	5.6%
razi-parjikolaei2016-red-macroalgae-metals (2016) · DK	Valuable Biomolecules from Nine North Atlantic Red Macroalga…	red macroalgae, dry matter	3	lognormal_mean_only	5.6%
hwang2010-dried-seaweed-korea-metals (2010) · KR	Total arsenic, mercury, lead, and cadmium contents in edible…	dried seaweed, dry weight	426	lognormal_mean_max	50.0%
almela2002-algae-food-arsenic-metals (2002) · ES	Heavy Metal, Total Arsenic, and Inorganic Arsenic Contents o…	algae food product, dry weight	18	uniform_range_only	33.3%

tHg

Published HMTc threshold: 100 ppb [as-sold] · 62.916 ppb (pre-rounding) **Cross-basis: ≈ 10 ppb [prepared] (÷7 (1:7 recon-**

stitution)) Rationale tag: `literature-baseline` **Regulatory ceiling reference:** no applicable cap loaded **Pool composition:** 1 sources contributing · 1 pseudo-samples in pool · 0 sample-level contributors **Bootstrap 95% CI: [12.02, 236.87] ppb Sensitivity: log-normal 12.24 · gamma 14.38 · nonparametric 12.03 ppb · spread 16% (stable)**

Defensibility. Pooled P97 from 1 contributing sources across multiple jurisdictions. No applicable regulatory ceiling for this analyte-matrix combination. The pooled value derives from the moment-matching methodology cited above; the contribution log lists every source's admission decision and basis conversion. Sensitivity band is stable across log-normal, gamma, and nonparametric parametric variants.

Brand legal read. The value is pooled P97 from 1 peer-reviewed and agency sources across multiple jurisdictions. Daubert defense rests on (a) the methodology is published, reproducible, and follows international precedent (WHO, EFSA, JECFA, ATSDR, Codex); (b) the source pool is global, not cherry-picked; (c) every numeric claim traces to a source page.

Retailer compliance read. A certified product tests at or below 100 ppb (native basis: [as-sold]) for this analyte. The certification mark on shelf indicates the brand has met every value in the Master Limit Table simultaneously. Lab COA comparisons should be done in the basis stated; the cross-basis equivalent is provided for reference.

Regulator read. Pooled methodology follows WHO/JECFA and EFSA precedent for total-diet-study pooling. 1 contributing sources across multiple jurisdictions. Sensitivity band stable.

Source	Title	Basis	N used	Reconstruction	Pool share
martinleon2021-edible-seaweeds-canary-metals-nitrates (2021) · ES, EU, JP, KR	Human Exposure to Toxic Metals (Cd, Pb, Hg) and Nitrates (NO…	edible algae, dry weight	4	lognormal_mean_only	100.0%

Adversarial Read – what a plaintiff's expert will attack, and the response

This section explicitly addresses the most likely attack vectors a plaintiff's expert, a competing standards body, or a journalist working an adversarial angle will use against this briefing. Each attack is named and answered.

Attack 1: "The pool is too small; the methodology cherry-picks." Response: The pool admits every source that contributes a usable statistic (sample-level data, percentile vector, mean, median, max, lb/ub mean, range). No source is excluded for being "summary-only." The contribution log per analyte lists every admission decision and every basis conversion. Per-source pool share is capped at 50% to prevent dominance by any single survey. The methodology is reproducible from the structured CSVs in `data/evidence/` and the engine in `tools/evidence/build-hmtc-aggregate-distributions.mjs`.

Attack 2: "Why P97 instead of P90? You're loosening the standard." Response: P97 per analyte is calibrated to deliver approximately 90% row-pass for the clean subcategory across the 10-analyte joint. Setting each analyte's limit at P90 of its own marginal would deliver a joint pass rate of 35% under independence and 50-70% under realistic correlation — well short of the Governing Principles' 90/30 row-pass target. P97 is the per-analyte percentile that delivers the row-level outcome the program promises. ALARA and the §80th-percentile ratchet trigger drive post-publication tightening.

Attack 3: "Why is your dirty subcategory P45 instead of P10?" Response: P45 of the dirty subcategory's distribution, applied per-analyte, delivers approximately 30% joint row-pass for dirty-subcategory brands. This is the design intent stated in the Governing Principles v2.0: the dirty subcategory standard must give dirty-platform brands a realistic path to certification so the program drives reformulation rather than excluding the platform. A P10 dirty standard would deliver <1% joint row-pass and the program would have no commercial uptake on the dirty side.

Attack 7: "Where does basis conversion happen and what authority do you cite?" Response: The 1:7 powder ↔ prepared-for-feeding conversion is the conservative reconstitution ratio (1 g powder + 6 g water = 7 g prepared) standard across infant-formula regulatory frame-

works (FDA Closer-to-Zero, EU 2023/915). Within-class conversions (as_sold ↔ dry_as_sold) pass through with factor 1. Every basis conversion is logged per source in the contribution log; unknown-conversion cases are flagged and not silently coerced.

Attack 8: "Show me one source you excluded." Response: No source contributing a usable statistic is excluded. Sources marked "no_usable_statistic" in the contribution log are LOD-censored entries with neither mean nor lb/ub recorded. Sources without a structured CSV row are documented in the routing audit and the gap report; those are extraction gaps, not exclusions, and they are tracked in Phase 1 backlog with target completion timelines.

Sources Cited

Every numeric claim in this briefing traces to one of the source pages below. Full source metadata lives at `wiki/sources/<source-id>.md` in the heavymetalindex.com repository.

Source ID	Title	Citation
`almela2002-algae-food-arsenic-metals` (A)	Heavy Metal, Total Arsenic, and Inorganic Arsenic Contents of Algae Food Products	Almela C; Algora S; Benito V; Clemente MJ; Devesa V; Súñer MA; Vélez D; Montoro R · 2002 · Journal of Agricultural and Food Chemistry · doi:10.1021/jf0110250
`aziz2021-gracilaria-seaweed-bangladesh` (A)	Potential of Gracilaria tenuistipitata var. liui grown in Nuniachara, Cox's Bazar, Bangladesh	Aziz A; Hassan A; Roy S K; Haque M Z; Saha B K; Ahmed S; Rahman M; Mohanta L C; Mashuk O F · 2021 · Bangladesh Journal of Scientific and Industrial Research · doi:10.3329/bjsir.v56i1.52694
`bralatei2017-seaweed-inorganic-arsenic-field-method` (A)	A field deployable method for a rapid screening analysis of inorganic arsenic in seaweed	Bralatei E; Nekrosiute K; Ronan J; Raab A; McGovern E; Stengel DB; Krupp EM; Feldmann J · 2017 · Microchim-

Source ID	Title	Citation
		ica Acta · doi:10.1007/s00604-017-2151-1
`fsanz2012-seaweed-inorganic-arsenic-au` (A)	Survey of Inorganic Arsenic in Seaweed and Seaweed-Containing Products Available in Australia	Food Standards Australia New Zealand · 2012 · Food Standards Australia New Zealand (FSANZ), Survey report · doi:null
`hwang2010-dried-seaweed-korea-metals` (A)	Total arsenic, mercury, lead, and cadmium contents in edible dried seaweed in Korea	Hwang YO; Park SG; Park GY; Choi SM; Kim MY · 2010 · Food Additives & Contaminants: Part B: Surveillance · doi:10.1080/1944
`llorente-mirandes2011-commercial-seaweed-arsenic` (A)	Determination of Water-Soluble Arsenic Compounds in Commercial Edible Seaweed by LC-ICPMS	Llorente-Mirandes T; Ruiz-Chancho MJ; Barbero M; Rubio R; López-Sánchez JF · 2011 · Journal of Agricultural and Food Chemistry · doi:10.1021/jf2040466
`martinleon2021-edible-seaweeds-canary-metals-nitrates` (A)	Human Exposure to Toxic Metals (Cd, Pb, Hg) and Nitrates (NO_3^-) from Seaweed Consumption	Martín-León V; Paz S; D'Eufemia PA; Plasencia JJ; Sagratini G; Marcantoni G; Navarro-Romero M; Gutiérrez ÁJ; Hardisson A; Rubio-Armendáriz C · 2021 · Applied Sciences · doi:10.3390/app11156934
`rakib2021-macroalgae-bangladesh-metals` (A)	Macroalgae in biomonitoring of metal pollution in the Bay of Bengal coastal waters of Cox's Bazar and surrounding areas	Rakib Md Refat Jahan; Jolly Y N; Dioses-Salinas Diana Carolina; Pizarro-Ortega Carlos Ivan; De-la-Torre Gabriel Enrique; Khandaker Mayeen Uddin; Alsubaie Abdullah; Almalki Abdulraheem S A; Bradley D A · 2021 · Scientific Reports · doi:10.1038/s41598-021-99750-7
`razi-parjikolaei2016-red-macroalgae-metals` (A)	Valuable Biomolecules from Nine North Atlantic Red Macroalgae: Amino Acids, Fatty	Behnaz Razi Parjikolaei; Annette Bruhn; Karin Loft Eybye; Martin Mork Larsen; Michael Bo

Source ID	Title	Citation
	Acids, Carotenoids, Minerals and Metals	Rasmussen; Knud Villy Christensen; Xavier C. Frette · 2016 · Natural Resources · doi:10.4236/nr.2016.74016

In addition to occurrence sources, this briefing cites:

- `docs/governance/hmtc-governing-principles.md` (v2.0, May 2026) — the five governing principles and the locked P97/P45 methodology
- `wiki/regulations/efsa-methylmercury-twi.md` — EFSA TWI 1.3 µg/kg-bw/week for MeHg
- `wiki/regulations/epa-iris-methylmercury-rfd.md` — EPA IRIS RfD 0.1 µg/kg-bw/day for MeHg
- `wiki/regulations/jecfa-methylmercury-ptwi.md` — JECFA PTWI 1.6 µg/kg-bw/week for MeHg
- WHO/JECFA pooled total-diet-study methodology
- EFSA Scientific Opinions on Pb (2010, 2024) and Cd (2009, 2024)
- ATSDR moment-matching guidance for summary-statistic sources
- Codex Alimentarius CAC/GL 50-2004 weighted pooling

Provenance Footer

This briefing is reproducible by:

1. Checking out commit `e51473a013e92827fd88e3716dd7713928984779` in the heavymetalindex.com repository.
2. Running `npm run evidence:pooling` to regenerate `data/evidence/hmtc_aggregate_distributions.csv`.
3. Running `npm run briefing -- --product=seaweed-kelp-foods` to regenerate this document.

Field	Value
Generated at	2026-06-18T13:34:12.086Z
Generator	`tools/standards/build-standards-briefing.mjs`
Pooling engine	`tools/evidence/build-hmtc-aggregate-distributions.mjs`

Field	Value
Methodology version	HMTc-10 / P97-P45 per CLAUDE.md Part 19 / Governing Principles v2.0
Git commit	`e51473a013e92827fd88e3716dd7713928984779`
Author	Karen Pendergrass, Standards Architect, Paleo Foundation
Status	Draft for stakeholder review

Disclosure: organizational role

Karen Pendergrass also serves as CEO of the Paleo Foundation, which operates the HMT&C certification program. This dual role is disclosed for transparency; the methodology in this briefing is reproducible from the open-source pooling engine and source data, independent of the author's organizational title.

Methodology implementation note

Methodology implementation: deterministic open-source pooling engine and document generator (this file's repository at the commit named in the Provenance Footer). Engine code and document templates were drafted with AI assistance (Claude, Anthropic) under the supervision and direction of the Standards Architect. All policy decisions, methodology choices, source-admission rules, basis conventions, and final values are the Standards Architect's; the AI assistance is a calculator-and-drafter, not a co-author.

End of briefing.

Part 4 — The Lot Testing Schedule

The Lot Testing Schedule is the operational manual that translates the Governance Policy's status framework (Part 2.3) and certification rules into the specific testing frequency, sampling methodology, speciation requirements, reflex triggers, and decision criteria that govern day-to-day Program operation. It is the companion document to Part 2 of this manual and is reproduced here in full so that Part 4 stands as the operational reference brand QA, certification auditors, and Program operations personnel use without needing to consult a separate document.

The Lot Testing Schedule is under independent version control as a controlled document. The version reproduced here is canonical as of the publication date of this manual edition. When the Schedule is revised between editions, the canonical version remains available at the Paleo Foundation's controlled-document registry; this manual edition's reproduction is authoritative as of binding, and the Program Operator will issue a Technical Clarification to resolve any operational discrepancy between this manual and a more recent Schedule revision.

The full content of the Lot Testing Schedule appears below, reproduced from `data/workbench/standards-book/source-documents/lot-testing-schedule.md`. The Schedule's fourteen sections cover: Purpose and Scope (§1); Product Risk Tiers and adjustments (§2); Testing Frequency by Status × Risk Tier (§3); Baseline Testing for Program Entry (§4); Status A, B, and C Surveillance Protocols (§5–7); Reflex Testing Triggers and Protocol (§8); Speciation Requirements (§9); Sampling Methodology and Chain-of-Custody (§10); Decision Rules and Status Assignment (§11); CAPA Timelines Summary (§12); Enhanced Controls and Best Practices (§13); and Document Review and Revision (§14).

[Reproduction of the Lot Testing Schedule begins on the next page. The full source document is at `data/workbench/standards-book/source-documents/lot-testing-schedule.md` *and is concatenated into the bound manual by the build pipeline; in this v0-pre-audit assembly the Schedule is referenced rather than inlined to keep the manual draft readable. The audit-strengthened v0 inlines the full Schedule content under the build's* `npm run manual` *flow.]*

Cross-references

- Status framework that drives testing frequency: Part 2.3.
- Decision rules under measurement uncertainty (applied to all lot results): Part 2.6.
- Reflex trigger from Supplier Intelligence Alerts: Part 2.9.
- CAPA initiation timelines correspondence: Part 2.8.
- Sampling, chain-of-custody, and laboratory requirements operational source: this Part.

Heavy Metal Tested & Certified, Seafood Standards Program

Companion Document to the HMTc Governance Policy

Document Owner: Program Operator / The Paleo Foundation **Effective Date:** May 2026 **Controlled Document:** Uncontrolled When Printed

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1. Purpose and Scope

This Lot Testing Schedule establishes the defensible, risk-tiered testing strategy for all products certified or enrolled under the HMTc Program. It defines the testing frequency, sampling methodology, speciation requirements, reflex triggers, and decision criteria for each certification status, product risk tier, and lot condition. This document operationalizes the governance framework set forth in the HMTc Governance Policy and is intended for use by quality teams, operations personnel, certification auditors, and the Program Operator.

The testing strategy is designed around three core principles: **legal defensibility** (every test, schedule, and decision rule must withstand scrutiny from regulators, auditors, and plaintiffs' attorneys); **scientific rigor** (testing protocols reflect current analytical best practices and are grounded in exposure science); and **commercial pragmatism** (testing frequency is proportionate to risk, avoiding unnecessary burden on compliant brands while intensifying oversight where needed).

2. Product Risk Tiers

Testing frequency under this Schedule is determined by both the product's certification status and its risk tier. Risk tiers are assigned based on the product's typical consumption frequency, volume relative to body weight, and the degree to which it may serve as a sole or dominant food source for the target population.

2.1 Risk Tier Definitions

2.1.1 High-Risk Categories. Products with the greatest potential to carry the seafood program's signature toxicants, methylmercury and inorganic arsenic, at the highest concentrations. A testing gap in these categories creates the highest exposure risk.

- Marine predatory fish (tuna, swordfish, shark, marlin)
- Seaweed and kelp foods (nori, wakame, kombu, dulse)

2.1.2 Moderate-Risk Categories. Products consumed regularly but typically as part of a diversified diet, at moderate volume relative to body weight.

- Non-predatory marine fish (cod, herring, sardine, anchovy, mackerel)
- Shellfish (bivalves and crustaceans)

2.1.3 Lower-Risk Categories. Products consumed occasionally or intermittently, at lower volumes relative to body weight, and not typically a dominant dietary component.

- Freshwater fish

2.2 Risk Tier Adjustments

The Program Operator may adjust a specific product's risk tier based on factors including but not limited to: production volume, historical testing results, known ingredient risk profiles, and consumption data. For example, a non-predatory fish product sourced from a region or species known to bioaccumulate methylmercury may be elevated to a higher risk tier for testing purposes. Risk tier adjustments are documented and communicated to the Licensee.

3. Testing Frequency by Status and Risk Tier

The following table defines the minimum testing frequency for routine surveillance. The Program Operator may increase frequency at any time based on risk signals, audit findings, or trend analysis. All testing follows the full heavy metals panel unless otherwise specified.

Status	High-Risk	Moderate-Risk	Lower-Risk	Hold & Release	Independent Sampling
Status A	Monthly	Quarterly	Quarterly	Not required	Yes (per schedule)
Status B	Every lot or monthly (whichever is more frequent)	Monthly	Monthly	Recommended	Yes, monthly
Status C (Low: 150–175%)	Every lot	Every lot	Every lot	Mandatory	Yes, every lot
Status C (High: 175–200%)	Every lot	Every lot	Every lot	Mandatory	Yes + 3rd-party sampling

3.1 Key Definitions

Independent Sampling: Samples are collected and submitted to the laboratory by a party independent of the brand's production team. This may be the Program Operator, a designated third-party auditor, or another party approved by the Program Operator. Independent sampling

ensures the integrity of the surveillance protocol and provides evidence that testing was not subject to selection bias.

Hold-and-Release: The finished product lot is physically quarantined and may not ship until laboratory results are received and the lot is confirmed to meet at minimum the applicable status criteria. Hold-and-release procedures must include documented quarantine location, responsible personnel, and release authorization records.

Third-Party Sampling (High Probation): In addition to the brand's own testing, the Program Operator directs independent sampling by a designated third party at the brand's expense. This provides a second, fully independent data point and is required when MaxPoL has reached 175–200% to ensure data integrity under heightened risk conditions.

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4. Baseline Testing (Program Entry)

All SKUs entering the HMTc Program must undergo baseline testing across three (3) production lots before any certification status is assigned. Baseline testing establishes the product's heavy metal profile, identifies which metals are closest to limits, and provides the data foundation for status assignment and ongoing surveillance design.

4.1 Baseline Testing Requirements

- Three lots must be tested, each representing a distinct production run (different dates, shifts, or ingredient lots where possible).
- Each lot is tested for the full heavy metals panel (all eight priority metals).
- Speciation testing is performed where required by the Standards methodology (methylmercury for all fish and marine products; inorganic arsenic for all seafood, with seaweed and kelp products carrying the highest total-arsenic burden; hexavalent chromium where a credible pathway exists).
- Sampling follows the composite method: each analytical sample is a composite drawn from multiple consumer units across the lot.
- Chain-of-custody documentation is required for all baseline samples.

4.2 Baseline Status Assignment

The three-lot baseline results are evaluated collectively. The initial certification status is assigned based on the worst-performing lot among the three, subject to the decision rules and Tier 1/Tier 2 metal classifications defined in the Governance Policy. If one lot is Status A and another is Status B, the initial assignment is Status B with the CAPA and elevated surveillance obligations that entails.

4.3 Expedited Baseline via Prior-Data Credit

A brand operating a mature, accredited lot-testing program may complete the three-lot baseline with a single new full-panel lot under the prior-data-credit pathway defined in Governance Policy §4.5. Up to two of the three baseline lots may be satisfied by qualifying prior results (ISO/IEC 17025-accredited, within the prior twelve months, structured and chain-of-custody documented, basis-comparable), credited per analyte only for the analytes those lots actually measured. The completing lot is sampled under this Schedule's independent-sampling and chain-of-custody requirements (§10) and run across the full panel, including any analyte not covered by the credited prior data. For an analyte established by the completing lot alone, the next two routine surveillance lots serve as confirmatory baseline points; in high-risk categories under monthly surveillance (§5.1), that confirmation completes within the first surveillance cycle. The pathway is not available on the confidential remediation track.

5. Status A: Routine Surveillance Protocol

Status A products have demonstrated full compliance (MaxPoL at or below 100% for all analytes). The surveillance protocol is designed to confirm continued compliance and detect any upward drift before it reaches actionable levels.

5.1 Testing Frequency

- **High-Risk Categories** (predatory fish, seaweed and kelp): Monthly testing via independent sampling.
- **Moderate-Risk Categories** (non-predatory fish, shellfish): Quarterly testing via independent sampling.

- **Lower-Risk Categories** (freshwater fish): Quarterly testing via independent sampling.

5.2 Trend Monitoring

The Licensee must perform a formal trend analysis of heavy metal results at least quarterly. This analysis should use control charts or equivalent statistical methods to identify any upward trajectory in concentrations, even if all results remain below limits. The Program Operator may request submission of trend data as part of routine oversight.

5.3 Internal Warning Trigger

If any analyte result on routine surveillance reaches or exceeds 80% of its HMTc limit (80% PoL), the following actions are triggered even though the result is technically compliant:

- Confirmatory testing of the next available lot for the affected analyte.
- Enhanced monitoring: the Licensee should investigate whether the elevated result reflects a trend or an isolated event.
- For Tier 1 metals approaching 80% of limits, the Program Operator should be notified and speciation testing (if applicable) should be performed regardless of whether the standard speciation trigger has been met.

5.4 Lot Release

Hold-and-release is not mandatory for Status A products, but it is strongly recommended that the Licensee hold any lot selected for testing until results are received. If a routine surveillance lot unexpectedly fails, the lot must be isolated and the noncompliance procedures activated immediately.

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6. Status B: Elevated Surveillance Protocol

Status B products have a transitional exceedance on one or more Tier 2 metals (MaxPoL above 100% but at or below 150%). The elevated surveillance protocol is designed to monitor the effectiveness of CAPA actions and confirm an improving trend.

6.1 Testing Frequency

- **High-Risk Categories:** Every lot or monthly, whichever is more frequent. If a brand produces more than one lot per month in a high-risk category, every lot is tested.
- **Moderate-Risk Categories:** Monthly independent sampling.
- **Lower-Risk Categories:** Monthly independent sampling.

6.2 Hold-and-Release

Hold-and-release is not mandated for Status B but is strongly recommended, particularly for the first several lots following CAPA implementation. The Licensee should coordinate production and testing so that results are available before product is widely distributed. If a tested lot exceeds the 150% transitional cap or involves a Tier 1 metal exceedance, the lot may not be released as certified.

6.3 Progress Monitoring

Every lot tested during Status B provides data to gauge improvement. The quality team must trend consecutive results and submit periodic progress reports to the Program Operator. The CAPA's interim reduction targets (e.g., "next lot will be at or below 120% PoL; within 6 months, at or below 100%") are tracked against actual results. The Program Operator may conduct interim reviews to verify that the CAPA is being executed and that results are trending downward.

6.4 Escalation

If any lot tested during Status B meets any of the following conditions, the SKU is immediately escalated to Status C:

- MaxPoL exceeds 150% for any analyte.
- Any Tier 1 metal exceeds 100% of its limit.
- Two or more consecutive lots show no improvement or worsening results relative to the baseline.
- The Licensee fails to initiate or execute CAPA within required timelines.

7. Status C: Probation Surveillance Protocol

Status C products are on probation due to significant exceedances, repeated non-improvement, or Tier 1 metal violations. The probation protocol is designed to provide maximum oversight and prevent any non-compliant product from reaching consumers while the brand executes corrective actions.

7.1 Testing Frequency

All Status C products, regardless of risk tier, are tested every lot. There are no exceptions to this requirement during probation. The Program Operator may additionally require split-sample testing (simultaneous analysis by two independent laboratories) for high-value or high-visibility products.

7.2 Hold-and-Release (Mandatory)

Hold-and-release is mandatory for all Status C products. No product may leave the quarantine area until laboratory results are received and the lot is confirmed to meet at least Status A or Status B criteria. The Probation Lot Release Plan (documented in the CAPA) must specify: the physical quarantine location and procedures; the designated personnel authorized to release lots; the laboratory being used and expected turnaround time; the pass criteria for release (Status A or B); and the fail handling procedures (rework, downgrade to non-certified, sticker-over, or withdrawal).

7.3 Graduated Probation Controls

As defined in the Governance Policy, Status C is differentiated by severity:

Low Probation (MaxPoL 150–175%): Every-lot testing with hold-and-release. The brand's own accredited laboratory sampling is permitted, subject to Program Operator audit rights. The Program Operator may conduct unannounced verification sampling.

High Probation (MaxPoL 175–200%): Every-lot testing with hold-and-release, plus mandatory independent third-party sampling directed by the Program Operator. The third-party sampler collects samples independently of the brand's quality team and submits them directly to a Program-designated laboratory. This dual-track approach ensures data integrity under heightened risk conditions.

7.4 Exit Criteria

To exit probation, the brand must satisfy both of the following conditions:

- Produce a minimum of five (5) consecutive lots that individually test at Status A or Status B levels, demonstrating that the product can consistently meet standards under real production conditions. The Program Operator may require more than five consecutive lots depending on the severity and nature of the original nonconformance.
- Complete all CAPA milestones and demonstrate Verification of Effectiveness through documented follow-up testing. The corrective and preventive measures must be shown to have achieved their intended result.

8. Reflex Testing Triggers and Protocol

Reflex testing is any test conducted outside the routine surveillance schedule in response to a defined trigger event. Reflex tests carry the same weight as routine surveillance results and are subject to the same decision rules and status implications.

8.1 Trigger Events

The following events require reflex testing of all affected SKUs:

Trigger Category	Specific Triggers	Testing Requirement
Ingredient Changes	New ingredient, new supplier, change in ingredient origin or grade	Full panel on next 2 lots after change
Water Source Changes	New municipal source, well change, treatment system modification	Full panel on next 2 lots after change
Equipment Changes	New processing equipment contacting product, equipment repairs involving metal surfaces	Full panel on next lot after change
Packaging Changes	New packaging material, new packaging supplier, change in liner or closure	Full panel on next 2 lots (to assess migration over time)
Formulation Changes	Any change in ingredient ratios, addition or removal of ingredients	Full panel on next 2 lots after change

Trigger Category	Specific Triggers	Testing Requirement
Process Changes	Changes in contact time, temperature, pH, mechanical stress, or processing sequence	Full panel on next lot after change
Upstream Supplier Notification	Supplier notifies of change in their own sourcing or processing	Full panel on next lot using affected ingredient
Surveillance Warning	Any Tier 1 metal result at or above 80% PoL on routine testing	Confirmatory test on next lot; speciation if applicable
Supplier Intelligence Alert	Program Operator issues anonymized alert regarding a supplier or ingredient category	Full panel on next lot using flagged ingredient or supplier

8.2 Reflex Testing Protocol

Reflex tests must use the full heavy metals panel and follow all standard sampling and chain-of-custody requirements. Results are evaluated under the same decision rules (including measurement uncertainty) as routine surveillance. If a reflex test reveals a nonconformance, the applicable status transition and CAPA requirements are triggered immediately, regardless of where the product currently sits in the routine testing cycle.

9. Speciation Requirements

Speciation analysis determines the specific chemical form of a metal present, which is critical for accurate risk assessment. The following speciation requirements apply:

Metal	Form of Concern	Mandatory Speciation	Reflex Speciation Trigger
Arsenic	Inorganic arsenic (iAs)	All seafood products; seaweed and kelp products carry the highest total-arsenic burden	Total As approaches or exceeds limit in any product
Mercury	Methylmercury (MeHg)	All fish and marine products	Total Hg approaches or ex-

Metal	Form of Concern	Mandatory Speciation	Reflex Speciation Trigger
			ceeds limit in any product
Chromium	Hexavalent chromium (Cr(VI))	Only if credible pathway exists (certain water sources, processing)	Total Cr is anomalously high or Program Operator directs
Lead, Cadmium, Nickel, Aluminum, Tin	Total element only	Not required	Not applicable

When speciation is performed, the speciated result (e.g., inorganic arsenic rather than total arsenic) becomes the compliance result for that metal, superseding the total element concentration. Failure to perform required speciation renders the lot ineligible for certification. The reflex speciation trigger of "approaches the limit" is operationally defined as reaching 50% of the applicable HMTc limit for total metal, consistent with the speciation triggers defined in the HMTc Seafood Standards. The Program Operator may lower this threshold further based on risk analysis but may not raise it above 50%.

10. Sampling Methodology and Chain-of-Custody

10.1 Lot-Based Composite Sampling

All certification testing is conducted on a lot basis. For each lot tested, the analytical sample must be a composite prepared from multiple consumer-unit samples drawn from across the lot. The number of sub-samples and method of compositing must be statistically robust, designed to represent the entire lot's average quality with 95% confidence that the lot meets requirements. The sampling plan is subject to approval by the Program Operator. Any alternative approach (stratified sampling, continuous batch sampling) requires prior written approval.

10.2 Chain-of-Custody

A complete chain-of-custody record is required for every sample, from the point of collection through laboratory receipt and analysis. The record

must document: who collected the sample and their affiliation; when and where the sample was collected; how the sample was sealed, labeled, and transported; who received the sample at the laboratory; and confirmation of sample integrity upon receipt. All personnel involved in sampling must be trained on these procedures.

Consequences of documented chain-of-custody breach. Any documented breach in chain-of-custody invalidates the result and is treated as a data integrity issue per the Governance Policy §8.2. Documented breaches include but are not limited to:

- **Custody gap:** any period during which sample location and possession is not recorded.
- **Broken seal:** sample arrives at the laboratory with seal damaged, missing, or showing evidence of tampering.
- **Non-conforming labeling:** sample identifiers at any handoff do not match the chain-of-custody record.
- **Transport non-conformance:** sample transported outside the documented protocol (e.g., refrigerated samples allowed to warm; samples transported under conditions inconsistent with the analytical method's stability requirements).
- **Receipt discrepancy:** laboratory documents the sample as received in a state inconsistent with the chain-of-custody record (mass, volume, container condition).

When a documented breach is identified, the lot's result is treated as invalid and the lot is subject to retest per a Program Operator-approved confirmatory protocol. Repeated breaches at the same Licensee or laboratory trigger data-integrity escalation per Governance Policy §8.2 and may result in Status D suspension.

Discretion for minor procedural lapses. The Program Operator may exercise discretion on minor procedural lapses where substantive sample integrity is preserved, for example, a timestamp recorded one hour late where the actual custody record (signatures, transfers) is intact. Such discretion is documented in the case record and is not exercised when sample integrity itself is in doubt. The default disposition for any chain-of-custody ambiguity is invalidation; discretion applies only to clearly procedural rather than substantive lapses.

10.3 Laboratory Requirements

All testing must be performed by laboratories accredited to ISO/IEC 17025 for the relevant analytical methods and food matrices. Methods must be fit for purpose, meaning they can reliably quantify target metals at or below the applicable HMTc limits. The laboratory's LOQ for each analyte must be at or below the HMTc action level. Results reported as non-detect (below LOQ) are treated as zero for compliance purposes, provided the LOQ meets this requirement. If an LOQ exceeds the Program limit, the result is treated as indeterminate and retesting with a more sensitive method is required.

10.4 Reporting Requirements

Laboratory reports must express results on a finished-product basis (as sold), clearly identifying each analyte and its form (total vs. speciated). Reports must include the LOQ for each analyte and the associated Measurement Uncertainty (MU). All data must be provided in structured, machine-readable format as required by the Governance Policy. PDF submission is not acceptable.

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11. Decision Rules and Status Assignment

11.1 Measurement Uncertainty Application

The decision rules incorporating Measurement Uncertainty are defined in the Governance Policy (Section 7) and apply to all lot test results under this Schedule. In summary: **Pass** if (Result + MU) is at or below the limit; **Fail** if (Result – MU) is above the limit; **Borderline** if neither condition is met. Borderline results require confirmatory action before status assignment.

11.2 PoL Calculation and Status Mapping

For each lot, the PoL is calculated for every analyte: PoL = (Result / Applicable Limit) × 100%. The MaxPoL (highest PoL among all analytes) determines the lot's status:

MaxPoL Range	Metal Tier	Status Assignment	Action Required
0–100%	Any	Status A	Routine surveillance
101–150%	Tier 2 only	Status B	CAPA (10 days), elevated testing
101–150%	Any Tier 1 metal	Status C	CAPA (5 days), every-lot hold-and-release
151–175%	Any	Status C (Low)	CAPA (5 days), every-lot hold-and-release
176–200%	Any	Status C (High)	CAPA (5 days), every-lot + 3rd-party sampling
> 200%	Any	Status D (Suspension)	Stop-ship, CAPA (2 days), mark removal

11.3 No Testing into Compliance

As stated in the Governance Policy, repeated ad hoc retesting to obtain a passing result is strictly prohibited. Any retesting must be part of a pre-approved confirmatory plan, scientifically justified, and documented. Undocumented resampling or lab-shopping is a data integrity violation subject to suspension or revocation.

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12. CAPA Timelines Summary

The following timelines govern CAPA initiation and execution across certification statuses. These timelines are maximum allowable periods; faster action is always expected.

Element	Status B	Status C	Status D	Status E
CAPA initiation	10 business days	5 business days	2 business days	As directed
Full plan submission	30 calendar days	15 business days	10 business days	N/A
Containment	As needed	Immediate	Within 48 hours	Immediate

Element	Status B	Status C	Status D	Status E
Target for Status A	12 months	As defined in CAPA	Full re-qualification	Re-application
VOE required	Yes	Yes	Yes	If re-applying
Exit requirement	Improving trend + meet A criteria	5+ consecutive passing lots + CAPA closure	Program Operator approval	Full re-application

13. Enhanced Controls and Best Practices

The following measures are recommended beyond the minimum requirements. While not all are mandatory, they provide additional assurance and may be expected by regulators as evidence of due diligence.

13.1 Risk-Tiering of Ingredients and Suppliers

Identify raw materials and suppliers that are high-risk for heavy metal contamination (predatory fish for methylmercury, bivalve shellfish for cadmium and lead, seaweed and kelp for inorganic arsenic, certain harvest regions and waters). Apply stricter incoming material specifications and testing for high-risk tiers. Require certificates of analysis for each ingredient lot from high-risk suppliers. Conduct periodic audits or verification testing of supplier materials.

13.2 In-Process Testing

Where feasible, implement in-process checks at critical control points (e.g., post-filtration water testing, intermediate stream sampling). Rapid screening tools (XRF analyzers, portable test kits) may be used in-house for early warning, with results periodically validated against accredited laboratory data.

13.3 Trend Analysis and Internal Warning Levels

Perform formal trend analysis at least quarterly using control charts or equivalent statistical methods. Define internal warning levels below the HMTc limits (e.g., 70% PoL) that trigger management review and proactive investigation when breached, even if the product remains fully compliant. Document all trend analyses and management responses.

13.4 Change Management

Maintain a formal change management process that captures all changes to ingredients, suppliers, equipment, packaging, formulation, water sources, and processing conditions. Every change should be evaluated for potential impact on heavy metal levels before implementation, and reflex testing should be planned accordingly.

13.5 Documentation and Transparency

Maintain comprehensive records of all testing (internal and external), CAPA actions, trend analyses, supplier evaluations, and communications with the Program Operator. Proactively inform the Program Operator of unusual results or emerging concerns rather than attempting to resolve them quietly. A complete, transparent paper trail is the foundation of legal defensibility and program integrity.

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14. Document Review and Revision

This Lot Testing Schedule shall be reviewed at least annually or upon any update to the HMTc Governance Policy, HMTc Seafood Standards, or relevant regulatory requirements. Revisions will incorporate new scientific insights, regulatory changes, improvements in testing technology, and lessons learned from program operations. All revisions are managed under document control procedures consistent with the Governance Policy. The version in effect at the time of a testing decision governs that decision.

Document Control

Version	Date	Changes
1.0	March 2026	Initial issuance (PDF).
2.0	May 2026	Markdown migration. No substantive change to testing protocols; reformatted for repo-based version control and cross-

Version	Date	Changes
		referencing with the Program Manual.
2.1	June 2026	Added §4.3 Expedited Baseline via Prior-Data Credit, the testing-schedule companion to Governance Policy §4.5. Completing lot uses independent sampling; new-analyte confirmation folds into routine surveillance.

5.1 How HMTc Standards Are Set

A reader who has worked through Parts 1, 2, 2.5, 3, and 4 has encountered every operational element of the HMTc Program: why it exists, how it is governed, how brands use the evidence base, what the per-product standards are, and how surveillance operates. What remains is the methodology that ties the operational program back to its underlying analytical work. This chapter specifies, in principle and at appropriate detail, how a specific number, say, the lead limit for a given product row, gets produced and what an adversarial reviewer can verify about it.

This chapter respects the wiki/HMTc firewall: the engine arithmetic stays workbench-side, where it is protected as a trade secret per the Colophon (Part 5.7) and where exposing it would let competing certifiers shortcut years of methodology development. The **principle** of standards-setting is in this chapter; the **mechanism** is in the Governing Principles; the **inputs** are on heavymetalindex.com and are publicly auditable.

The evidence base, and what auditable means

Every concentration limit in Part 3 of this manual was derived from a pooled distribution of source pages on heavymetalindex.com for the relevant product-by-metal cell. The pooling math (the per-analyte percentile calculations, the source-level bootstrap analysis, the methodology stability checks across alternative parametric assumptions) lives on the workbench-side infrastructure under the wiki/HMTc firewall established by the HMTc Governing Principles (Principle 4) and described in Part 1.5 of this manual. But the **inputs** to the pooling, the source pages contributing to each cell, are public.

An adversarial reader who wants to audit a specific HMTc limit can follow this path:

1. Locate the product chapter in Part 3 of this manual. The Master Limit Table gives the HMTc limit, the applicable regulatory cap, and the basis note for each analyte.
2. Read the chapter's Sources Cited section. Each citation resolves to a source page on heavymetalindex.com.
3. Read the source page. Each source page summarizes the underlying document's findings with concentrations, sample sizes, methods, and basis (as-sold, as-consumed, dry weight). The source page links to the original document where available.
4. Check the routing audit (`data/evidence/product_source_routing_audit.csv`) to confirm that no relevant source has been omitted from the cell.
5. For the pooled distribution itself, qualified reviewers may request the workbench-side aggregate distribution data under controlled terms (see the controlled-terms request process below).

This design preserves the firewall while making the evidence inputs auditable. **The failure mode this design prevents is the "we set this limit because we said so" attack.** HMTc cannot make that argument because the source attribution is public.

Controlled-terms request process

Qualified reviewers include:

- **Regulatory bodies acting under official inquiry**, FDA, EFSA, state attorneys general, equivalent international agencies, who require methodology audit access for regulatory analysis or rulemaking purposes.
- **Plaintiff's or defense counsel requesting access for active litigation** in which an HMTc-certified product is at issue or in which HMTc certification or methodology has been raised by either party.
- **Certified or pending HMTc Licensees auditing the methodology** that produces their own product category's standards. Licensees have a contractual audit right under the License Agreement (see Part 5.6).
- **Academic researchers conducting peer-track review** with institutional approval and a documented research protocol that includes confidentiality undertakings.

Requests are submitted to the Program Operator via the contact channels in the Colophon (Part 5.7). The Program Operator responds within **30 business days** with one of:

- For regulatory bodies under official inquiry: the requested aggregate distribution data directly, without a controlled-access NDA.
- For non-regulatory requestors: a controlled-access Non-Disclosure Agreement that preserves the trade-secret status of the pooling engine implementation while making the engine's inputs (the contributing source pages and their summary statistics) and outputs (the per-analyte percentile values, the row-pass calibration math, and the methodology's stability under alternative parametric assumptions) auditable.
- A documented refusal with the specific basis, where the request falls outside the qualified-reviewer categories. Refusal decisions are reviewable by the Independent Appeals Committee (Part 2.12 §14.1).

The 30-business-day commitment is an operational commitment by the Program Operator. The Program Operator publishes annual statistics on controlled-terms request volume, response times, and disposition in the transparency report.

The standards-setting principle in three layers

Layer 1: The 90/30 row-pass intent

The HMTc Standards are calibrated to deliver the following row-level outcomes at initial publication:

- Approximately **90%** of clean-platform brands pass row-level certification (achieve Status A) on day one.
- Approximately **30%** of dirty-platform brands pass row-level certification on day one.

These targets are the policy choice. They sit at the intersection of all five Governing Principles. Too tight (e.g., 50/10) violates Principles 3 and 5 because brands cannot recruit and the program cannot scale. Too loose (e.g., 99/80) violates Principles 1 and 3 because the mark loses meaning and clean brands lose differentiation. 90/30 is the policy answer to all five simultaneously.

Adversarial readers attacking "why P97 and not P95" must instead attack "why 90% and not 85%." The answer to that question traces directly to the Governing Principles. The principle-level argument is the strongest possible form of the methodology defense because it converts a technical argument into a policy argument, and policy arguments are won or lost on the principles they serve.

Clean vs. dirty platform classification rule

A *platform* is a subcategory of products within a parent product category, for example, predatory versus non-predatory marine fish, finfish versus shellfish, fresh/frozen versus canned forms. Platform classification determines whether per-analyte standards use the P97 (clean) or P45 (dirty) percentile, so the classification rule is load-bearing.

The classification rule is data-driven: a platform is classified *clean* when its empirical median across the literature-pooled distribution sits at or below the parent-category median for the analyte; *dirty* when meaningfully above. "Meaningfully above" is operationalized as p50(subcategory) > 1.5 × p50(parent category) for at least one Tier 1 metal. Platforms not meeting the dirty threshold are classified clean by default.

The classification is reproducible from public data. Any reviewer can verify a specific platform's classification by examining the pooled distributions on heavymetalindex.com via the product-category and subcategory pages. Reclassification (clean ↔ dirty) is data-driven and follows the rule above; it is not a discretionary decision by the Program Operator.

Layer 2: The per-analyte percentile mechanism

To deliver the 90/30 row-level intent, per-analyte limits are set as follows:

- **Per-analyte standard for a clean subcategory** = P97 of that subcategory's marginal occurrence distribution for that analyte.
- **Per-analyte standard for a dirty subcategory** = P45 of that subcategory's marginal occurrence distribution for that analyte.

The reasoning is the row-pass calibration: a brand certifies a product (one row, ten analytes simultaneously), not one analyte at a time. Under analyte independence, per-analyte limits set at the raw P90 of the clean distribution would deliver a joint row-pass rate of approximately 0.9^10 = 35%. Under realistic analyte correlation (around 0.6, since supply-chain hygiene drives multiple metals together), the joint rate sits around 50–70%. To deliver the 90% row-pass intent for clean-platform brands,

per-analyte limits are uplifted from the raw P90 to P97; the analogous uplift for dirty-platform brands moves from raw P30 to P45.

The analyte-correlation estimate of approximately 0.6 is consistent with cross-analyte correlation observed in published surveillance datasets (FDA Total Diet Study[1] EFSA scientific opinions[8, 10] on Pb, Cd, and iAs in food). It is an estimate from external surveillance data, not a measurement from HMTc-internal lot data. The 12-month post-launch methodology review will measure empirical analyte correlation from HMTc lot results and may revise the per-analyte percentile uplift if the empirical correlation diverges materially from the assumed value.

The uplift is uniform across all analytes today. Per-analyte differentiation, for example, lead tighter than nickel based on regulatory pressure or toxicology, is a Tier 2 calibration deferred until HMTc has accumulated 6–12 months of lot-testing data to measure actual joint pass rates empirically.

Layer 3: Pool admission, basis, regulatory cap, and rationale tagging

Every source loaded for a (row, metal) cell contributes to the pooled distribution unless explicitly dispositioned out with a stated reason. Sample-level data uses raw values; summary-only sources (mean, median, max, SD, lower/upper bound, IQR) contribute via moment-matching pseudo-sample expansion under log-normal or gamma assumptions following the standard practice in environmental exposure assessment (WHO/JECFA pooled TDS methodology[13] EFSA Scientific Opinions on Pb and Cd in food[8, 11]).

Sources are weighted by their reported sample size, capped at twice the median per-source N. The cap prevents any single large survey, for example, a 5,000-sample FDA TDS sub-extract, from dominating the pooled distribution for a cell where most contributing sources have N=50–200. The 2× ratio preserves large-N influence proportionally while preventing pool concentration in a single source. Without the cap, a single large survey could effectively determine the per-analyte percentile for a cell regardless of what other contributing sources report; with the cap, the percentile reflects the weighted consensus of the corpus rather than the dominant single survey.

Each row's standard is published in the native basis of the product form: fish, shellfish, and seaweed rows in as-sold (edible muscle or tissue), and where a product is sold canned or otherwise prepared, in the as-consumed form. Brands ship and test in their product's native form; the certification mark matches that form. Sources reporting in non-native bases are converted into the row's native basis during pooling via documented factors (the 1:7 reconstitution ratio for dried-to-prepared conversions; within-class identity for as-sold to dry-as-sold). The cross-basis equivalent of every published value is shown in the Master Limit Table for ease of reference by audiences operating in different bases. The cross-basis equivalent of every published value is shown in the Master Limit Table for ease of reference by audiences operating in different bases.

The final HMTc value is `min(per-analyte pooled percentile, lowest applicable regu` with both expressed in the row's native basis. The cap is converted into native basis if it was published in a different one. The rationale tag is one of:

- **literature-baseline**, the pooled percentile binds. This is the typical case where HMTc's methodology produces a value at or below the regulatory floor.
- **regulatory-alignment**, the cap binds. The pooled percentile is higher than the regulatory ceiling, and the ceiling governs.
- **precautionary**, a standard set lower than both the pooled percentile and the cap when staff document a precautionary rationale (vulnerable-population concern, cumulative exposure, scientific uncertainty).
- **feasibility-driven**, a standard set higher than the pooled percentile only via a documented feasibility-driven rationale paired with a sunset plan to tighten as analytical methods or supply-chain practices improve. **The gap between literature-baseline and the final value, if any, is stated honestly.**

Layer 4: Ratchet trigger (post-publication, distinct from initial limit-setting)

After initial publication, the Standards Ratchet specified in Part 2.10 operates on HMTc-internal lot-testing data. The 80th-percentile-of-Status-A trigger and the breadth-condition are independent of the limit-setting math: limit-setting uses external survey data (FDA TDS, EFSA opinions, peer-reviewed studies, EU basket surveys); the ratchet uses

program-internal lot-testing data accumulated under the structured-data requirement in Part 2.7.

The two mechanisms operate at different times, on different data sources, and serve different functions. **The 80% ratchet trigger and the 97% / 45% limit-setting percentiles are not redundant; they are complementary.**

Why this methodology is defensible

A plaintiff's expert reading this chapter could advance three categories of attack: that the methodology is opaque (we can't reproduce it), that the policy choice is arbitrary (the 90/30 numbers are made up), or that the inputs are biased (the source pages were selected to support a predetermined conclusion).

The first attack fails because the principle is documented here, the mechanism is documented in the Governing Principles, and qualified reviewers can request the engine outputs under controlled terms. The methodology is opaque in the sense that the engine code is a trade secret; it is not opaque in the sense that the reasoning is unverifiable.

The second attack fails because the 90/30 policy choice traces to the five Governing Principles. An attacker who wants to argue that a different policy choice would better serve the Governing Principles can do so on the merits, and that argument is exactly the kind of public discourse a credible standards program welcomes. The program does not claim that 90/30 is the only defensible calibration; it claims that 90/30 is the calibration that best serves the five principles simultaneously, and welcomes principled counterarguments.

The third attack fails because the source pages are public. An attacker who believes a relevant source has been omitted can check the routing audit and the source's frontmatter; an attacker who believes an irrelevant source has been included can apply the same check. Selection bias arguments must point to specific source pages whose inclusion or exclusion is in dispute; they cannot be advanced in the abstract because the corpus is open.

The Program-Level Adversarial Read in Part 5.2 enumerates these and other attack vectors in detail.

Cross-references

- Governing Principles (the policy source): Part 5.1 source document is `docs/governance/hmtc-governing-principles.md` v2.1.
- The corpus and the wiki/HMTc firewall: Part 1.5.
- Standards Ratchet (post-publication tightening): Part 2.10.
- Per-product master limit tables: Part 3.
- The published methodology preprint: `Pendergrass, K. Certification as a Framewo` `Exposure in Infant and Child Foods. doi: 1` The preprint's 80%-achievability calibration framing reflects the program's March 2026 state; the current operational framing is the 90/30 row-pass intent specified above.

5.2 Program-Level Adversarial Read

This chapter enumerates the categories of attack a plaintiff's expert, a hostile regulatory reviewer, a competing standards body, or a defense attorney preparing cross-examination would advance against HMTc as a whole, and documents the program's response to each on the merits. The chapter is the output of the Pre-publication Adversarial Audit conducted on the drafted Program Manual; each attack vector was named, located in the manual, and either confirmed as adequately addressed or used to drive a manual revision before this chapter was written. Per-product adversarial reads appear in each Part 3 chapter; this chapter addresses program-level attacks that apply across all product categories.

The chapter is organized by attack category. For each: the attack as a hostile reviewer advances it; the program's response with manual location.

Methodology underspecification

Attack. "The manual asserts that limits are calibrated via P97 clean / P45 dirty per-analyte percentiles to deliver 90/30 row-pass calibration. But the boundary between clean and dirty subcategories is undefined; the analyte-correlation assumption of 0.6 is asserted but not derived; the 2× median per-source N cap is named but not motivated. An independent analyst cannot replicate the calibration from this manual."

Response. Part 5.1 §"Layer 1" now specifies the clean-vs-dirty platform classification rule: a platform is clean when its empirical median across the literature-pooled distribution sits at or below the parent-category median for the analyte; dirty when p50(platform) > 1.5 × p50(parent category) for at least one Tier 1 metal. The classification is data-driven and

reproducible from public corpus data on heavymetalindex.com. Part 5.1 §“Layer 2” anchors the 0.6 analyte-correlation estimate to cross-analyte correlation observed in FDA TDS[1] and EFSA surveillance datasets, and commits to an empirical revision at the 12-month post-launch methodology review if HMTc-internal lot data diverges materially from the assumed value. Part 5.1 §“Layer 3” motivates the 2× median per-source N cap: it prevents any single large survey from dominating the pooled distribution while preserving large-N influence proportionally.

The principle of standards-setting is specified at sufficient depth in Part 5.1 for independent analysts to understand the calibration architecture. The engine implementation that operationalizes the principle is held as a trade secret per the firewall in Part 5.7 (Colophon); qualified reviewers may request the engine’s inputs and outputs under the controlled-terms request process in Part 5.1.

Toxicological anchor language

Attack. “The Tier 1 classification rationale in Part 2.2 uses shorthand framings, ‘no safe threshold,’ ‘genotoxic carcinogen’, that are stronger than the cited authorities’ actual language. EFSA’s actual conclusion on lead, for example, was that ‘a threshold for the critical effects could not be identified,’ which is not identical to ‘no safe threshold.’ The manual overstates the toxicological consensus.”

Response. Part 2.2 Tier 1 and Tier 2 rationale columns have been revised to use the cited authorities’ actual language. The lead entry now reads: “EFSA (2010) concluded that ‘a threshold for the critical effects of lead in children’s neurodevelopment could not be identified’ (BMDL01 = 12 µg/L blood lead for one-point IQ decrement). JECFA (2011) withdrew its PTWI on the grounds that ‘conventional tolerable intake values were no longer health-protective’ for endpoints without an identifiable threshold.” Analogous revisions apply to cadmium, arsenic, and methylmercury. The substantive toxicological positions are unchanged; the framing now matches the cited authorities’ actual statements.

Chain-of-custody enforceability

Attack. "The Lot Testing Schedule §10.2 says chain-of-custody breach 'may invalidate' results, discretionary. The Governance Policy §8.2 names chain-of-custody breach as a data-integrity violation triggering Status D, categorical. The tension is exploitable: defense experts argue 'may' is discretionary; plaintiff's experts argue §8.2 makes it categorical."

Response. The Lot Testing Schedule §10.2 has been revised. Any documented breach invalidates the result and is treated as a data-integrity issue per Governance Policy §8.2. Five specific breach types are enumerated (custody gap, broken seal, non-conforming labeling, transport non-conformance, receipt discrepancy). Discretion is preserved only for clearly minor procedural lapses where substantive integrity is preserved (e.g., a one-hour-late timestamp where actual custody record is intact); the default disposition for any chain-of-custody ambiguity is invalidation. The discretion-vs-categorical tension is resolved in favor of categorical invalidation with narrow, documented procedural-discretion exceptions.

Ratchet gaming

Attack. "The ratchet trigger is gameable. A dominant brand could withhold its best-performing SKUs from Status A by delaying CAPA closure, keeping the trigger percentile up. A coalition of large brands could manipulate the breadth condition through SKU consolidation. A brand could exit and re-enter the program to reset its grandfathering status. The rule is structurally vulnerable to three documented gaming pathways."

Response. Part 2.10 §12.3 has been revised with three anti-gaming clauses. (1) The trigger pool now combines Status A and Status B lot results, making withholding gaming impossible, keeping a SKU at Status B no longer keeps its data out of the trigger calculation. (2) A single-Licensee cap limits any one Licensee to contributing no more than 25% of the SKUs in the trigger pool for a given analyte; SKUs above the cap are randomly excluded from the trigger calculation to prevent dominant-brand manipulation. (3) Part 2.10 §12.7 adds the exit-and-re-enter prohibition: a Licensee that withdraws and re-enrolls within 24 months is evaluated against the current standard at re-entry, not the standard in effect at original exit. Grandfathering applies only to continuous certification.

Each gaming pathway is now structurally prevented rather than merely identified.

Institutional conflict of interest

Attack. "The Standards Architect is also CEO of the Paleo Foundation. The dual role is disclosed in the colophon, but disclosure is not mitigation. The appeals process is internal to the Program Operator, which is the entity with financial interest in upholding the determination. A Licensee challenging a status determination is appealing to the entity whose revenue depends on the determination being upheld. The §14.4 individual-conflict provision addresses reviewer conflicts but not the institutional one."

Response. Part 2.12 §14.1 now specifies a **two-tier appeals process**. Tier 1 is the internal Program Operator review (the existing process). Tier 2 is an **independent Appeals Committee** composed of three external reviewers, a toxicologist or food-safety scientist, a food-industry quality professional, a legal advisor with regulatory-affairs background, none of whom are employed by or under contract to the Paleo Foundation. Committee members serve two-year staggered terms, sit on no more than three appeals per calendar year, and recuse from cases involving Licensees they have relationships with. **Tier 2 reversals are binding on the Program Operator.** The Committee operates under a funding structure that prevents financial dependence on case outcomes. Anonymized appeal outcomes are published in the annual transparency report.

The two-tier structure is the program's structural answer to the institutional conflict. A Licensee dissatisfied with the Program Operator's internal review has binding recourse to a body independent of the Program Operator. The disclosure of the Standards Architect / CEO dual role remains in the Colophon as a transparency element; the independent Appeals Committee is the mitigation that disclosure alone does not provide.

Definitional completeness

Attack. "The manual uses 'platform' (clean vs. dirty) as the load-bearing classification term but never defines it. 'Subcategory,' 'achievability

threshold,' 'controlled document,' and 'Technical Committee' are used in the manual without Glossary entries. The most consequential vocabulary in the methodology is undefined."

Response. Part 5.3 Glossary has been extended with entries for *platform* (with the full classification rule from Part 5.1), *subcategory* (cross-referenced to *platform*), *achievability threshold* (with the distinction between standards-setting external data and ratchet-trigger internal data), *controlled document* (with the version-in-effect rule), and *Technical Committee* (with its advisory role distinguished from the binding-decision authority of the Program Operator and the Independent Appeals Committee).

Overclaim risk

Attack. "Part 2.5 §6.2 prohibits 'heavy metal free' and 'safe' marketing claims. But Part 1.4 says the program creates 'a documented record' that is 'the difference between exposure and defense.' Part 1.3′s ROI framework describes 'risk reduction' in language that could be read as a safety claim. The manual maintains the prohibition in operational rules but uses risk-reduction language in legal-defensibility arguments that approach the prohibited claim."

Response. Parts 1.3 and 1.4 have been revised with explicit hedging language. Part 1.3 now states: "The model quantifies *risk-event* reduction (probability and severity of recalls, delistings, and litigation triggers), not safety. A brand with HMTc certification is in a better defensive posture than a brand without it; both brands operate in a contamination environment that cannot be eliminated by any program. The certification mark is an action-level compliance signal under surveillance, not a safety guarantee." Part 1.4 adds: "These improvements reduce *probability* and *severity* of contamination events; they do not produce safety guarantees. The mark certifies action-level compliance under surveillance, not the absence of risk." Internal consistency is maintained between the operational prohibition in Part 2.5 §6.2 and the framing language in Parts 1.3 and 1.4.

Citation completeness

Attack. "The manual makes specific numeric claims, '$2.0 million per major case,' '$28,000 Prop 65 settlement,' '7.8 µg lead/day aggregate', without inline citation. A reviewer cannot move from a numeric claim to the source page in one step."

Response. Key numeric claims in Parts 1, 2, and 5 now carry inline bracketed citations to their primary sources (e.g., *Federal Judicial Center, Litigation Cost Survey of Major Companies*; *People v. Promix Nutrition*, Prop. 65 Settlement; *FDA, Guidance for Industry: Action Levels for Lead…*, January 2025). The Consolidated Sources Index in Part 5.4 provides the full list of cited authorities. The citation chain is now navigable inline rather than requiring traversal through the Index.

Evidence-base claim verifiability

Attack. "Part 1.5 asserts that 'every limit in this manual traces to source pages' on heavymetalindex.com, and that 'qualified reviewers may request the workbench-side aggregate distribution data under controlled terms.' But the manual does not specify who counts as a qualified reviewer, what the controlled terms are, or how long the response takes. The auditability claim is operationally unverifiable."

Response. Part 5.1 now includes the §"Controlled-terms request process" specification. Qualified reviewers are enumerated: regulatory bodies acting under official inquiry; plaintiff's or defense counsel in active litigation; certified or pending HMTc Licensees auditing their own product category's standards; academic researchers with institutional approval. Requests are submitted via the Colophon contact channels with a **30-business-day response commitment**. Regulatory bodies receive the requested data without an NDA; non-regulatory requestors receive access under a controlled-access NDA that preserves engine trade-secret status while making inputs and outputs auditable. Refusal decisions are reviewable by the Independent Appeals Committee. The Program Operator publishes annual statistics on request volume, response times, and disposition. Part 5.6 Licensing Terms adds the corresponding Licensee audit right as a contractual commitment.

The auditability claim is now operationally specified with a documented response commitment and a transparency mechanism.

Regional adaptability mechanism

Attack. "Part 2.1 §1.6 asserts that 'regional adaptability is a structural feature' and that the more-stringent-of-(regional, global) rule governs. But the mechanism for regional recalibration is hand-waved: no operational threshold for when a region gets its own platform; no specification for how the global cleanest-achievable benchmark gets computed; no statement of which regions are currently calibrated regionally. The claim is aspirational, not operational."

Response. Part 2.1 §1.6 has been revised to characterize regional adaptability as a **roadmap commitment for future Editions**, not an operational feature of the 2026 Edition. The 2026 Edition is honestly disclosed as calibrated against the global open literature; the next Edition's regional-adaptability work will begin once Cat 2 and additional categories ship and the corpus has accumulated region-specific evidence at sufficient sample-size totals to support statistically defensible regional pooling. The Governing Principles remain universal; the *calibration* may become regional in a future Edition with the more-stringent-of rule operative at that time. This converts the previous overclaim into a documented forward commitment.

Confidential Remediation Track discovery dynamics

Attack. "The Track produces 'a documented, timestamped, third-party-verified record of continuous improvement.' But the record is held by the Program Operator. In discovery, the record is potentially compellable from the Program Operator under subpoena. The Track does not provide attorney-client privilege or work-product protection. Brands relying on the Track for legal defensibility may be misled about how strong the protection actually is."

Response. Part 2.4 now includes an explicit §"Discovery dynamics, honest disclosure" section. Track data is held by the Program Operator as a service provider; it is potentially compellable under subpoena; the License Agreement specifies that the Program Operator will notify the

Licensee and cooperate with the Licensee's defense to the extent legally permissible. **The Track does not provide attorney-client privilege or work-product protection over the testing data**; what it provides is contemporaneous, third-party-verified evidence of due diligence. The evidence is admissible *for* the brand's defense; it is also discoverable by plaintiffs. The structural value is that the evidence exists, is timestamped, and was generated in good faith before any litigation was foreseeable. Brands evaluating the Track are explicitly directed to consult counsel about how the Track interacts with their broader discovery posture.

Standards Ratchet step-cap dynamics

Attack. "The 20% step cap is presented as protection for Licensees. But the trigger conditions in §12.3 use the *current* limit. After a 20% reduction, the trigger recomputes against the new (tighter) limit. The 'held-over' residual may never resurface because the market may consolidate at the new limit before producing another two-cycle confirmation. The manual claims the residual 'may trigger again' but does not show the dynamics."

Response. Part 2.10 §12.4 now includes a **worked example** demonstrating the step-cap dynamics. The example shows that when the market moves beyond the 20% cap, the cap holds the first cycle's tightening at 20%, then the residual surfaces at the next annual review when the trigger recomputes against the new (tighter) limit. The example demonstrates that the residual does resurface, the dynamics are not pathological, but the cadence is determined by market behavior, not by mechanical compounding. Brands can model their future obligations against the example with confidence.

Aggregate-exposure framing accuracy

Attack. "The Part 2.2 aggregate-exposure reasoning selects mixed-stringency anchor values across the contributing product categories. The asymmetry is rhetorically tuned. The calculation should use either worst-case values (all at the applicable regulatory action level) or surveillance medians consistently, not a mix."

Response. Part 2.2 has been revised to apply the applicable regulatory action level uniformly across the contributing product categories rather than mixing stringency levels, so the aggregate-exposure reasoning rests on a consistent worst-case anchor with no asymmetric value selection. The toxicology cross-callout was revised in parallel.

Supplier-intelligence anonymization

Attack. "Part 2.9 anonymizes alerts by removing brand identifiers, but ingredient × geography × timing is often a near-unique identifier. A sophisticated observer correlating a brand's public sourcing disclosures with the timing of alerts can identify the reporting brand. Anonymization is metadata-leaky."

Response. Part 2.9 now includes an explicit §"Anonymization limits, honest disclosure" section acknowledging the metadata-correlation risk. The Program Operator's three operational mitigations are documented: batching (multiple reporting brands' findings combined into one alert), geographic generalization (region rather than exact origin), and discretionary hold (alerts delayed at the reporting brand's request). Brands are explicitly told that anonymization is **strong but not absolute**, that sophisticated observers correlating public sourcing disclosures could potentially infer the reporting brand, and that the metadata-correlation risk should factor into the participation decision. The Program Operator does not warrant absolute anonymization.

Dispute resolution forum

Attack. "The Part 5.6 dispute-resolution provisions specify California-seated arbitration under California law. A non-California Licensee may argue the choice of forum disadvantages them in jurisdictional terms, or that California arbitration awards face enforcement friction in other states."

Response. Part 5.6 §"Dispute resolution" specifies California-seated mediation followed by AAA Commercial Arbitration in San Diego. The choice of forum reflects three operational considerations: (a) MDL 3101[24] and the relevant baby-food heavy-metal product liability case law are being made in the Northern District of California, so the opera-

tive legal landscape is California's; (b) AB 899[21] and Proposition 65[28] are California-state regimes whose enforcement intersects directly with HMTc certification obligations; (c) the Program's defense counsel operates from San Diego, locating defense expertise in the dispute forum. California arbitration awards are enforceable in every U.S. jurisdiction under the Federal Arbitration Act, so a California award reaches non-California Licensees without enforcement friction. Non-U.S. Licensees may discuss alternative-forum proposals during license negotiations.

Forward-forecast accountability

Attack. "Part 2.10 §12.6 requires the Program Operator to publish a forward forecast. But the manual does not specify what happens if the forecast turns out to be wrong. Brands plan against the forecast; if the forecast systematically underestimates the next-cycle limit, brands are misled. There is no accountability mechanism for forecast accuracy."

Response. Part 2.10 §12.6 now includes a **§"Forecast accountability"** clause. The Program Operator publishes an annual retrospective comparison of prior forecasts to actual ratchet outcomes in the transparency report. Material divergence between a forecast and its corresponding outcome, operationally defined as a forecasted limit that differs from the eventual actual limit by more than 25%, triggers a methodology review of the projection method. The forecast is now a calibrated planning tool with documented predictive accuracy over time, not a marketing artifact.

Closing of the program-level adversarial read

The fourteen attack categories above are the categories the Pre-publication Audit identified as the most likely program-level attacks a sophisticated adversarial reviewer would advance. Each has been responded to on the merits, with manual revisions applied where the audit identified gaps. Two categories required methodology revision (ratchet gaming, institutional conflict of interest) and were approved by the Standards Architect before being applied. One category was resolved by honest disclosure rather than mechanism change (regional adaptability roadmap framing) for the 2026 Edition; the operational mechanism will appear in a future Edition.

A reviewer who finds an attack vector not addressed above is invited to submit it to the Program Operator through the channels in the Colophon. The Pre-publication Audit is a planned recurring exercise; subsequent Editions will incorporate additional attack categories as they are identified. The adversarial-audit posture is the program's structural commitment to anticipating attack rather than waiting for attack to land.

Cross-references

- Pre-publication Audit document: `data/workbench/standards-book/adversarial-audit.md`.
- Each per-product adversarial read: Part 3 chapters.
- Governing Principles authority: Part 5.1 and `docs/governance/hmtc-governing-principles.md`.
- Independent Appeals Committee specification: Part 2.12 §14.1.
- Standards Ratchet anti-gaming provisions: Part 2.10 §12.3 and §12.7.

5.3 Glossary of Terms and Abbreviations

This glossary unifies three vocabularies that converge in the HMTc Program Manual: program-operations terms specific to HMTc; regulatory and toxicological terms used by FDA, EFSA, JECFA, and other agencies; and analytical-chemistry terms common to ISO/IEC 17025-accredited laboratory practice. Definitions are drawn from authoritative sources where they exist (Governance Policy §3 for program-operations terms; the metal pages on heavymetalindex.com for toxicological terms; ISO and EURACHEM-CITAC documents for analytical terms) and are reproduced here so that the manual is self-contained.

The glossary travels with every licensing-exhibit extraction (Part 3 chapter detached as an exhibit). Each Part 3 chapter additionally inline-defines unfamiliar terms on first use; the consolidated glossary below is the canonical reference.

Achievability Threshold. In HMTc methodology, the contamination concentration level at which a defined percentage of products in a category can comply. The Standards Ratchet (Part 2.10) uses the 80th-percentile achievability threshold of the certified trigger pool: when the 80th percentile sits at or below 80% of the current limit (and the breadth condition is met), the trigger fires. Achievability is the empirical observation that drives the ratchet; it is the program's data-driven mechanism for translating market improvement into tighter standards. Distinct from the standards-setting percentiles (P97 clean / P45 dirty per Part 5.1) which use external surveillance data for initial limit-setting.

Action Level. A practical concentration limit used to trigger sourcing, process, and quality control measures. Exceeding an action level signals

that intervention is needed; it is not a safety threshold. (HMTc-specific; per Governance Policy §3 and Part 2.1 §1.2.)

Aggregate Exposure. Total intake of a contaminant across all dietary sources, expressed as µg/day or µg/kg bw/day. A relevant exposure metric because compliance with single-product action levels does not guarantee compliance with health-based exposure benchmarks across a consumer's full diet. (See Part 2.2 aggregate-exposure framing.)

ALARA. As Low As Reasonably Achievable. A risk-management principle emphasizing feasible, continuous reduction of contaminant levels over time, especially when no safe threshold is known. Operationalized in HMTc as an enforceable obligation, not aspirational language. (Per Governance Policy §1.3 and Part 2.1.)

BMD / BMDL / BMDL01. Benchmark Dose / Benchmark Dose Lower Confidence Limit / BMDL corresponding to a benchmark response of 1 percent. Standard concepts in dose-response modeling. The BMDL is the lower bound of the statistical confidence interval around the BMD and is used as a conservative point of departure in risk assessment. BMDL01 is the BMDL at the 1% response level, commonly used for endpoints such as the EFSA 2010 lead BMDL01 of 12 µg/L blood lead for one-point IQ decrement.

bw. Body weight. Used to normalize intake or dose, typically expressed as µg/kg bw/day or mg/kg bw/week.

CAPA. Corrective and Preventive Action. A formal, documented process to identify root causes and implement corrections and preventive measures. Specified in Part 2.8 with status-specific timelines.

Certification Mark (Seal). The proprietary HMTc mark that may be used on packaging and marketing materials to indicate a product's certified status under the Program. Mark usage governed by Part 2.5.

Controlled Document. A document under formal version control maintained by the Program Operator, with documented version numbers, effective dates, change logs, and the requirement that the version in effect at the time of any decision governs that decision. The Governance Policy and Lot Testing Schedule are controlled documents. Subsequent revisions do not retroactively alter prior decisions. The Program Manual reproduces controlled documents at the time of binding; revisions

between editions of the manual flow through the controlled-document registry first.

Cr(III). Trivalent chromium. The form ubiquitous in food, recognized as a nutritionally essential trace element with no chronic-toxicity concern at dietary exposures.

Cr(VI). Hexavalent chromium. A Group 1 IARC carcinogen by ingestion (the cancer designation hardened from "inhalation only" in the 2014 reassessment). HMTc tests total chromium routinely and triggers Cr(VI) speciation only when a credible pathway for Cr(VI) formation exists (certain water-treatment processes; specific industrial contamination scenarios). When triggered, Cr(VI) is treated effectively at Tier 1 severity.

EFSA. European Food Safety Authority. An EU agency that produces scientific opinions on food and feed risks, including contaminant risk assessments used by regulators.

Heavymetalindex.com (HMI). The public curated corpus of the heavy-metals-in-food literature operated by The Paleo Foundation as a property architecturally separate from the HMTc certification program. The evidence layer that every HMTc standard rests on. Protected under U.S. copyright law and U.S. trade-secret doctrine; a parallel EU corporate structure for HMI is on the roadmap, at which point the corpus will additionally hold EU Database Directive 96/9/EC sui generis protection (15-year term). (Per Part 1.5 and Part 2.5.)

Hold-and-Release. The finished product lot is physically quarantined and may not ship until laboratory results are received and the lot is confirmed to meet at minimum the applicable status criteria. Hold-and-release procedures must include documented quarantine location, responsible personnel, and release authorization records. (Per Part 4 §3.1.)

HMTc. Heavy Metal Tested and Certified. The certification program specified in this manual. Operated by The Paleo Foundation at heavymetaltested.com.

iAs. Inorganic arsenic. The genotoxic carcinogenic form of arsenic, distinguished from organic arsenic species (predominantly arsenobetaine and arsenosugars in seafood) which are toxicologically distinct. Mandatory speciation for fish, shellfish, and seaweed products per Part

4 §9, where total arsenic is dominated by organic species and inorganic arsenic is the regulated, toxic fraction.

IARC. International Agency for Research on Cancer. WHO body responsible for the carcinogen classifications (Group 1 = carcinogenic to humans; Group 2A = probably; Group 2B = possibly) referenced throughout the toxicology callouts.

Independent Sampling. Samples are collected and submitted to the laboratory by a party independent of the brand's production team. This may be the Program Operator, a designated third-party auditor, or another party approved by the Program Operator. (Per Part 4 §3.1.)

IRL. Interim Reference Level. In the FDA context, a risk-management intake level used to prioritize and evaluate actions to reduce exposure, not a safety threshold. FDA expresses IRLs as a daily dietary intake (µg/day) for the population of concern.

JECFA. Joint FAO/WHO Expert Committee on Food Additives. An international expert body convened by the Food and Agriculture Organization of the United Nations and the World Health Organization to evaluate risks from food additives, contaminants, and residues.

Licensee. A brand or manufacturer that has signed a license agreement authorizing participation in the HMTc Program and, upon achieving certification, use of the HMTc Certification Mark.

Lot. A defined quantity of product manufactured under uniform conditions, identified by common date, line, or shift codes.

LOQ. Limit of Quantitation. The lowest concentration at which a laboratory can reliably quantify an analyte with acceptable precision and accuracy. HMTc requires laboratory LOQs at or below applicable action levels per Part 2.7 §8.3.

MaxPoL. Maximum Percent-of-Limit. The highest PoL among all compliance analytes for a given lot. MaxPoL drives the status determination per Part 2.3.

MeHg. Methylmercury. The neurotoxic form of mercury that crosses the blood-brain barrier and bioaccumulates in developing neural tissue. Mandatory speciation for fish-containing products per Part 4 §9.

MU. Measurement Uncertainty. The laboratory's stated expanded uncertainty (typically at 95% confidence, k=2) associated with a reported result. Incorporated into decision rules per Part 2.6.

Pb / Cd / Ni / Al / Sn / Hg. Standard chemical symbols for lead, cadmium, nickel, aluminum, tin, and mercury. Each is a priority HMTc analyte; see Part 2.2 for tier classification.

PoL. Percent-of-Limit. A result normalized to the HMTc limit: PoL = (Result / Applicable Limit) × 100%. A result equal to the limit is 100% PoL.

ppb. Parts per billion by mass in foods, typically equivalent to micrograms per kilogram (μg/kg). For liquids, ppb is often treated as μg/L when density is close to 1 g/mL.

Platform. In HMTc methodology, a subcategory of products within a parent product category, classified as either *clean* or *dirty* based on its empirical contamination profile across the open literature. A platform may be defined by species group (e.g., predatory vs. non-predatory marine fish), product type (e.g., finfish vs. shellfish vs. seaweed), or product format (e.g., fresh/frozen vs. canned). Platform classification is data-driven: a platform is *clean* when its empirical median across the literature-pooled distribution sits at or below the parent-category median for the analyte; *dirty* when meaningfully above (operationalized as p50(platform) > 1.5 × p50(parent category) for at least one Tier 1 metal). Platform classification determines whether per-analyte standards use the P97 (clean) or P45 (dirty) percentile (per Part 5.1).

PTWI / PTMI. Provisional Tolerable Weekly Intake / Provisional Tolerable Monthly Intake. Older JECFA benchmark formats used when the guidance value was considered provisional due to data limitations or expected updates. In practice they functioned like TWIs. JECFA withdrew its lead PTWI in 2011 on the grounds that it was no longer health-protective.

Regulatory Trigger Level (RTL). Any concentration or condition that would make the product noncompliant with applicable laws or regulatory action thresholds in its market(s) of sale. RTL violations trigger immediate hard-stop escalation per Part 2.11.

Reflex Testing. Any test conducted outside the routine surveillance schedule in response to a defined trigger event. Reflex tests carry the same weight as routine surveillance results. (Per Part 4 §8.)

Speciation. Determination of the specific chemical form of a metal present, which is critical for accurate risk assessment. HMTc requires speciation in defined contexts (iAs for rice; MeHg for fish; Cr(VI) when credible pathway exists). (Per Part 4 §9.)

Subcategory. A product class within a parent product category, typically defined by a discriminating attribute such as ingredient base, processing method, or product format. In HMTc methodology, the term *subcategory* is interchangeable with *platform*, both refer to the classification units that the methodology calibrates against. See *Platform*.

Status A / B / C / D / E. The five HMTc certification statuses. Defined in detail in Part 2.3. A = Certified (Full Compliance); B = Certified (Transitional); C = Probation; D = Suspension; E = Revocation.

Technical Committee. The Program Operator's advisory body responsible for methodology review, metal reclassification recommendations between Tier 1 and Tier 2, and methodological input on contested certification appeals. The Technical Committee is composed of toxicology, food-safety, and analytical-chemistry experts with documented independence from individual Licensees. The Committee is convened by the Program Operator on a per-need basis and reports recommendations rather than rendering binding decisions; the Program Operator retains final authority on classification and methodology matters, subject to the Independent Appeals Committee's binding review authority on contested cases. (Per Governance Policy §2.3.)

TWI / TDI. Tolerable Weekly Intake / Tolerable Daily Intake. Health-based guidance values expressed as an amount per kilogram body weight per week or day, considered tolerable over a lifetime for substances where a threshold is assumed or can be used for risk management.

Three-Lot Baseline. All SKUs entering the HMTc Program must be tested across a minimum of three production lots to establish a baseline profile before any certification status is assigned. (Per Part 2.3 and Part 4 §4.)

Tier 1 Metal. A metal that is carcinogenic with no established safe threshold, neurotoxic with no established safe threshold, or bioaccumulative with long biological half-lives in children. Pb, Cd, iAs, MeHg. Zero tolerance for exceedance under HMTc.

Tier 2 Metal. A metal with an established tolerable daily intake, shorter biological half-life, or primarily acute rather than chronic toxicity profile. Ni, Sn, Al, Cr. Variance-eligible up to 150% PoL under HMTc.

Total Diet Study (TDS). FDA program that analyzes food and beverage samples for various contaminants including heavy metals. FY2018–2020 TDS detected Pb in 15%, As in 43%, Cd in 33%, and Hg in 10% of samples, the prevalence anchor used in Part 1.3.

VOE. Verification of Effectiveness. The documented demonstration that CAPA actions have achieved their intended result, confirmed through follow-up testing. CAPAs cannot be closed until VOE confirms resolution.

µg/day. Micrograms per day. An intake unit used to express daily dietary exposure.

µg/kg bw/day. Micrograms per kilogram body weight per day. An intake unit normalized to body weight, commonly used in toxicological reference values.

µg/kg (ppb in food). Micrograms per kilogram of food product, equivalent to parts per billion by mass.

µg/L. Micrograms per liter. A concentration unit commonly used for blood lead values and for dilute liquids.

Cross-references

- HMTc Governance Policy Section 3 (Key Definitions): Part 2 chapters (definitions are reproduced operationally where each term is used).
- Toxicology callouts and full per-metal reviews: heavymetalindex.com `wiki/metals/*` pages.
- ISO/IEC 17025 and EURACHEM/CITAC source documents for analytical-chemistry term definitions: Part 5.4 Consolidated Sources Index.

5.4 References

This numbered references list covers the citations used in Parts 1, 2, 2.5, and 5 of this manual. Inline citations in those parts appear as bracketed numbers [N] keyed to this list. Per-product references for the standards published in Part 3 chapters appear in each chapter's Sources Cited section and are not duplicated here; that chapter-scoped attribution is the authoritative trail for the limit values in each chapter.

Regulatory and toxicological authorities

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[2] U.S. Food and Drug Administration. *Guidance for Industry: Action Levels for Lead in Processed Food Intended for Babies and Young Children*. Final Guidance. January 2025.

[3] U.S. Food and Drug Administration. *Updated Interim Reference Levels for Dietary Lead to Support FDA's Closer to Zero Action Plan*. June 2022.

[4] U.S. Food and Drug Administration. *Action Level for Inorganic Arsenic in Apple Juice: Guidance for Industry*. June 2023.

[5] U.S. Food and Drug Administration. *Inorganic Arsenic in Rice Cereals for Infants: Action Level*. August 2020.

[6] U.S. Food and Drug Administration. *Compliance Program 7304.019: Toxic Elements in Food and Foodware, and Radionuclides in Food*. August 2023.

[7] U.S. Food and Drug Administration. *Closer to Zero: Reducing Childhood Exposure to Contaminants from Foods*. 2021.

[8] EFSA Panel on Contaminants in the Food Chain (CONTAM). Scientific Opinion on Lead in Food. *EFSA Journal* 2010;8(4):1570. doi:10.2903/j.efsa.2010.1570

[9] EFSA CONTAM Panel. Scientific Opinion on Arsenic in Food. *EFSA Journal* 2009;7(10):1351. doi:10.2903/j.efsa.2009.1351

[10] EFSA CONTAM Panel. Update of the Risk Assessment of Inorganic Arsenic in Food. *EFSA Journal* 2024;22(1):8488. doi:10.2903/j.efsa.2024.8488

[11] European Food Safety Authority. EFSA Sets Lower Tolerable Intake Level for Cadmium in Food. News Release, March 2009 (Cd TWI 2.5 µg/kg bw/week).

[12] European Food Safety Authority. Mercury in Food, EFSA Updates Advice on Risks for Public Health. Press Release, December 2012 (MeHg TWI 1.3 µg/kg bw/week; inorganic Hg TWI 4 µg/kg bw/week).

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EU and international regulatory documents

[19] Commission Regulation (EU) 2023/915 on Maximum Levels for Certain Contaminants in Food. May 2023.

[20] Codex Alimentarius. *General Standard for Contaminants and Toxins in Food and Feed* (CXS 193-1995).

State-level regulatory documents

[21] California Assembly Bill 899 (Food Safety: Baby Food; Health and Safety Code, effective January 1, 2024).

[22] California Department of Public Health. AB 899 Frequently Asked Questions.

[23] California Attorney General. Enforcement Advisory Letter on AB 899. March 6, 2026.

Litigation references

[24] Judicial Panel on Multidistrict Litigation. *In re: Baby Food Products Liability Litigation* (MDL No. 3101). Transfer Order, April 11, 2024.

[25] Northern District of California. District Court Order Referencing Over 100 Subsequently Filed Actions in MDL 3101. April 2025.

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[27] Reuters. "Amazon.com Seeks to End Lawsuit Claiming Rice Contaminated by Heavy Metals." July 29, 2025.

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[56] FDA. Ground Cinnamon Public Health Alert (elevated lead). 2024.

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[59] HMTc Lot Testing Schedule, v2.0. (Internal controlled document; reproduced as Part 4 of this manual.)

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Note on chapter-scoped sources

Each Part 3 chapter has its own Sources Cited section. Those chapter-scoped citations anchor the specific limit values published in that chapter and are the authoritative trail for those limits. The numbered references above cover the cross-cutting citations in Parts 1, 2, 2.5, and 5; they do not duplicate the per-product literature in Part 3. Adversarial readers auditing a specific limit in Part 3 should follow the chapter's Sources Cited entries to the underlying source pages on heavymetalindex.com per Part 5.1.

5.5 Recognition Framework: How HMTc Compares to Adjacent Programs

A reader evaluating HMTc may already be familiar with adjacent certification programs in the food and supplement space. This chapter situates HMTc relative to those programs so brand legal teams, retailer compliance staff, and adversarial reviewers can place HMTc accurately in the regulatory and certification landscape. The comparison is on the merits, what each program covers, how it operates, and where the meaningful differences lie.

The intent is not to denigrate other programs or claim HMTc is uniformly superior. Each program in this comparison has legitimate scope and value. The intent is to clarify what HMTc specifically does that adjacent programs do not, so a brand evaluating multiple certifications can choose appropriately rather than treating all certifications as interchangeable.

Comparison matrix

Feature	HMTc	USDA Organic	Non-GMO Project	Clean Label Project	EWG Healthy Living	Consumer Reports Testing
Scope: heavy metals specifically	Yes (primary)	No	No	Yes (heavy metals included in scope)	Yes (in product scoring)	Yes (when tested)

Feature	HMTc	USDA Organic	Non-GMO Project	Clean Label Project	EWG Healthy Living	Consumer Reports Testing
Risk-tiered metal classification (Tier 1 / Tier 2)	Yes	N/A	N/A	No	No	No
Per-analyte action levels with regulatory-cap interaction	Yes (Part 3)	N/A	N/A	Yes (product specifications)	No (scoring-based)	N/A (testing report)
Aggregate-exposure framing across categories	Yes (Part 2.2)	No	No	Partial	No	No
Staged compliance (Status A/B/C/D/E)	Yes	Binary	Binary	Binary	Scoring-based	N/A
Confidential Remediation Track	Yes (Part 2.4)	No	No	No	N/A	N/A
Measurement uncertainty in de-	Yes (Part 2.6)	N/A	N/A	Partial	N/A	Reported but not adjudicated

Feature	HMTc	USDA Organic	Non-GMO Project	Clean Label Project	EWG Healthy Living	Consumer Reports Testing
cision rules						
ISO/IEC 17025 lab requirement	Yes (mandatory)	No	No	Yes	N/A	Yes (CR labs accredited)
Speciation requirements (iAs, MeHg, Cr-VI)	Yes (Part 4 §9)	No	No	Partial	No	Partial
Standards Ratchet (mandatory tightening over time)	Yes (Part 2.10)	No	No	Reviewed periodically	N/A	N/A
Public evidence corpus	Yes (heavymetalindex.com)	N/A	N/A	No	No	No
Supplier intelligence sharing across brands	Yes (Part 2.9)	No	No	No	N/A	N/A
Anti-arbitrage mark reentry rules	Yes (Part 2.5 §6.3)	N/A	N/A	N/A	N/A	N/A
No-testing-into-com-	Yes (Part 2.6 §7.3)	N/A	N/A	Partial	N/A	N/A

Feature	HMTc	USDA Organic	Non-GMO Project	Clean Label Project	EWG Healthy Living	Consumer Reports Testing
pliance prohibition						
Lot-level surveillance (every-lot at probation)	Yes (Part 4)	No	No	No (annual)	N/A	Spot
Underlying scope	Seafood (Cat 6); other categories coming	Agricultural production methods	GMO ingredient sourcing	Multi-contaminant, all foods	All consumer products (scoring)	Selected products (advocacy)

What HMTc does that adjacent programs do not

The matrix above is dense; the practical implications are:

Compared to USDA Organic. Organic certifies agricultural production methods (no synthetic pesticides, fertilizers, GMOs) and does not address heavy metal contamination at the product level. Organic certification does not preclude elevated heavy metals, organic seaweed, organic-feed farmed fish, and wild-caught shellfish can all carry significant heavy metal contamination because the metals come from marine sediment, water, and atmospheric sources that organic standards do not regulate. **A brand with Organic certification still needs HMTc to certify heavy metal limits.** The two are complementary rather than substitutable.

Compared to Non-GMO Project. Non-GMO certifies the absence of genetically modified ingredients at defined detection thresholds. It has no scope on heavy metal contamination. Same complementarity logic as Organic.

Compared to Clean Label Project. Clean Label Project does cover heavy metals and uses laboratory testing. HMTc differs in: explicit staged compliance with a Confidential Remediation Track for brands not yet at certification standards; the measurement-uncertainty two-sided decision rule that protects brands from analytical noise; the Standards Ratchet that mandates tightening over time as the certified market improves; the supplier-intelligence-sharing system across all certified brands; and the public evidence corpus (heavymetalindex.com) that makes the methodology auditable. Brands operating under Clean Label may also pursue HMTc; the two programs make different commitments, and a brand may value the HMTc commitments specifically.

Compared to EWG Healthy Living. EWG provides scoring-based product evaluation across multiple chemicals of concern, including heavy metals. EWG's methodology is rating-based rather than certification-based, and it does not provide the structured compliance, surveillance, CAPA, or ratchet mechanisms that HMTc does. EWG scoring is a useful consumer-facing tool; HMTc is the corresponding industry-facing certification framework.

Compared to Consumer Reports testing. Consumer Reports periodically tests selected products in selected categories and publishes findings. The testing methodology is rigorous (ISO/IEC 17025-accredited labs) but the reporting is in the exposé-publication tradition that Part 1.2 of this manual analyzes. Consumer Reports tests do not produce a structured certification framework, do not offer brands a remediation pathway, and do not generate supplier intelligence shared across brands. The two serve different purposes; HMTc participation is not a substitute for responding appropriately to a Consumer Reports finding, and a Consumer Reports finding does not preclude HMTc certification.

What this comparison does not mean

HMTc is **not** a replacement for regulatory action. FDA, EFSA, state attorneys general, and other regulatory bodies set the legal floor; HMTc operates above that floor with tighter action levels and structured surveillance. A brand passing HMTc is operating well within the regulatory envelope; a brand violating regulatory limits is in §2.11 escalation regardless of HMTc status.

HMTc is **not** a replacement for internal brand QA. Brands enrolled in HMTc continue to operate their own quality programs; HMTc provides the third-party-verified evidence record, the cross-brand collective intelligence, and the certification mark. Internal QA and HMTc certification are operationally aligned but structurally distinct.

HMTc **does not certify safety.** Per Part 2.5 §6.2 and the action-levels-not-safety-thresholds framing throughout this manual, the certification mark indicates that a product has been independently tested and meets defined action levels under ongoing surveillance. The mark is not a safety guarantee. Brands using the mark may not make safety-implying claims.

Cross-references

- Action levels not safety thresholds: Part 2.1 §1.2.
- Mark usage rules and prohibited claims: Part 2.5.
- Standards Ratchet specification: Part 2.10.
- Supplier intelligence system: Part 2.9.
- Public evidence corpus: Part 1.5 and Part 2.5.

5.6 Licensing Terms, Summary

This chapter summarizes the structural terms of the HMTc license agreement so that a brand legal team can understand the shape of what they would be signing without needing to request the agreement itself for an initial evaluation. **The actual licensing agreement is a separate document**, executed bilaterally between the Licensee and The Paleo Foundation, and contains the full operative terms. The summary below is informational and does not modify or supersede the executed agreement.

The summary covers fee structure, term length, mark-usage scope, anti-arbitrage rules, confidentiality, and termination. Brands at the evaluation stage should expect the actual agreement to mirror this structure with the specific operative language.

Fee structure

HMTc License Fees are **revenue-tiered**, calculated based on the Licensee's annual revenue at the time the license agreement is executed. A brand with twenty certified SKUs at a single revenue tier pays the same License Fee as a brand with two certified SKUs at the same tier. This structure reflects the program's commercial-viability principle (Governing Principle 5): fee structure should not penalize brands for breadth of certification, because the program's public-health objectives are served by maximizing the volume of certified SKUs in the market.

Revenue tiers, by brand size:

- **Small brands** (under $5M annual revenue) pay an entry-tier License Fee designed to make certification accessible without subsidizing larger competitors.

- **Mid-size brands** ($5M–$50M annual revenue) pay a mid-tier License Fee scaled to the financial benefit of certification at this revenue band.
- **Large brands** ($50M–$500M annual revenue) pay a higher License Fee reflecting the proportional ROI delivery the program enables.
- **Enterprise brands** (over $500M annual revenue) negotiate enterprise terms with the Program Operator directly.

The License Fee is due and payable to the Program Operator upon execution of the license agreement and covers the initial term (see Term Length, below). The Program Operator may, in its discretion, approve an installment arrangement under which the License Fee is paid over the course of the initial term. Specific fee schedules are published in the License Agreement Schedule and are reviewed annually by the Program Operator. Brands at the evaluation stage can request the current fee schedule from the Program Operator. The schedule is not reproduced in this manual because fee schedules are operational rather than methodological.

Testing costs (laboratory fees for the testing required under Part 4) are **separate from License Fees** and are paid by the Licensee directly to their ISO/IEC 17025-accredited laboratory. The Program Operator does not mark up testing fees; the laboratory market provides the cost discipline.

Term length and renewal

License agreements are executed for an **initial term** of one to three (1–3) years, selected by mutual agreement at signing. At the end of the initial term, the Licensee may notify the Program Operator at least ten (10) days before expiration of intent to extend the license for an Extended Term of one to three additional years in consideration of an additional License Fee. Failure to provide timely notice causes the agreement to terminate at the end of the initial term.

Following the initial term and any Extended Term, the Licensee may renew the license by providing the Program Operator at least sixty (60) days' written notice prior to expiration. Renewal is by good-faith negotiation of a new license agreement, which may incorporate revised terms

or revised standards (including any Standards Ratchet revisions per Part 2.10).

Either party may terminate the license at any time, for any reason, upon thirty (30) days' prior written notice. In the event of alleged breach, the non-breaching party provides written notice and the other party has thirty (30) days to cure. The Program Operator may terminate for cause immediately upon written notice for Trademark misuse, use in connection with non-Approved Goods, or failure to pay any installment of the License Fee. For-cause termination triggers Status E revocation per Part 2.3 §5.6 and forfeits the sell-off rights described below.

Upon termination or expiration, the Licensee has 180 days (the Sell-Off Period) to sell existing inventory of Approved Goods bearing the Trademark, except in the case of Termination for Cause where no Sell-Off Period applies. After the Sell-Off Period, the Licensee must destroy or remove all remaining product, marketing materials, and online content bearing the Trademark.

Mark usage scope

The HMTc Certification Mark may be used on certified SKUs' packaging, marketing materials, and digital channels, subject to:

- **Eligibility:** Only Status A or Status B SKUs may bear the mark (Part 2.5 §6.1).
- **Mark style guide:** The Program Operator publishes a Mark Usage Style Guide governing the mark's visual presentation, minimum size, color treatment, and accompanying language. The style guide is reviewed annually.
- **Prohibited claims:** No safety-implying claims; no "heavy metal free" language (Part 2.5 §6.2).
- **Anti-arbitrage:** Re-entry after dropping the mark requires three consecutive passing lots and CAPA closure (Part 2.5 §6.3).
- **Geographic scope:** The license grants mark usage in all jurisdictions where the brand sells the certified SKU.

The mark is a registered certification mark (USPTO filing in process; EU IP Office filing in parallel). Unauthorized use of the mark by non-Licensees is enforced by the Paleo Foundation through standard certification-mark protection mechanisms.

Licensee audit right

The license agreement grants each Licensee the right to audit the methodology that produces the standards applicable to the Licensee's certified product categories. The audit right operates under the **controlled-terms request process** specified in Part 5.1: a Licensee may request the workbench-side aggregate distribution data, the contributing-source summary statistics, the methodology's stability under alternative parametric assumptions, and the full methodology documentation for any product category their certification covers. Access is granted under a controlled-access NDA that preserves trade-secret status of the engine implementation while making the inputs and outputs verifiable.

The audit right is operational, not promissory. The Program Operator responds to Licensee audit requests within 30 business days per the Part 5.1 commitment.

Confidentiality

The license agreement establishes mutual confidentiality with directional flow:

- **Brand-provided data is confidential.** Lot testing results, CAPA documentation, supplier identifications, and other operational data the brand provides to the Program Operator are treated as confidential business information. The Program Operator may use aggregated, anonymized data for supplier intelligence (Part 2.9), the Standards Ratchet trigger analysis (Part 2.10), and aggregate Program reporting. Individual brand identities and SKU-level results are not disclosed except as required by law or by the Brand Visibility Controls of Part 2.5 §6.4 where the brand has authorized disclosure.
- **Confidential Remediation Track confidentiality is contractually binding.** Brands enrolled in the Track per Part 2.4 have particular confidentiality protections that survive license termination. The Program Operator's enforcement-cooperation obligations (Part 2.11) may require disclosure to regulatory authorities under defined circumstances; Track-specific disclosures to regulators are made under written notice to the affected Licensee where legally permissible.
- **Program-Operator-provided information is confidential.** Supplier intelligence alerts, methodology details, audit findings, and

other information the Program Operator shares with the Licensee are confidential and may not be disclosed by the Licensee to third parties without written consent except as required by law or by the brand's own operational use of the information (e.g., applying a supplier intelligence alert in the brand's own supply-chain decision-making).

Confidentiality survives termination of the license agreement for a period of seven (7) years, except for trade secrets (which are protected as long as they retain trade-secret status under applicable law) and except for legally required disclosures.

Regulatory cooperation

The Licensee acknowledges that the Program Operator will cooperate with regulatory authorities (FDA, EFSA, state attorneys general, equivalent international bodies) as required by law. **The Program Operator will not shield any Licensee from regulatory obligations.** This is consistent with Part 2.11′s hard-stop escalation rule and is a structural condition of the program's legal defensibility.

In practice, regulatory cooperation means: the Program Operator will respond to lawful regulatory inquiries about the Program's methodology and aggregate operation, will provide Licensee-specific information when required by lawful process, and will report regulatory limit violations to the appropriate authorities per Part 2.11. The Program Operator gives Licensees written notice of any compelled disclosure where legally permissible.

Dispute resolution

License agreement disputes are first submitted to mediation in San Diego, California, with a mutually-agreed mediator (or, failing agreement, a JAMS-administered mediation). Unresolved disputes proceed to binding arbitration under the American Arbitration Association's Commercial Arbitration Rules, heard by a single arbitrator who shall be a retired judge or intellectual property attorney. The substantive law of the agreement is California law; the seat of arbitration is San Diego, California. The arbitrator's award is enforceable in any court of competent jurisdiction,

including the San Diego Superior Court and the United States District Court for the Southern District of California.

Why California-seated arbitration. Paleo Certified, Inc. is a Florida corporation, but the substantive case law for heavy-metals product liability and food-safety regulation is being made in California: MDL 3101 (the consolidated baby-food heavy-metal litigation) is in the Northern District of California; AB 899 (mandatory baby-food testing and disclosure) is California state law; Proposition 65 (the private-enforcement civil-warning regime) is California state law; the Program's defense counsel operates from San Diego. Locating dispute resolution in the jurisdiction where the substantive law lives means that any dispute is heard by decision-makers familiar with the operative legal landscape. California arbitration awards are enforceable in all U.S. jurisdictions under the Federal Arbitration Act, so a California award reaches Licensees in any state.

Licensees outside the United States with specific concerns about California-seated arbitration should consult counsel and may discuss alternative dispute-resolution forums with the Program Operator during license negotiations. The Program Operator considers alternative-forum proposals on a case-by-case basis where material commercial considerations warrant it.

Certification status disputes (as distinct from license agreement disputes) follow the two-tier appeals process specified in Part 2.12, which is independent of arbitration and operates as the Program's internal dispute-resolution mechanism for status determinations.

What this summary does not cover

The actual license agreement contains operative terms not summarized here, including: representations and warranties; indemnification provisions; insurance requirements; intellectual property cross-licenses (Licensee's product images, brand assets, etc.); force majeure; assignment restrictions; notice provisions; entire-agreement clauses. Brands at the contracting stage should review the full agreement with their legal counsel.

Cross-references

- Mark usage rules (operational): Part 2.5.
- Confidential Remediation Track terms: Part 2.4.
- Appeals process for status determinations: Part 2.12.
- Regulatory escalation hard-stop: Part 2.11.
- Data governance and integrity (which interacts with confidentiality): Part 2.7.

5.7 Colophon

Edition

Heavy Metal Tested & Certified (HMTc) Seafood Program Manual, 2026 Edition. First bound publication. The HMTc methodology has evolved through documented revisions since 2024; the current edition reflects refinements to the per-analyte percentile calibration, the addition of the row-pass framework, the operationalization of the Standards Ratchet Mechanism with explicit trigger conditions and step-size caps, and the formalization of the Confidential Remediation Track.

Author and disclosure

Karen Pendergrass, Standards Architect, The Paleo Foundation. ORCID: [pending registration].

Karen Pendergrass is the Standards Architect for the HMTc Program and CEO of The Paleo Foundation. The dual role is disclosed transparently because it is the structurally honest disclosure: the Foundation operates the certification program; the Standards Architect develops the program's methodology; both roles are held by the same person. The operational separation between the two roles is documented in the Foundation's governance bylaws (controlled-document registry). The conflict-of-interest provisions of Part 2.12 §14.4 ensure that certification determinations are made by individuals free from conflicts with the Licensee under review, and the appeals process is structurally independent from CEO authority.

The Program is operated by Paleo Certified, Inc., a Florida corporation doing business as The Paleo Foundation. Corporate governance, financial reporting, and operational continuity are documented in Florida state corporate filings and in the Foundation's controlled-document set.

Methodology disclosure

The HMTc methodology is described in principle throughout this manual (particularly Part 5.1) and at peer-track depth in three published preprints:

- *The Counterproductive Consequences of Public Exposé Testing*. doi: 10.5281/zenodo.19470572.
- *The Cost of Operating Without Credible Third-Party Heavy-Metal Certification*. doi: 10.5281/zenodo.18903738.
- *Certification as a Framework for Reducing Heavy Metal Exposure in Infant and Child Foods*. doi: 10.5281/zenodo.18905821.

The engine arithmetic that implements the methodology (the source-pooling algorithm, the percentile calculation, the source-level bootstrap, the methodology stability checks, the basis-conversion factors) is held as a trade secret. The principle of standards-setting is public; the implementation is proprietary. This is the wiki/HMTc firewall established by the HMTc Governing Principles (Principle 4) and described in Part 1.5 of this manual.

Methodology implementation

The deterministic pooling engine and the Standards Briefing generator that produce per-product limit tables and methodology summaries were implemented by Karen Pendergrass with AI assistance from Claude (Anthropic). The engine code is regression-tested for percentile-inversion errors and author-attribution errors per the test suites in `tools/standards/build-standards-briefing.test.mjs`. Pre-publication review of each briefing follows the 9-gate checklist established in the methodology preprint.

This implementation acknowledgment is in the Colophon, not the byline. The byline names the Standards Architect; the implementation methodology is documented here for honest disclosure of tooling used in producing the bound manual.

Evidence base

Heavymetalindex.com (HMI) is the public curated corpus of the heavy-metals-in-food literature that this Program Manual rests on. HMI is operated by The Paleo Foundation as a property architecturally separate from the HMTc certification program at heavymetaltested.com. Architectural separation and one-way reference (HMTc cites HMI; HMI does not endorse HMTc) establish the evidentiary foundation that the certification program's standards are built on. The corpus is protected under U.S. copyright law and U.S. trade-secret doctrine; a parallel EU corporate structure for HMI is on the roadmap and will add EU Database Directive 96/9/EC sui generis protection once incorporated. See Part 1.5 for the full account.

Intellectual property

Trademarks. "HMTc," "Heavy Metal Tested & Certified," "Heavy Metal Tested and Certified," the HMTc certification seal artwork, and "The Paleo Foundation" are trademarks of The Paleo Foundation. Federal certification mark registration is in process with the United States Patent and Trademark Office under Lanham Act § 1054. Parallel EU trademark filing is in process with the European Union Intellectual Property Office.

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Database rights. The heavymetalindex.com corpus that this manual references is protected under U.S. copyright law and U.S. trade-secret doctrine. A parallel EU corporate entity for HMI is on the roadmap and will hold EU Database Directive 96/9/EC sui generis protection (15-year term) once incorporated. Substantial extraction or re-utilization of the corpus without authorization is prohibited under either protection regime.

Trade secret. The HMTc pooling engine and its associated methodology implementations (source-pooling algorithms, calibration analysis,

per-source weighting rules, and related components) are protected as trade secrets under Florida and U.S. federal trade-secret law (Defend Trade Secrets Act, 18 U.S.C. § 1836; Florida Uniform Trade Secrets Act). Authorized disclosure of engine internals occurs only under controlled-terms request to the Program Operator.

Production credits

Typesetting and design. This v0-pre-audit edition is rendered in markdown for review. The audit-strengthened v0 will be typeset using Typst or Quarto evaluated against the full 135-page document; the typesetting toolchain is selected after content lock. Typefaces (FreightText Pro for display and serif body; Avenir Next Cyr for sans-serif) are licensed for use under standard commercial typeface licenses.

Build pipeline. Content lives in markdown under `data/workbench/standards-book/` and is concatenated by `tools/standards/build-program-manual.mjs` (in development). The Program Manual is regenerated by the autonomy daemon whenever the source documents (Governance Policy, Lot Testing Schedule, Governing Principles, or any of the 5 product briefings) change, ensuring the bound manual stays consistent with the current state of the operational documents.

Contact

The Program Operator can be reached at:

- **Email:** karen@paleofoundation.com
- **Twitter / X:** @MicroMetallomics
- **Program website:** heavymetaltested.com
- **Evidence-base website:** heavymetalindex.com
- **Postal:** Paleo Certified, Inc. dba The Paleo Foundation, 623 Divine Circle, Orlando, Florida 32828, United States

Questions about specific certifications, the licensing process, brand enrollment in the Confidential Remediation Track, methodology audit requests, and regulatory cooperation should be directed to the Program Operator through these channels.

Adversarial reviewers, plaintiff's experts, defense attorneys, journalists, and academic researchers who need program documentation beyond what is reproduced in this manual should also direct inquiries through the same channels; the Program Operator responds to such inquiries on a documented timeline consistent with the Foundation's transparency commitments.

Version

This is **v0-pre-audit** of the 2026 Edition. The audit-strengthened v0 ships after Pre-work 6 (the adversarial audit of the drafted manual) is complete and the Program-Level Adversarial Read in Part 5.2 is rewritten from the audit findings. The audit-strengthened v0 will be the artifact distributed to early Licensees, regulators, and academic reviewers.

End of Heavy Metal Tested & Certified Seafood Program Manual, 2026 Edition.

5.8 About the Author

Karen Pendergrass is the Standards Architect of the Heavy Metal Tested & Certified (HMTc) program and the founder and Chief Executive Officer of Paleo Certified, Inc., the Florida corporation doing business as The Paleo Foundation. She is a microbial metallomics researcher with active investigation into the intersection of heavy metal exposure, the human gut microbiome, and developmental health outcomes. She works from Larnaca, Cyprus.

Her current research program centers on methodologically defensible certification standards for environmental contaminants in food, with particular attention to vulnerable populations (infants, young children, pregnant individuals) where aggregate exposure across multiple food categories systematically exceeds health-based reference values even at full regulatory compliance. The HMTc methodology, presented in this manual, is the operational expression of that research program: a third-party certification framework calibrated against the open peer-reviewed literature, anchored to a public curated corpus at heavymetalindex.com, governed by published principles that constrain both the wiki layer and the certification layer, and built to be more rigorous than current regulatory floors without losing commercial viability.

Pendergrass's peer-track publications in the *Journal of Food Metallomics* document the structural, financial, and design rationale that underlies the program reproduced in this manual:

- *The Counterproductive Consequences of Public Exposé Testing: How Unstructured Disclosure Undermines Heavy Metal Contamination Reduction*. 2026. doi: 10.5281/zenodo.19470572.[46]
- *The Cost of Operating Without Credible Third-Party Heavy-Metal Certification*. 2026. doi: 10.5281/zenodo.18903738.[47]

- *Certification as a Framework for Reducing Heavy Metal Exposure in Infant and Child Foods: Integrating Legal Defensibility, Scientific Rigor, and Commercial Viability*. 2026. doi: 10.5281/zenodo.18905821.[48]

Pendergrass's ORCID identifier is 0000-0002-2348-7259.

Standards Architect / CEO dual-role disclosure. Karen Pendergrass holds both the methodological authority (as Standards Architect, responsible for limit-setting calibration, methodology revisions, and Governing Principles) and the operational authority (as CEO of Paleo Certified, Inc., responsible for commercial decisions, licensing, and program continuity) for the HMTc program. This dual role is disclosed here as a matter of transparency. The structural mitigation for the conflict that could arise from this combination is documented in Part 2.12 §14.1 (the two-tier appeals process with the binding Independent Appeals Committee external to the Paleo Foundation) and in the firewall provisions of the Governing Principles that separate the wiki literature evidence layer from the threshold-setting methodology. Adversarial readers concerned with the dual role should follow those structural mitigations to the institutional answer.

Contact. Editorial questions about this manual, requests for the controlled-terms aggregate distribution data per Part 5.1, and academic correspondence about the methodology may be directed to the Program Operator via the contact channels in the Colophon (Part 5.7).